

Chapter 8: Introduction to Spring Cloud

Eureka

localhost:8761

spring Eureka

HOME LAST 1000 SINCE STARTUP

System Status

Environmenttest

Data centerdefault

Current time2023-03-27T06:30:21 +0000

Uptime00:46

Lease expiration enabledtrue

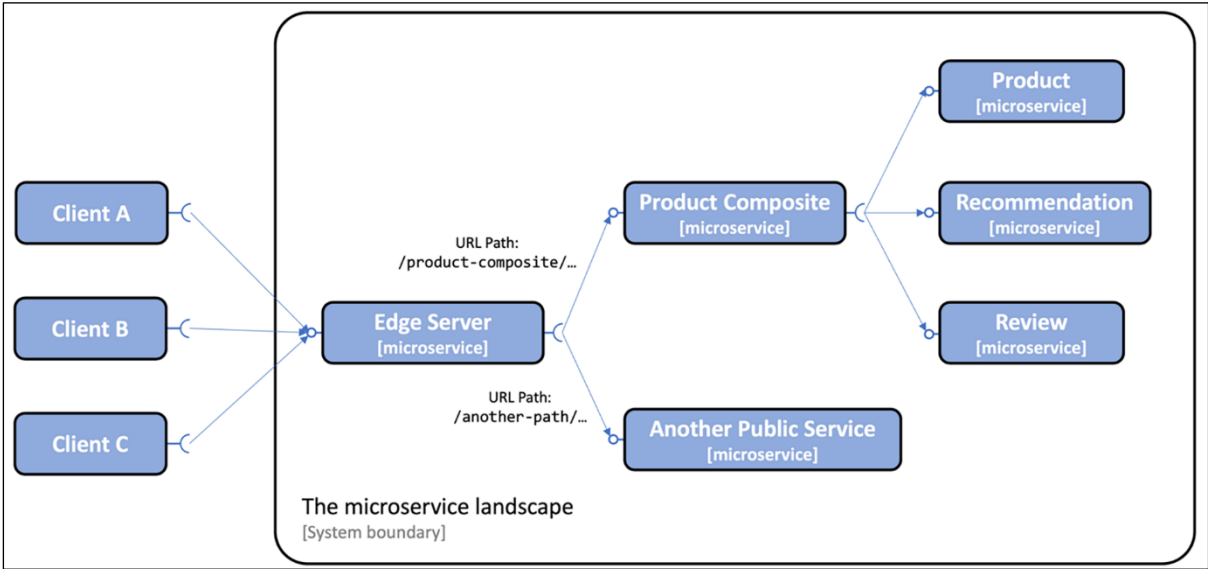
Renews threshold11

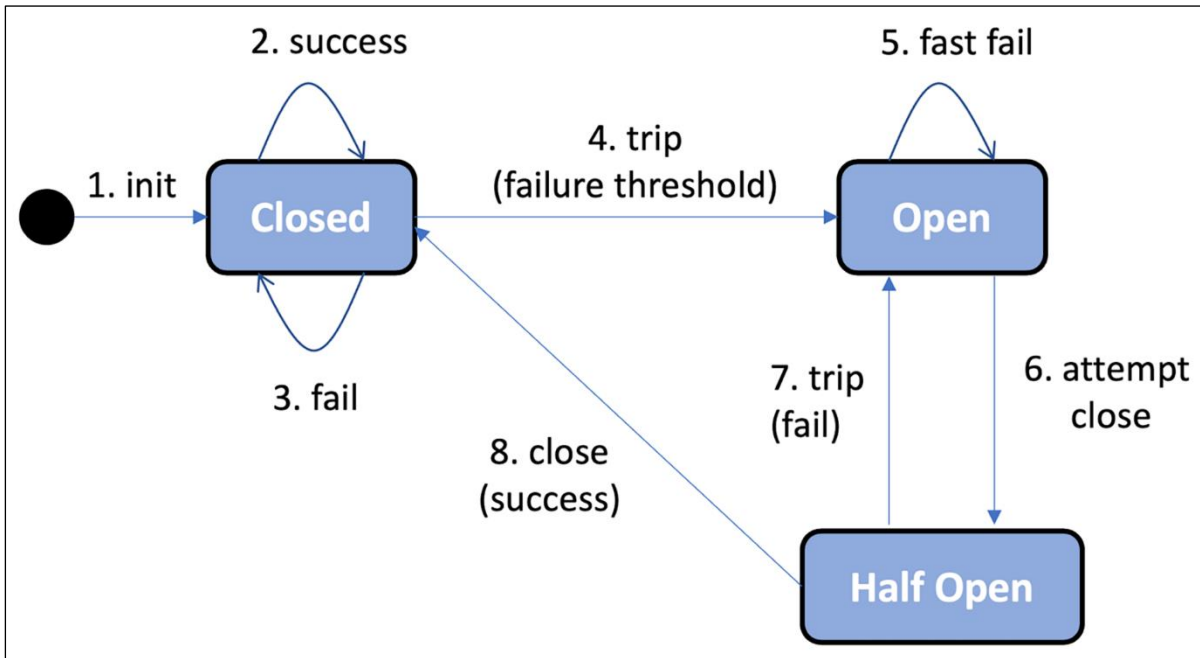
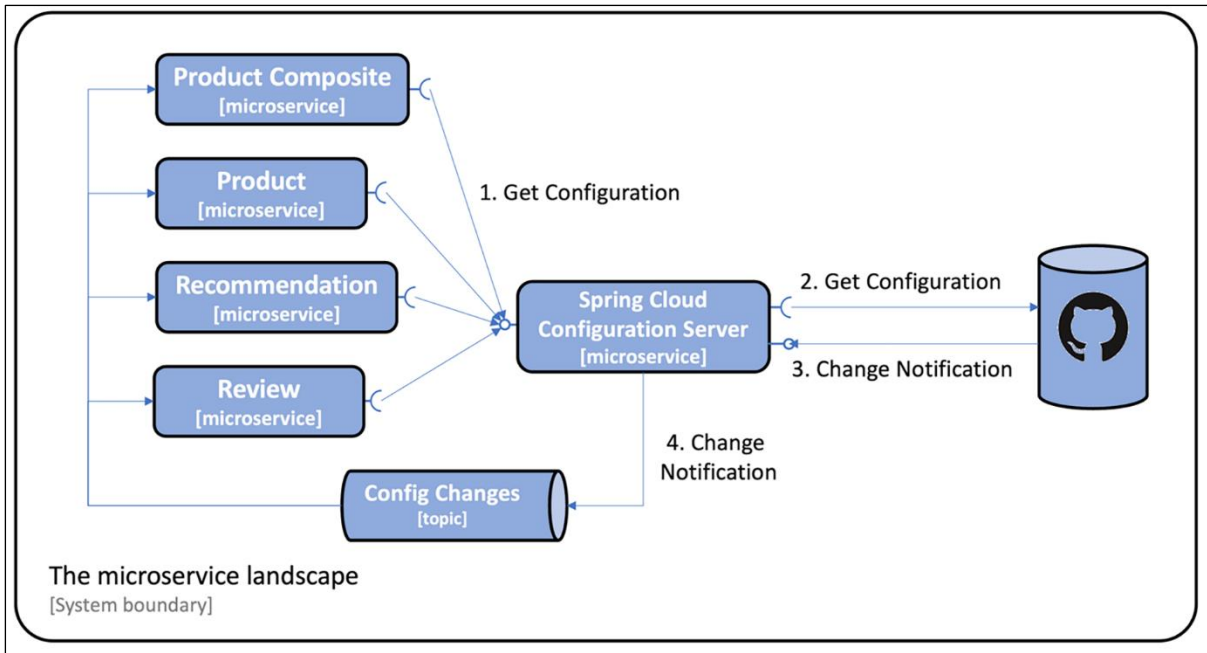
Renews (last min)48

DS Replicas

Instances currently registered with Eureka

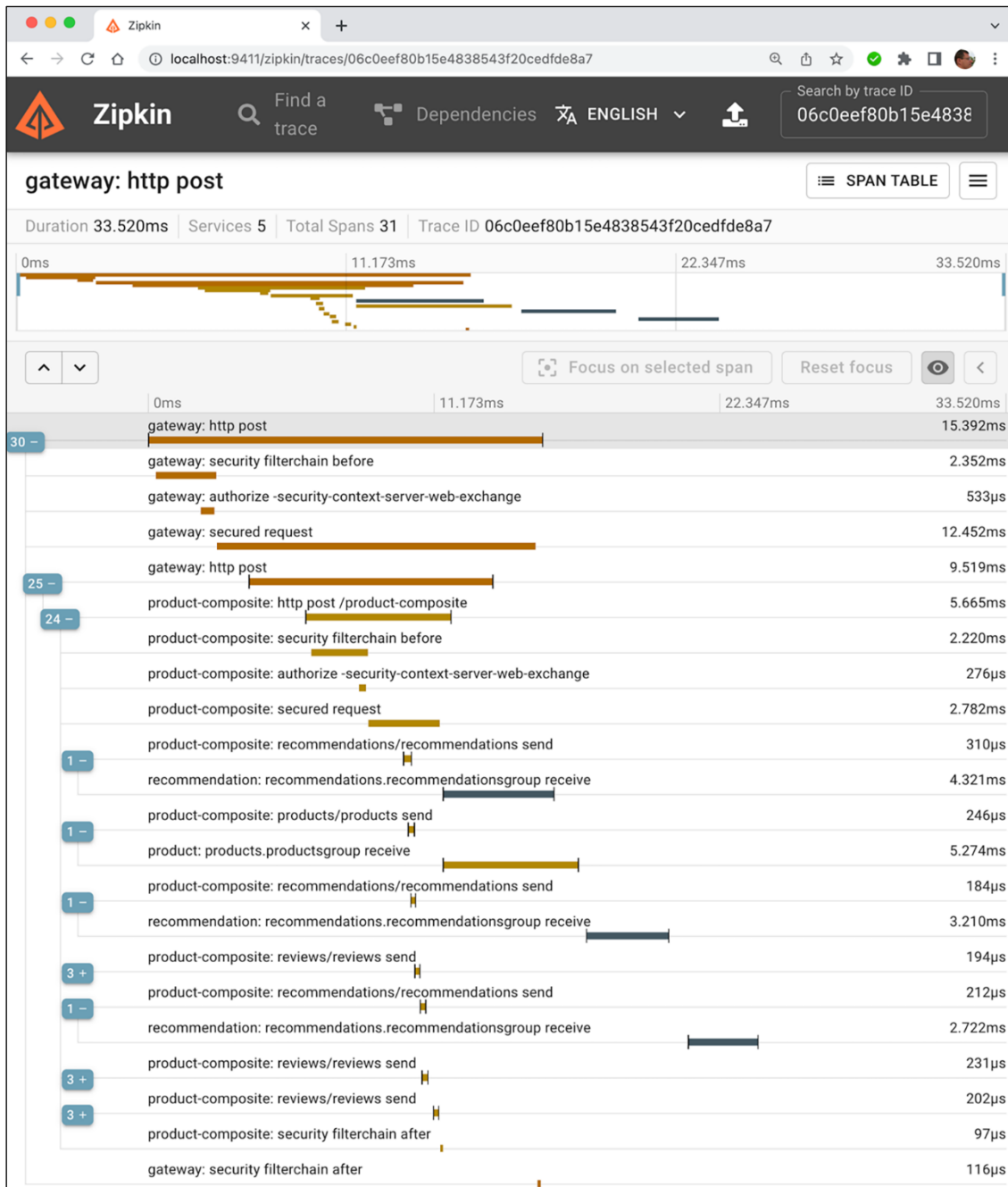
Application	AMIs	Availability Zones	Status
PRODUCT	n/a (1)	(1)	UP (1) - f864341e9cd7:product:8080
PRODUCT-COMPOSITE	n/a (1)	(1)	UP (1) - 4ac5b5b3f93b:product-composite:8080
RECOMMENDATION	n/a (1)	(1)	UP (1) - 509f6850f0d0:recommendation:8080
REVIEW	n/a (3)	(3)	UP (3) - 1b1a31d24e37:review:8080 , 68b4467ae262:review:8080 , 23f769e72255:review:8080



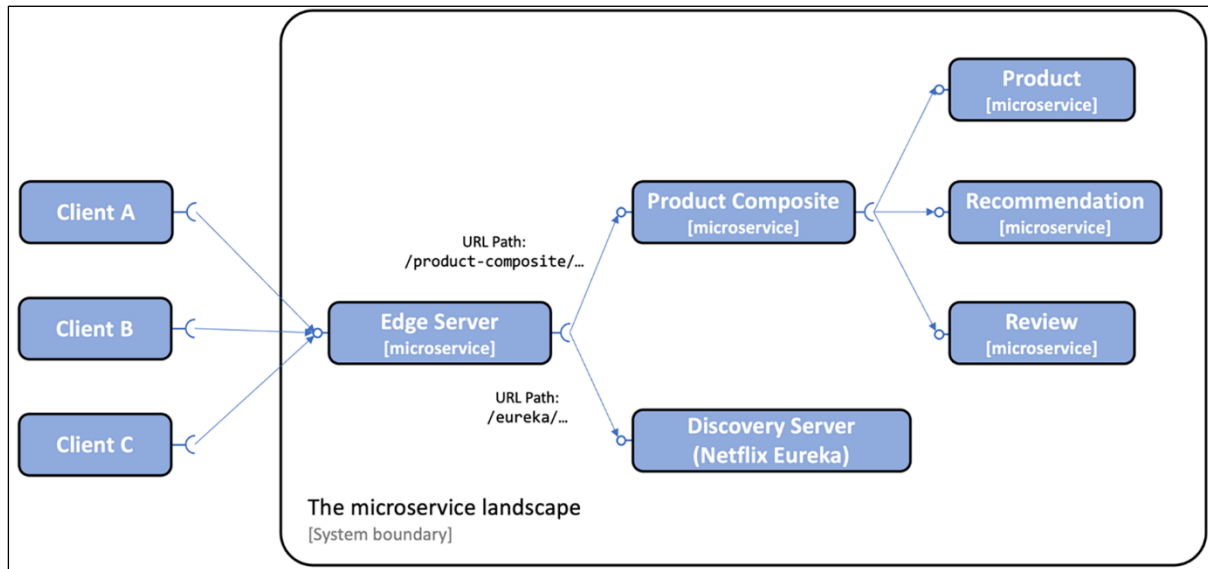


```
bash
{
  "status": "UP",
  "details": {
    "product": {
      "status": "UP",
      "details": {
        "failureRate": "-1.0%",
        "failureRateThreshold": "50.0%",
        "slowCallRate": "-1.0%",
        "slowCallRateThreshold": "100.0%",
        "bufferedCalls": 0,
        "slowCalls": 0,
        "slowFailedCalls": 0,
        "failedCalls": 0,
        "notPermittedCalls": 0,
        "state": "CLOSED"
      }
    }
  }
}
```

```
bash
{
  "status": "UNKNOWN",
  "details": {
    "product": {
      "status": "CIRCUIT_OPEN",
      "details": {
        "failureRate": "60.0%",
        "failureRateThreshold": "50.0%",
        "slowCallRate": "0.0%",
        "slowCallRateThreshold": "100.0%",
        "bufferedCalls": 5,
        "slowCalls": 0,
        "slowFailedCalls": 0,
        "failedCalls": 3,
        "notPermittedCalls": 3,
        "state": "OPEN"
      }
    }
  }
}
```

Chapter 10: Using Spring Cloud Gateway to Hide Microservices behind an Edge Server



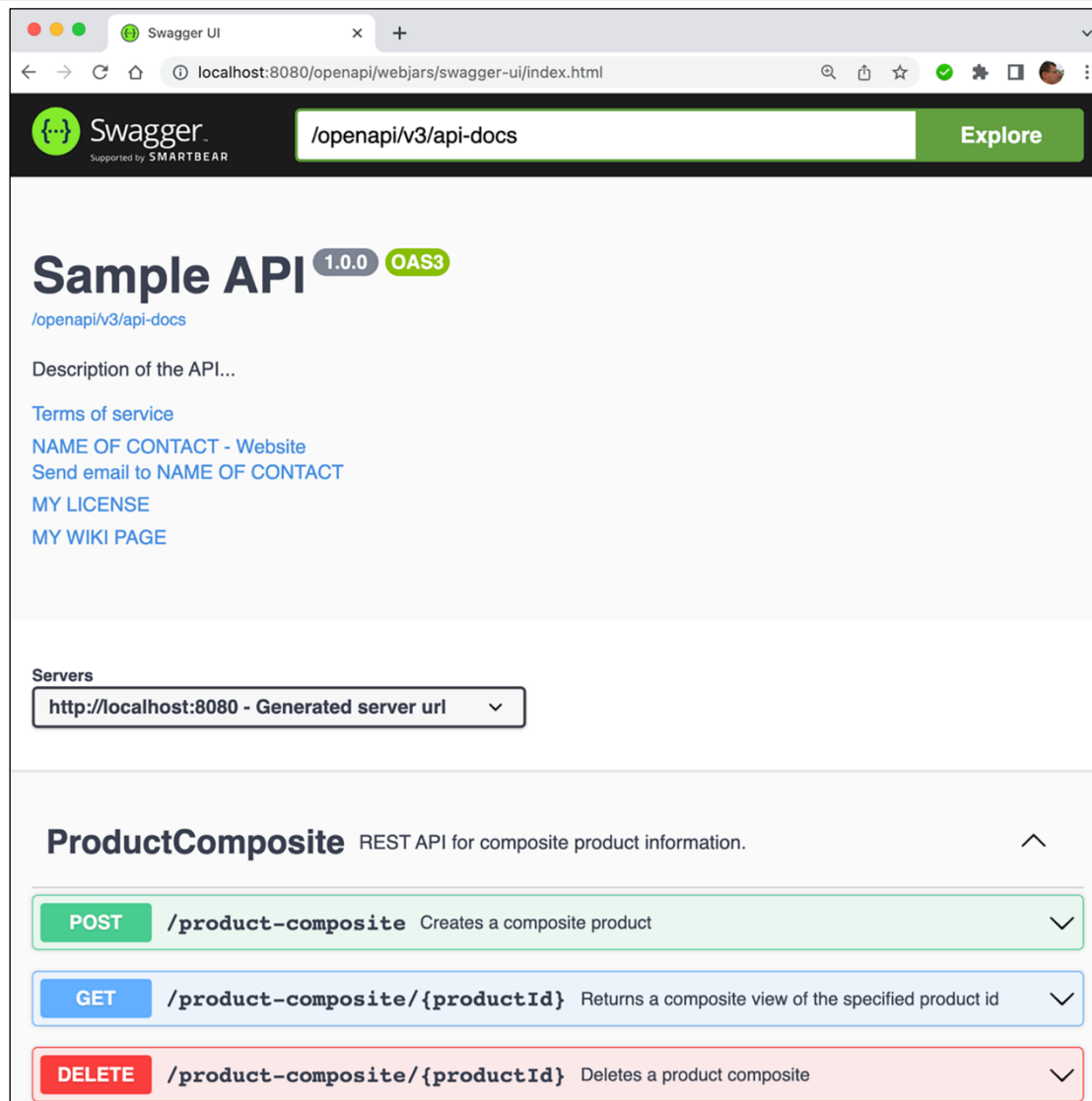
```
bash
$ ./test-em-all.bash start
Start Tests: Wed Mar 29 15:42:42 CEST 2023
...
:: Container chapter10-gateway-1          Started
...
Wait for: curl http://localhost:8080/actuator/health... DONE, continues...
...
Test OK (actual value: http://localhost:8080)
Test OK (HTTP Code: 200)
End, all tests OK: Wed Mar 29 15:43:10 CEST 2023
$
```

```
bash
NAME                                SERVICE    STATUS    PORTS
chapter10-eureka-1                  eureka     running   8761/tcp
chapter10-gateway-1                 gateway     running   0.0.0.0:8080->8080/tcp
chapter10-product-1                 product     running   8080/tcp
chapter10-product-composite-1       product-composite running   8080/tcp
chapter10-recommendation-1          recommendation running   8080/tcp
chapter10-review-1                  review      running   8080/tcp
$
```

```
bash
"product-composite": "lb://product-composite"
"product-composite-swagger-ui": "lb://product-composite"
"eureka-api": "http://eureka:8761"
"eureka-web-start": "http://eureka:8761"
"eureka-web-other": "http://eureka:8761"
"host_route_200": "http://httpstat.us:80"
"host_route_418": "http://httpstat.us:80"
"host_route_501": "http://httpstat.us:80"
```

```
bash
{"productId":1, ... "recommendations":[...],"reviews":[...],"serviceAddresses":{"...}}
$
```

```
bash
Pattern "/product-composite/**" matches against value "/product-composite/1"
Route matched: product-composite
...
LoadBalancerClientFilter url chosen: http://2911a48d86ec:8080/product-composite/1
```



The image shows the Swagger UI for a 'Sample API' (version 1.0.0, OAS3) running on localhost:8080. The interface includes a search bar with the path '/openapi/v3/api-docs' and an 'Explore' button. Below the header, there's a description of the API and links for 'Terms of service', 'NAME OF CONTACT - Website', 'Send email to NAME OF CONTACT', 'MY LICENSE', and 'MY WIKI PAGE'. A 'Servers' section shows a dropdown menu with 'http://localhost:8080 - Generated server url'. The main content area displays the 'ProductComposite' REST API for composite product information, listing three endpoints: POST /product-composite (Creates a composite product), GET /product-composite/{productId} (Returns a composite view of the specified product id), and DELETE /product-composite/{productId} (Deletes a product composite).

```
bash
d94d651b2d0b:gateway:8080
2911a48d86ec:product-composite:8080
9e0000bdd56e:product:8080
8a0eef3e8fbd:review:8080
84a68e56553b:recommendation:8080
$
```

Eureka

localhost:8080/eureka/web

HOME LAST 1000 SINCE STARTUP

System Status

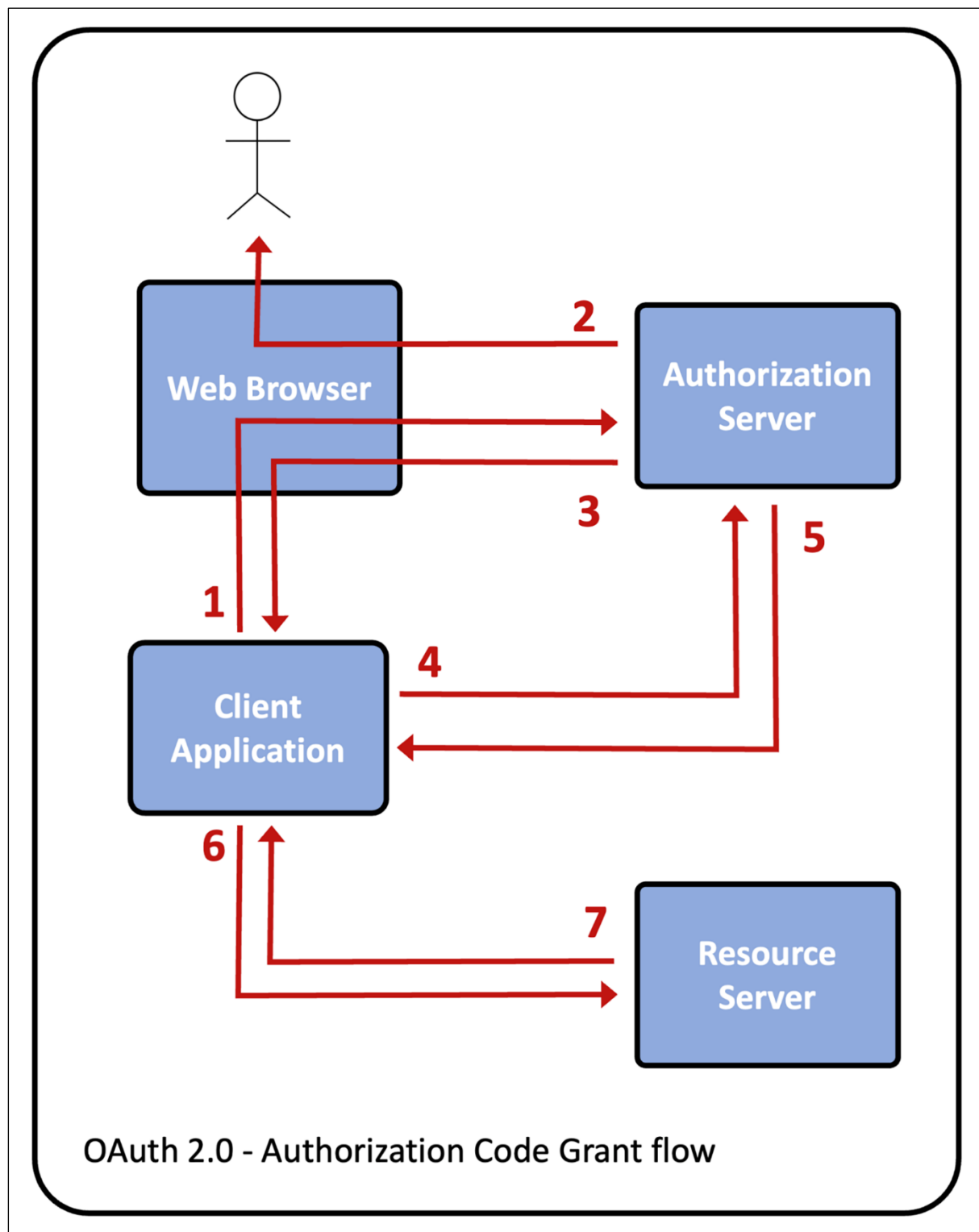
Environment	test	Current time	2023-03-29T15:03:16 +0000
Data center	default	Uptime	00:12
		Lease expiration enabled	true
		Renews threshold	10
		Renews (last min)	60

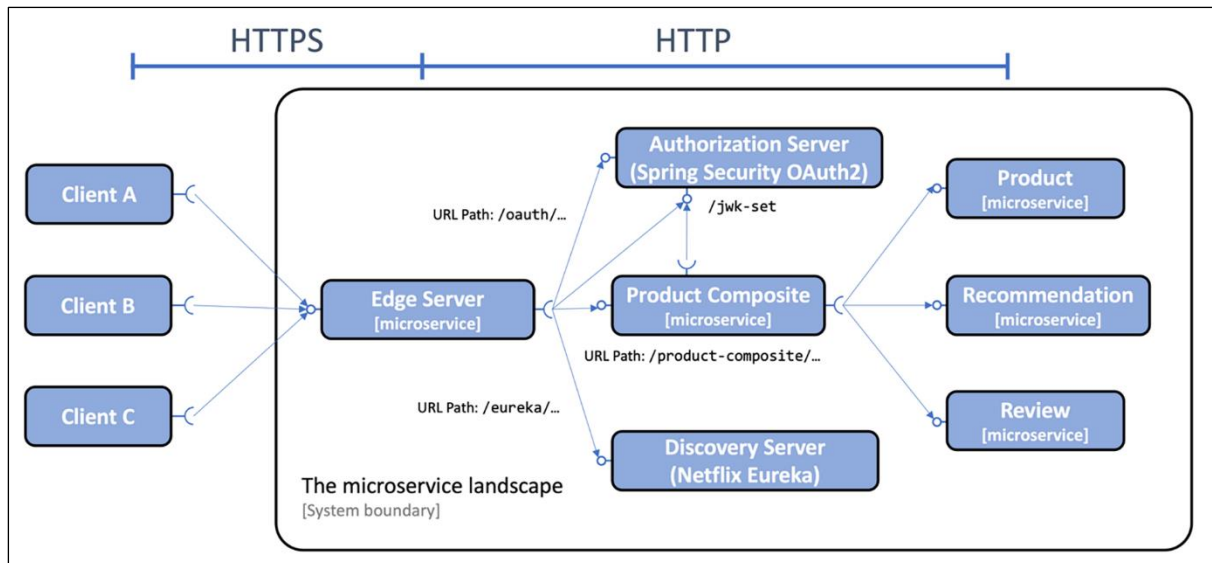
DS Replicas

Instances currently registered with Eureka

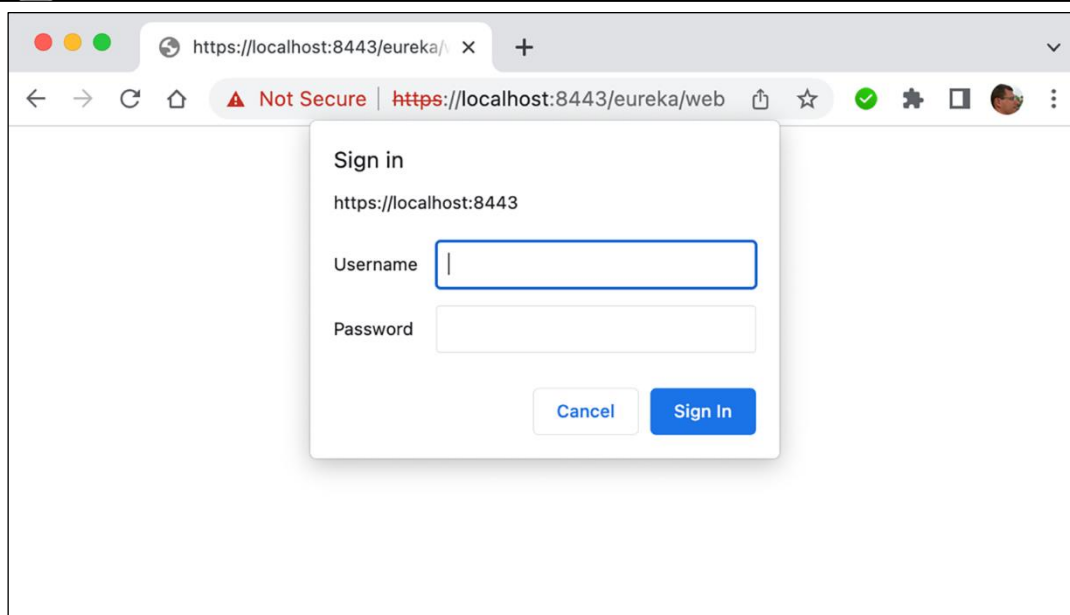
Application	AMIs	Availability Zones	Status
GATEWAY	n/a (1)	(1)	UP (1) - d94d651b2d0b:gateway:8080
PRODUCT	n/a (1)	(1)	UP (1) - 9e0000bdd56e:product:8080
PRODUCT-COMPOSITE	n/a (1)	(1)	UP (1) - 2911a48d86ec:product-composite:8080
RECOMMENDATION	n/a (1)	(1)	UP (1) - 84a68e56553b:recommendation:8080
REVIEW	n/a (1)	(1)	UP (1) - 8a0eef3e8fbd:review:8080

Chapter 11: Securing Access to APIs





```
bash
b4b0637cf8c6:gateway:8443
55a237832856:auth-server:9999
90a30474d7cb:product-composite:8080
697f17159678:product:8080
6d49c12522bb:review:8080
251a152c737c:recommendation:8080
$
```



Eureka

Not Secure | <https://localhost:8443/eureka/web>

Instances currently registered with Eureka

Application	AMIs	Availability Zones	Status
AUTH-SERVER	n/a (1)	(1)	UP (1) - 781f8298fd2f:auth-server:9999
GATEWAY	n/a (1)	(1)	UP (1) - dbf6372880b2:gateway:8443
PRODUCT	n/a (1)	(1)	UP (1) - 9f1deef92da2:product:8080
PRODUCT-COMPOSITE	n/a (1)	(1)	UP (1) - fc578652b403:product-composite:8080
RECOMMENDATION	n/a (1)	(1)	UP (1) - 00bcc1a9954e:recommendation:8080
REVIEW	n/a (1)	(1)	UP (1) - 552632decc14:review:8080

bash

```
{
  "access_token": "eyJr...jtDA",
  "scope": "product:write product:read",
  "token_type": "Bearer",
  "expires_in": 3599
}
```

\$

Please sign in

Not Secure | <https://localhost:8443/login>

Please sign in

u

.

Sign in

Consent required

Not Secure | https://localhost:8443/oauth...

Consent required

reader wants to access your account **u**

The following permissions are requested by the above app.
Please review these and consent if you approve.

☒ product:read

[Submit Consent](#)

[Cancel](#)

Your consent to provide access is required.
If you do not approve, click Cancel, in which case no information will be shared with the app.

my.redirect.uri

my.redirect.uri/?code=7XBshhce7M6tTNxB07...



This site can't be reached

my.redirect.uri's server IP address could not be found.

Try:

- Checking the connection
- [Checking the proxy, firewall and DNS configuration](#)

ERR_NAME_NOT_RESOLVED

[Details](#) [Reload](#)

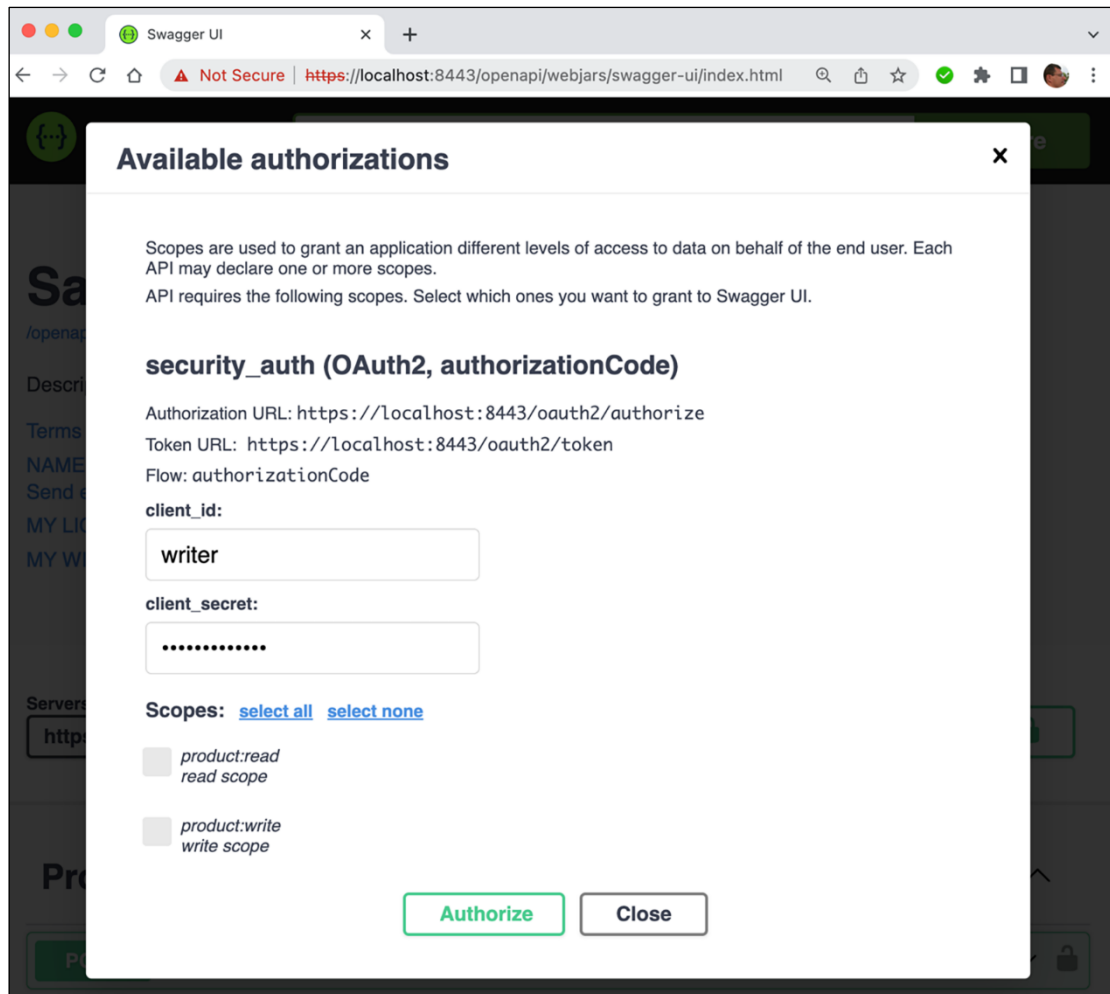

```
bash
{
  "access_token": "eyJr...ciwA",
  "refresh_token": "iCRP...HMvh",
  "scope": "product:read",
  "token_type": "Bearer",
  "expires_in": 3599
}
$
```

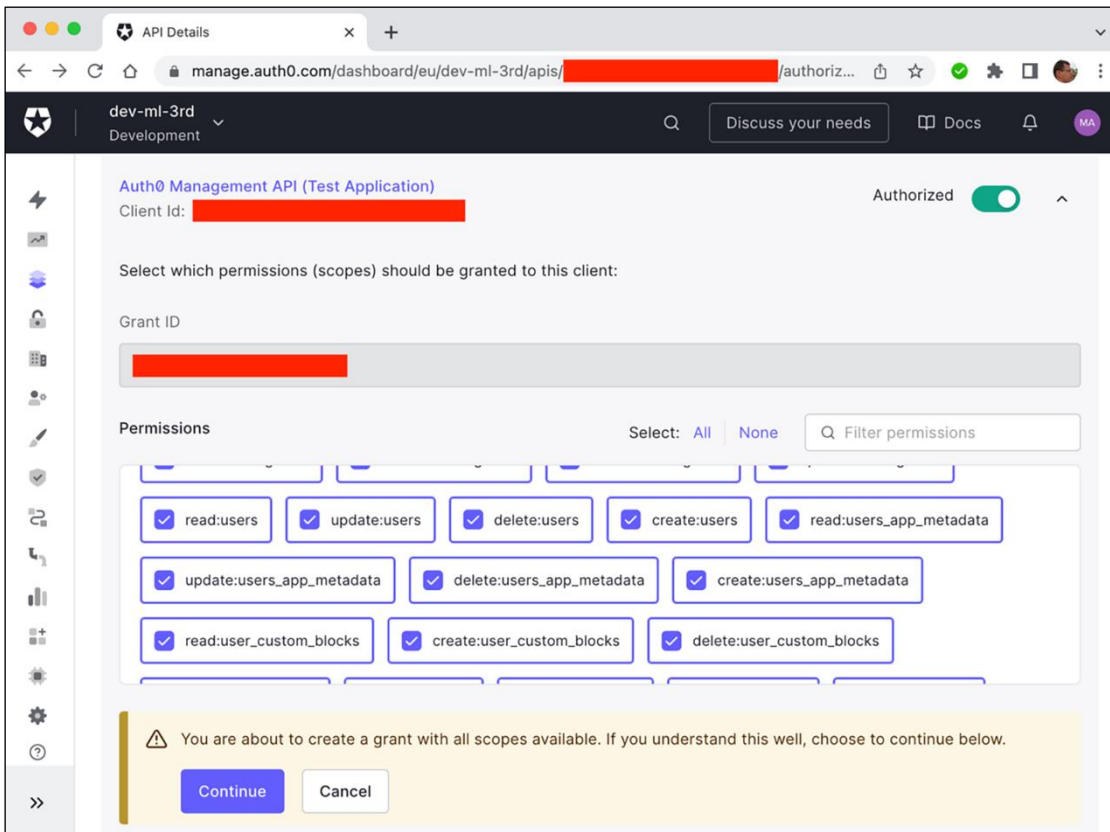
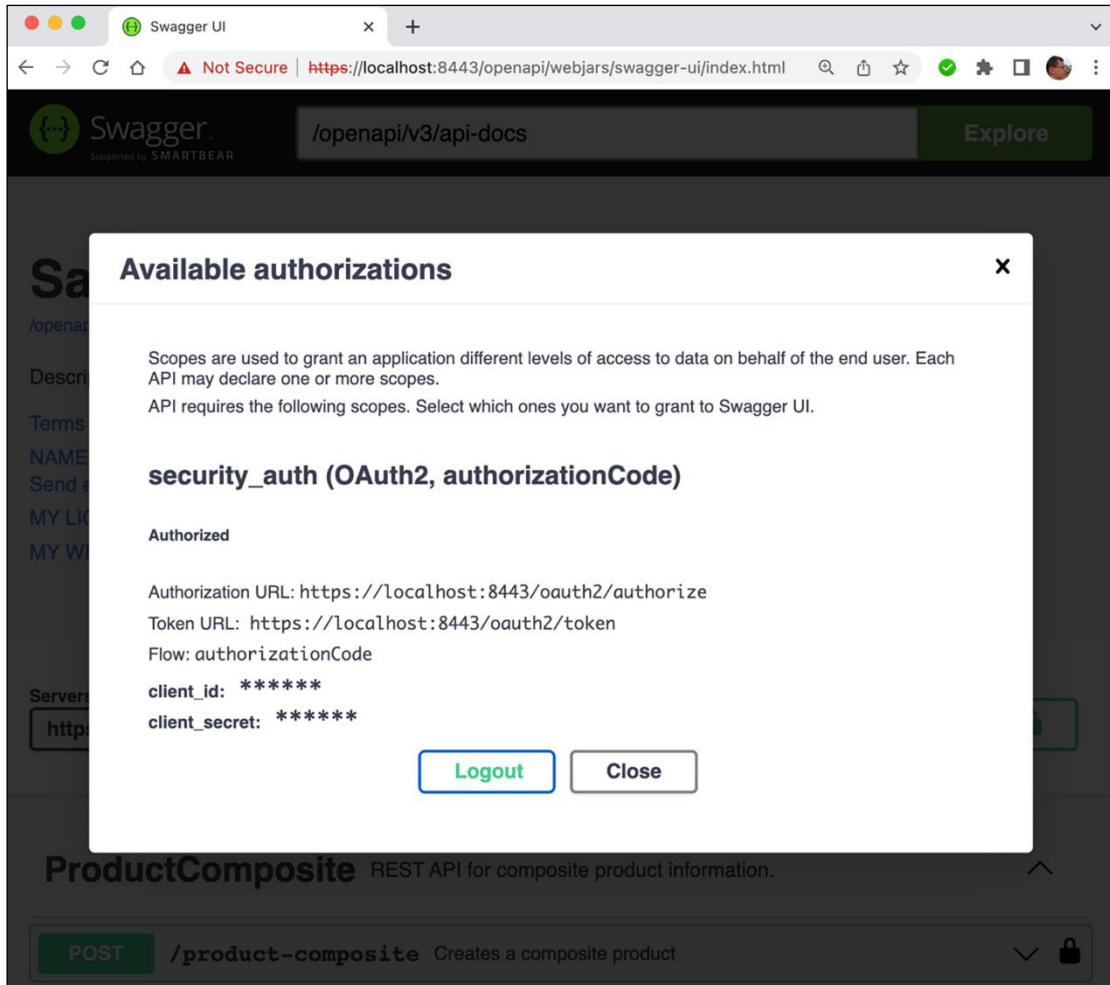
```
bash
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Bearer error="invalid_token",
```

```
bash
HTTP/1.1 200 OK
...
{"productId":1, ...}
$
```

```
bash
HTTP/1.1 403 Forbidden
WWW-Authenticate: Bearer error="insufficient_scope", error_description="The request requires higher privileges than provided by the access token.", error_uri="https://tools.ietf.org/html/rfc6750#section-3.1"
$
```

```
bash
chapter11-product-composite-1 | ... Authorization info: Subject: u,
scopes: [product:write, product:read], expires 2023-04-07T05:27:38Z:
issuer: http://auth-server:9999, audience: [writer]
$
```





Application Details

manage.auth0.com/dashboard/eu/dev-m1-3rd/applications/...

dev-m1-3rd
Development

← Back to Applications

Auth0 Management API (Test Application)

Machine to Machine Client ID

Quick Start Settings APIs

⚠ This feature is included in your current plan for up to 1,000 tokens per month. [Upgrade your subscription](#) or [contact enterprise sales](#) for additional monthly quota.

Basic Information

Name *
Auth0 Management API (Test Application)

Domain
dev-m1-3rd.eu.auth0.com

Client ID
[REDACTED]

Client Secret
[REDACTED]

The Client Secret is not base64 encoded.

```
bash
Update the tenant, set its default connection to a user dictionary...
...
Creates reader client app...
...
Creates writer client app...
...
Creates API product-composite (https://localhost:8443/product-composite)...
...
Creates user with email NNN...
...
Create client grant for the reader app to access the product-composite API...
...
Create client grant for the writer app to access the product-composite API...
...
Auth0 - OAuth2 settings:

export TENANT=...
export WRITER_CLIENT_ID=...
export WRITER_CLIENT_SECRET=...
export READER_CLIENT_ID=...
export READER_CLIENT_SECRET=...
$
```

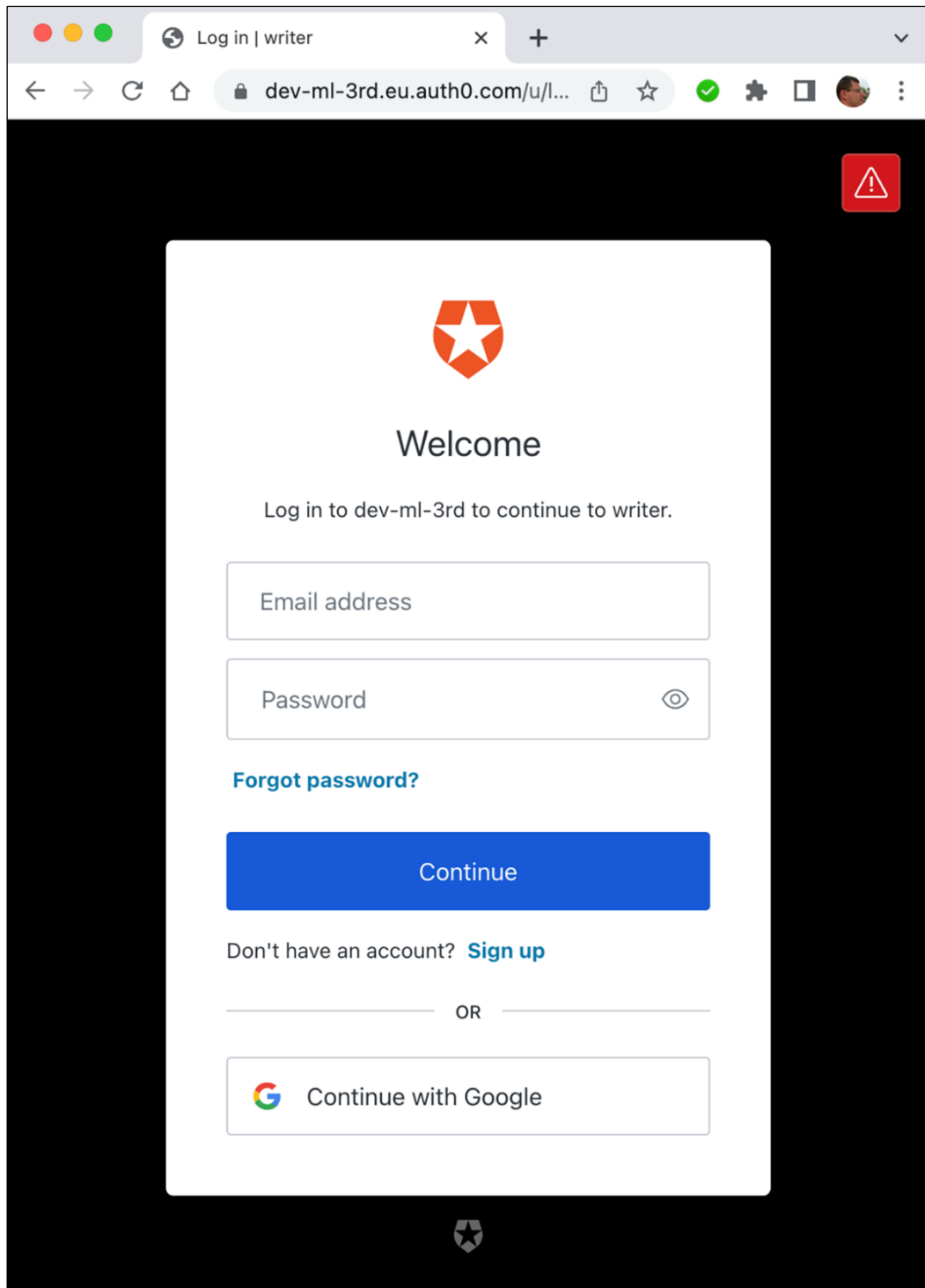
```
-bash ㉿%1
Update the tenant, set its default connection to a user dictionary...
...
Reader client app already exists
Writer client app already exists
API product-composite (https://localhost:8443/product-composite) already exists
User with email NNN already exists
Client grant for the reader app to access the product-composite API already exists
Client grant for the writer app to access the product-composite API already exists

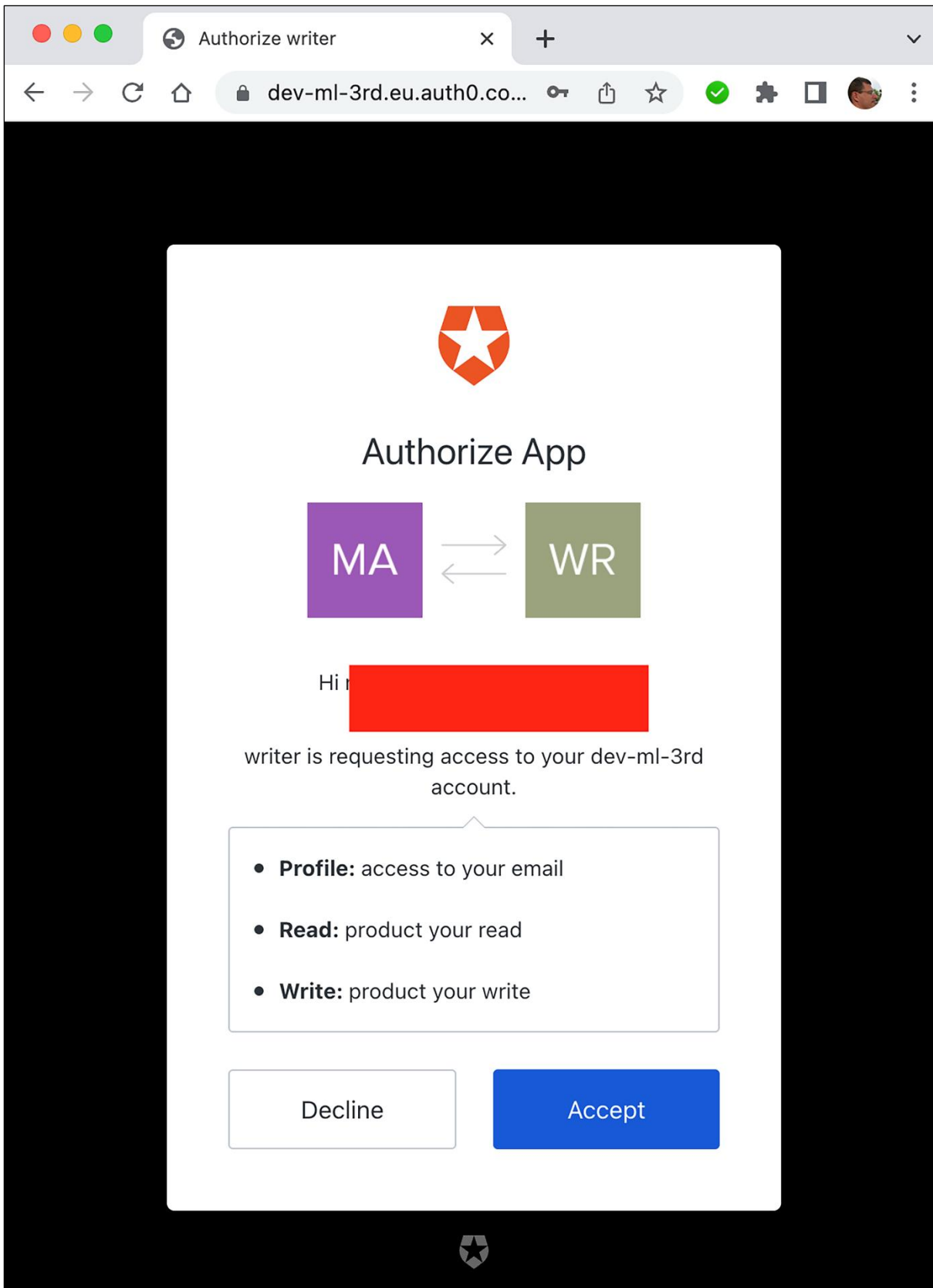
Auth0 - OAuth2 settings:

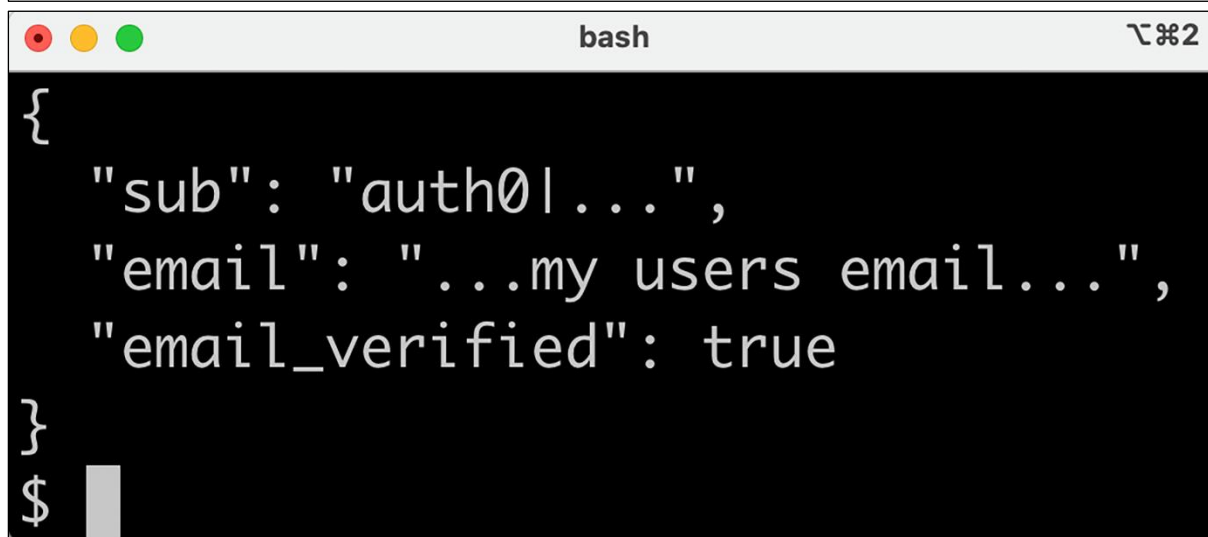
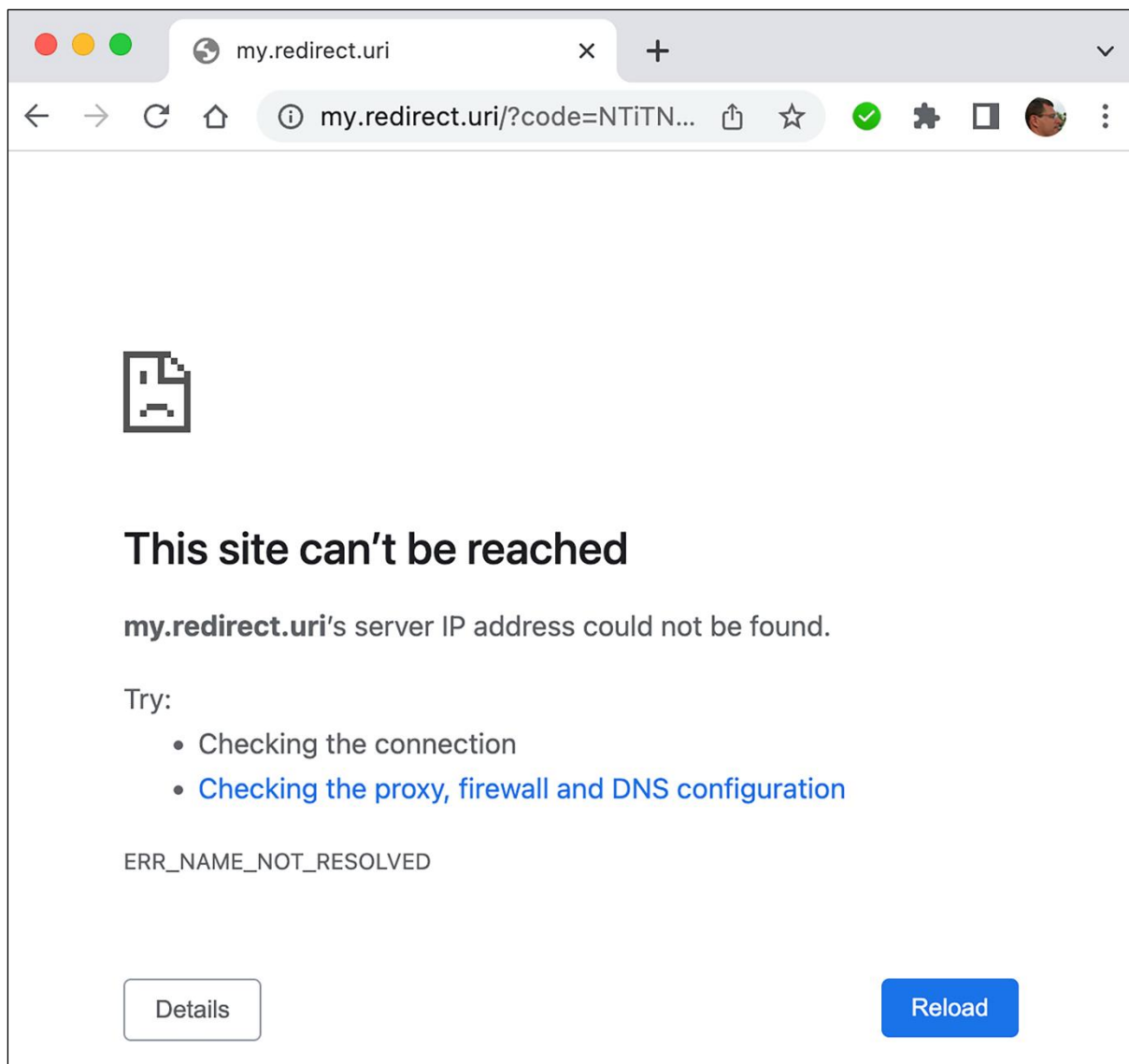
export TENANT=...
export WRITER_CLIENT_ID=...
export WRITER_CLIENT_SECRET=...
export READER_CLIENT_ID=...
export READER_CLIENT_SECRET=...
$
```

```
bash ㉿%2
chapter11-product-composite-1 | ... Authorization info:
Subject: ...@clients,
scopes: product:read product:write,
expires 2023-04-08T06:09:45Z:
issuer: https://dev-m1-3rd.eu.auth0.com/,
audience: [https://localhost:8443/product-composite]
$
```

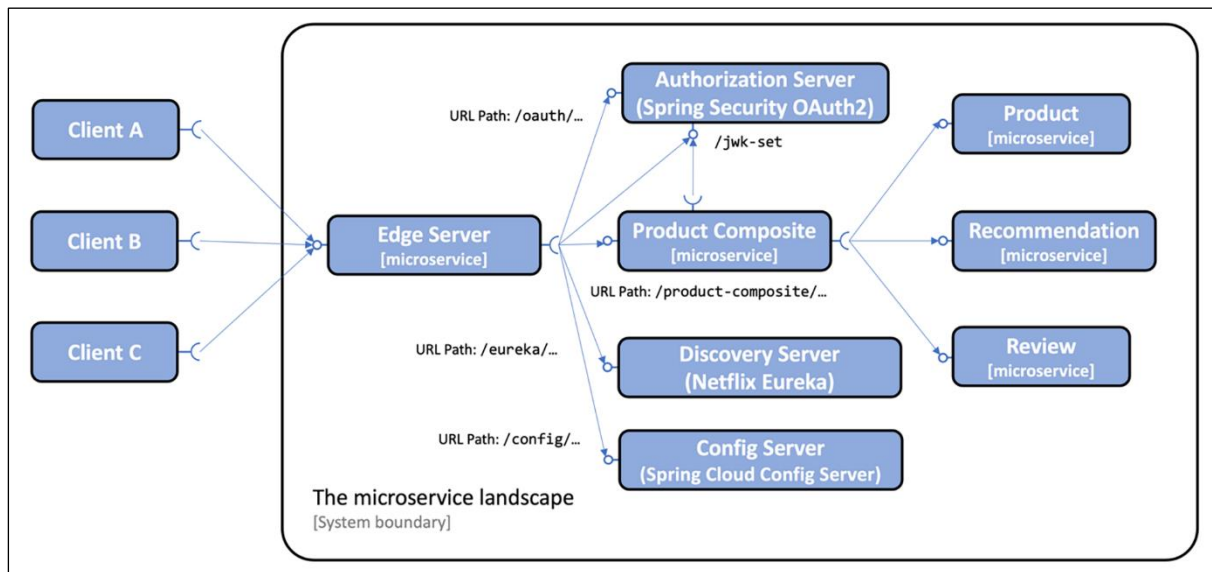
```
bash ㉿%2
chapter11-product-composite-1 | ... Authorization info:
Subject: ...@clients,
scopes: product:read,
expires 2023-04-08T06:09:49Z:
issuer: https://dev-m1-3rd.eu.auth0.com/,
audience: [https://localhost:8443/product-composite]
$
```







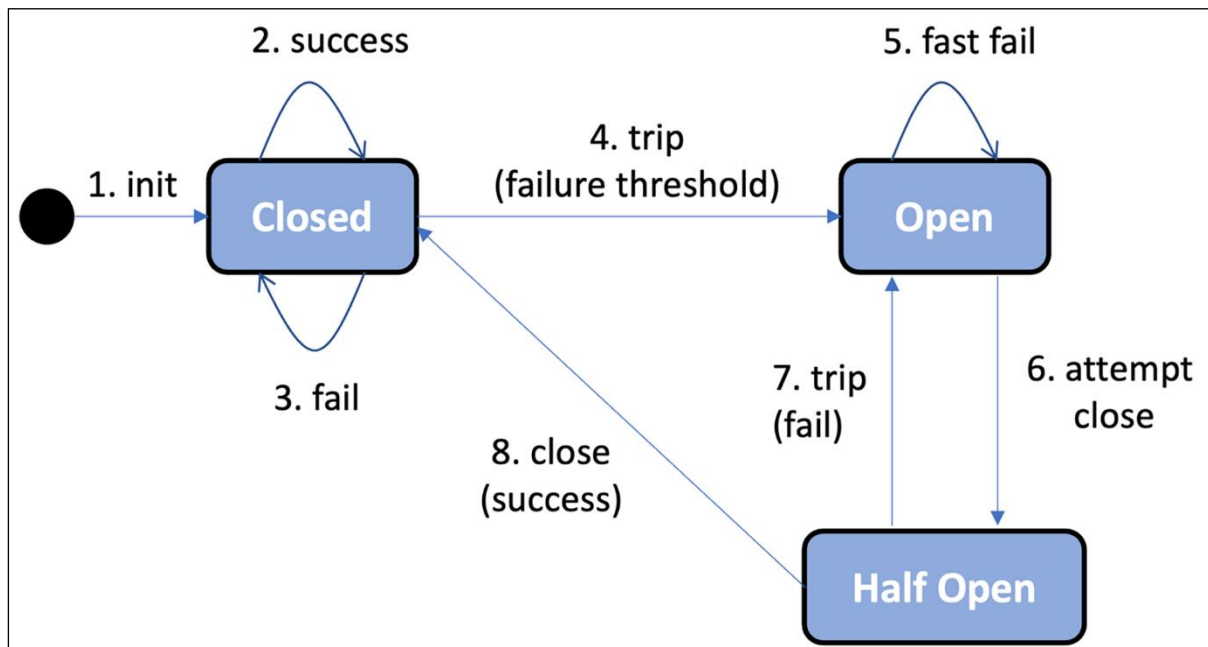
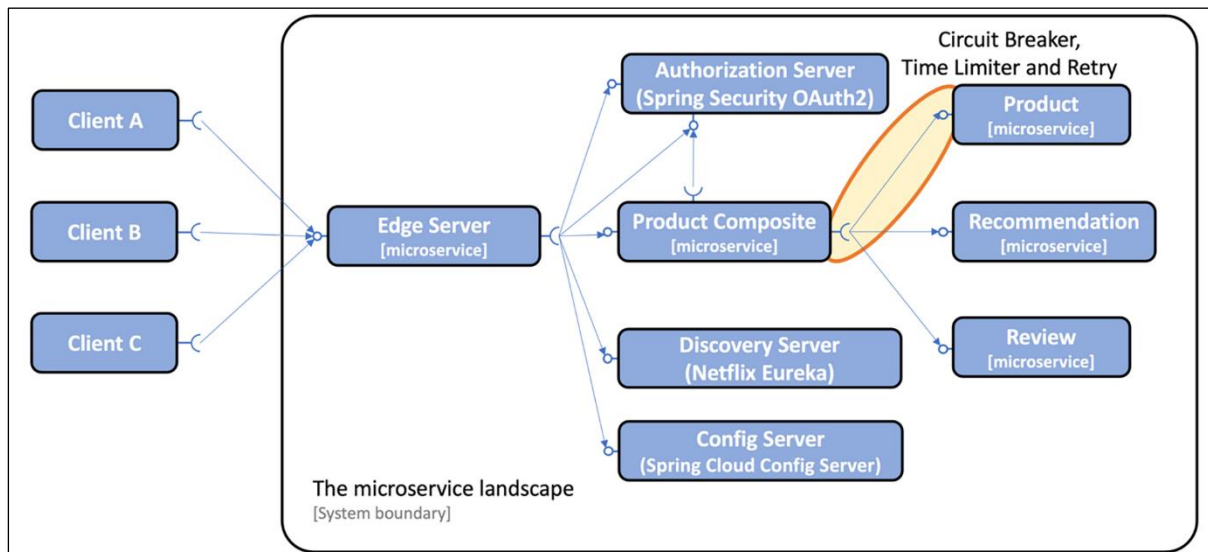
Chapter12: Centralized Configuration



```
bash
d91001603dcdf3eb1392ccbd40ff201cdf7b9af2fcaab3da39e37919033b206$
$
```

```
bash
hello world$
$
```

Chapter13: Improving Resilience Using Resilience4j



```
bash
{
  "timestamp": "2023-04-10T07:34:03.548+00:00",
  "path": "/product-composite/1",
  "status": 500,
  "error": "Internal Server Error",
  "message": "Did not observe any item or terminal signal within 2000ms in 'onErrorResume' (and no fallback has been configured)",
  "requestId": "28bc9b3f-42"
}
```

```
bash  ~%2
{
  "productId": 1,
  "name": "Fallback product1",
  "weight": 1,
```

```
bash  ~%2
CLOSED_TO_OPEN
OPEN_TO_HALF_OPEN
HALF_OPEN_TO_CLOSED
$
```

```
bash  ~%2
200

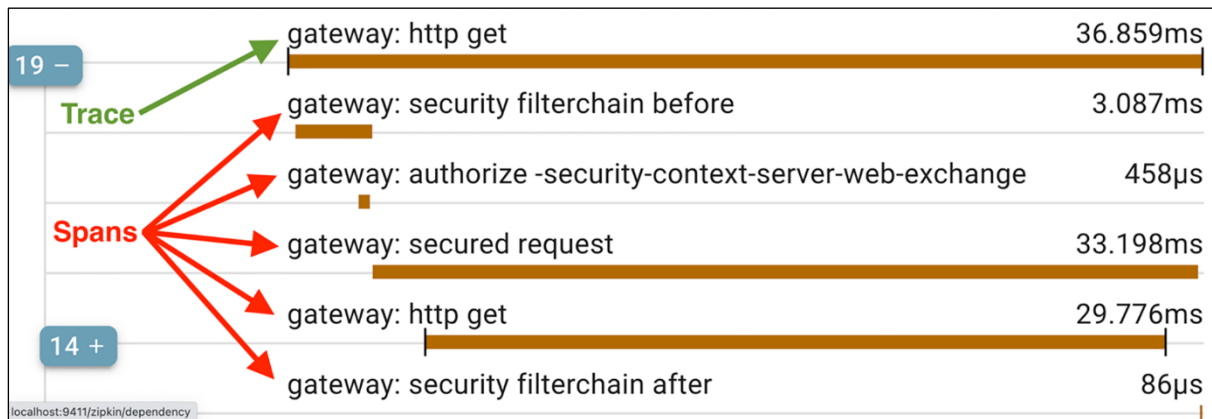
real    0m0.066s
```

```
bash  ~%2
200

real    0m1.072s
```

```
bash ㉿%2
{
  "retryName": "product",
  "type": "RETRY",
  "creationTime": "2023-04-10T07:39:51.136929251Z[Etc/UTC]",
  "errorMessage": "org.springframework.web.reactive.function.client.Web
ClientResponseException$InternalServerError: 500 Internal Server Error
from GET http://c4a5a8bbf39d:8080/product/1?delay=0&faultPercent=25",
  "numberOfAttempts": 1
}
{
  "retryName": "product",
  "type": "SUCCESS",
  "creationTime": "2023-04-10T07:39:52.153598627Z[Etc/UTC]",
  "errorMessage": "org.springframework.web.reactive.function.client.Web
ClientResponseException$InternalServerError: 500 Internal Server Error
from GET http://c4a5a8bbf39d:8080/product/1?delay=0&faultPercent=25",
  "numberOfAttempts": 1
}
$
```

Chapter 14: Understanding Distributed Tracing



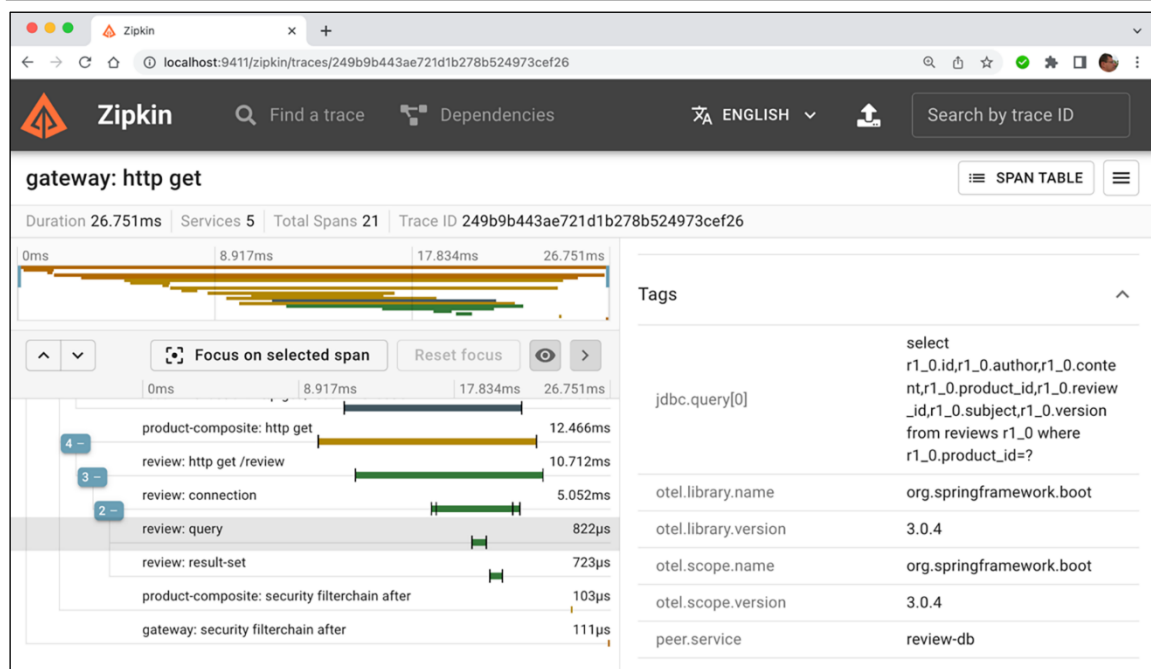
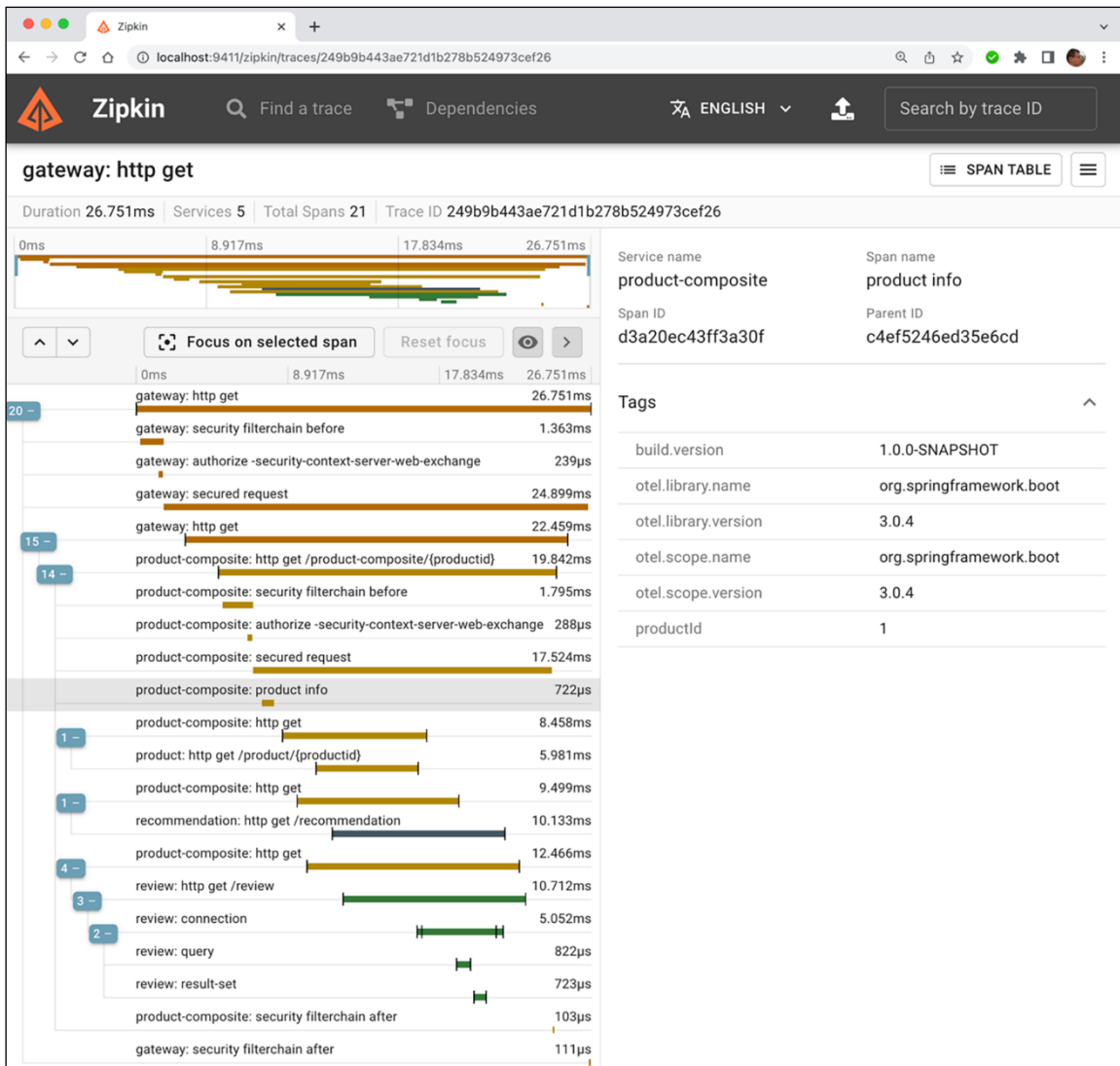
Zipkin

Find a trace Dependencies ENGLISH Search by trace ID

serviceName gateway x + a b RUN QUERY

10 Results EXPAND ALL COLLAPSE ALL Service filters

Root	c ↓ Start Time	Spans	Duration
gateway: http get	a minute ago (04/23 11:13:30:849)	21	26.751ms SHOW
gateway: http post	a minute ago (04/23 11:13:20:613)	12	22.687ms SHOW
gateway: http get	2 minutes ago (04/23 11:12:17:377)	20	37.461ms SHOW



Zipkin

Find a trace Dependencies

ENGLISH

Search by trace ID

serviceName gateway

RUN QUERY

7 Results

EXPAND ALL COLLAPSE ALL Service filters

Root	Start Time	Spans	Duration
gateway: http get	10 minutes ago (04/23 11:31:15:780)	21	2.038s
gateway: http get	11 minutes ago (04/23 11:30:25:735)	19	32.853ms

gateway: http get 2.053s

gateway: security filterchain before 10.254ms

gateway: authorize -security-context-server-web-exchange 420µs

gateway: secured request 2.042s

gateway: http get 2.037s

product-composite: http get /product-composite/(productid) 2.030s

product-composite: security filterchain before 2.997ms

product-composite: authorize -security-context-server-web-exchange 485µs

product-composite: secured request 2.015s

product-composite: product info 922µs

Tags

build.version 1.0.0-SNAPSHOT

error Did not observe any item or terminal signal within 2000ms in 'onErrorResume' (and no fallback has been configured)

otel.library.name org.springframework.boot

otel.library.version 3.0.4

otel.scope.name org.springframework.boot

Zipkin

Find a trace Dependencies

ENGLISH

Search by trace ID

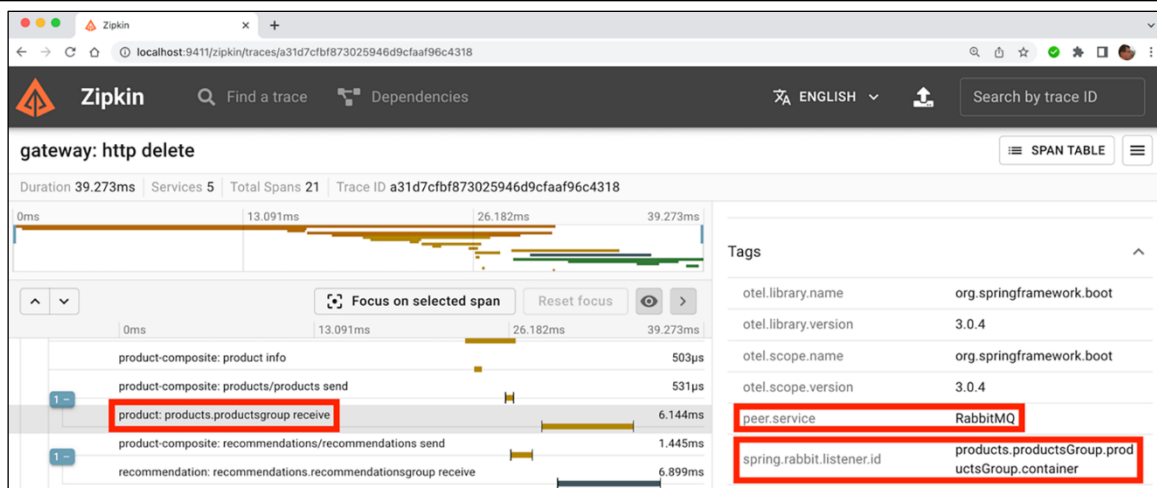
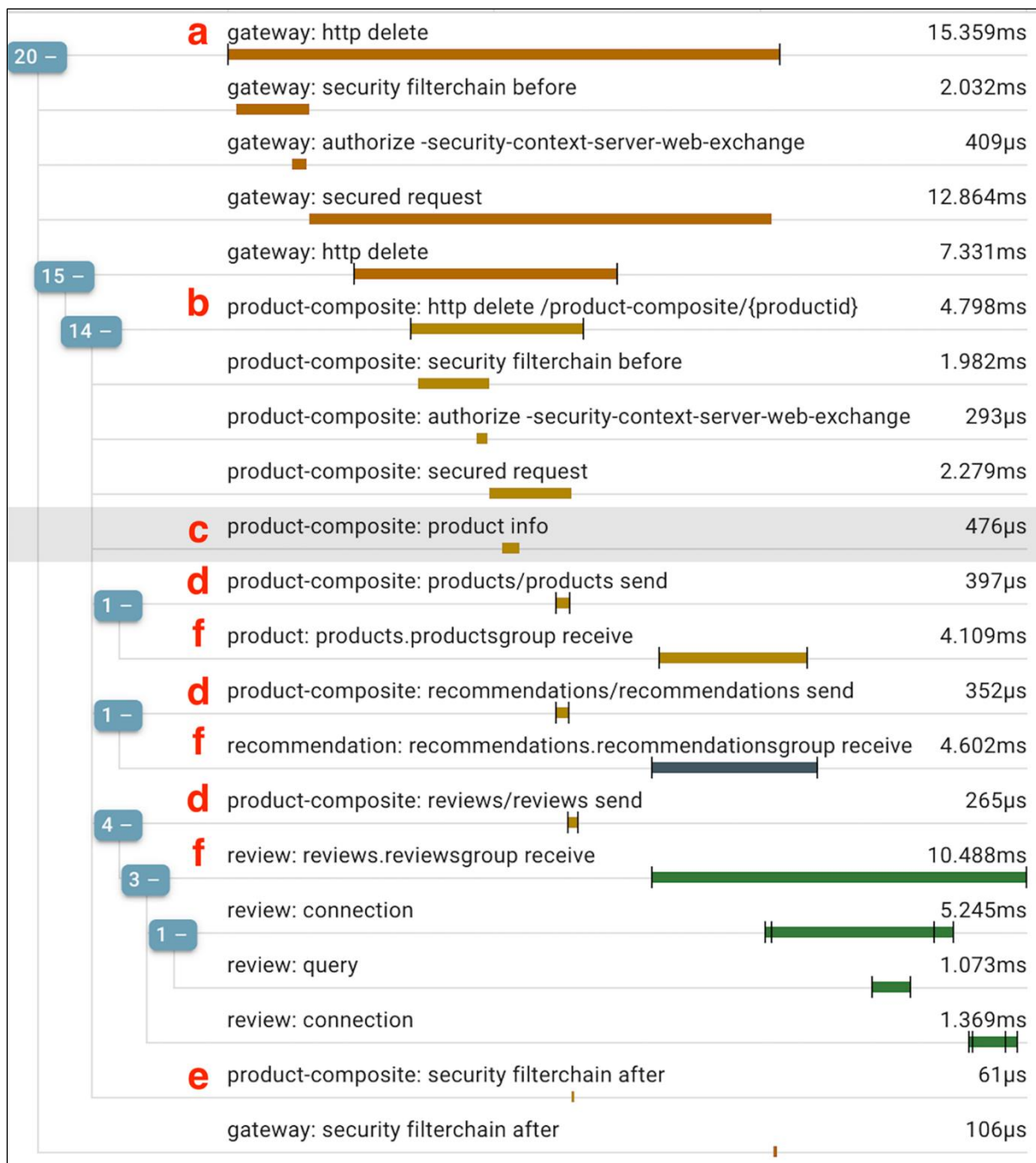
serviceName gateway

RUN QUERY

10 Results

EXPAND ALL COLLAPSE ALL Service filters

Root	Start Time	Spans	Duration
gateway: http delete	a minute ago (04/23 11:42:54:382)	21	39.273ms
gateway: http get	12 minutes ago (04/23 11:32:25:960)	21	2.027s



RabbitMQ Management

localhost:15672/#/queues/%2F/products.auditGroup

RabbitMQ

RabbitMQ 3.11.8 Erlang 25.2.2

Refreshed 2023-04-23 11:50:19

Refresh every 5 seconds

Virtual host All

Cluster rabbit@c7b9f5a0fcf5

User guest Log out

Overview

Connections

Channels

Exchanges

Queues

Admin

Get Message(s)

Message 1

The server reported 12 messages remaining.

Exchange

Routing Key

Redelivered

Properties

Payload

products

products

•

timestamp: 1682241111

message_id: fb695a1c-1ede-d667-7b51-33d714604b29

priority: 0

delivery_mode: 2

headers:

partitionKey: 113

target-protocol: amqp

traceparent: 00-99a9f2501e4d454643184c6b1cb0a232-7458430fe56d6df1-01

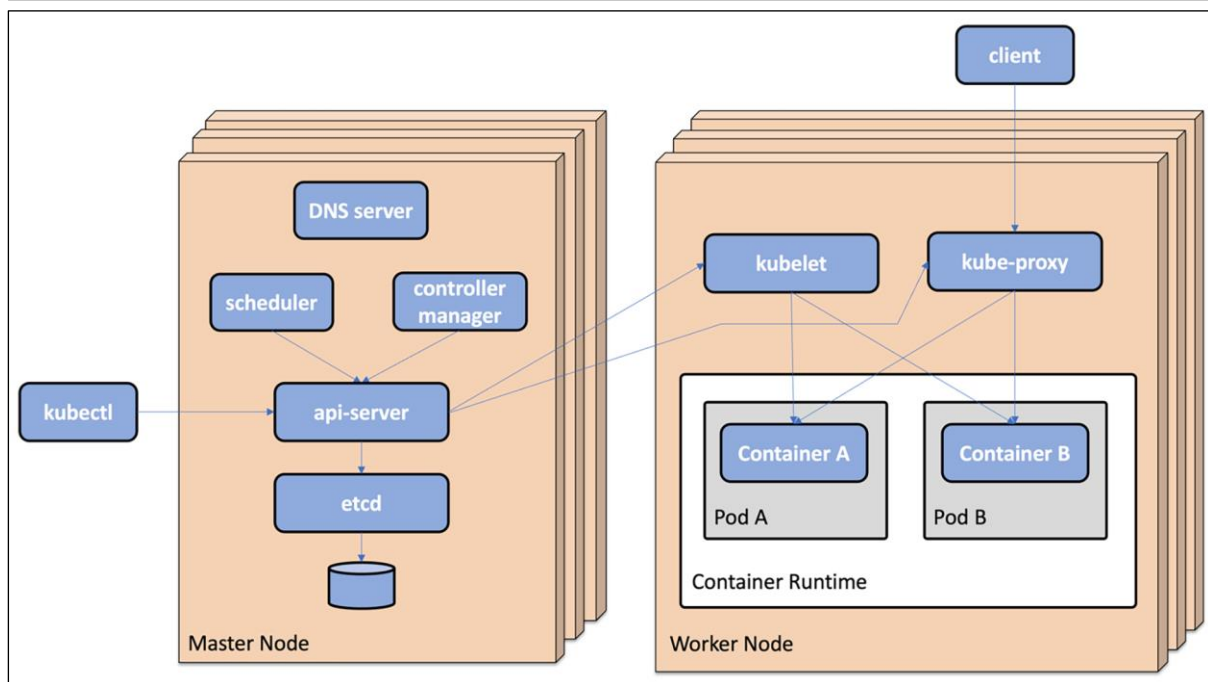
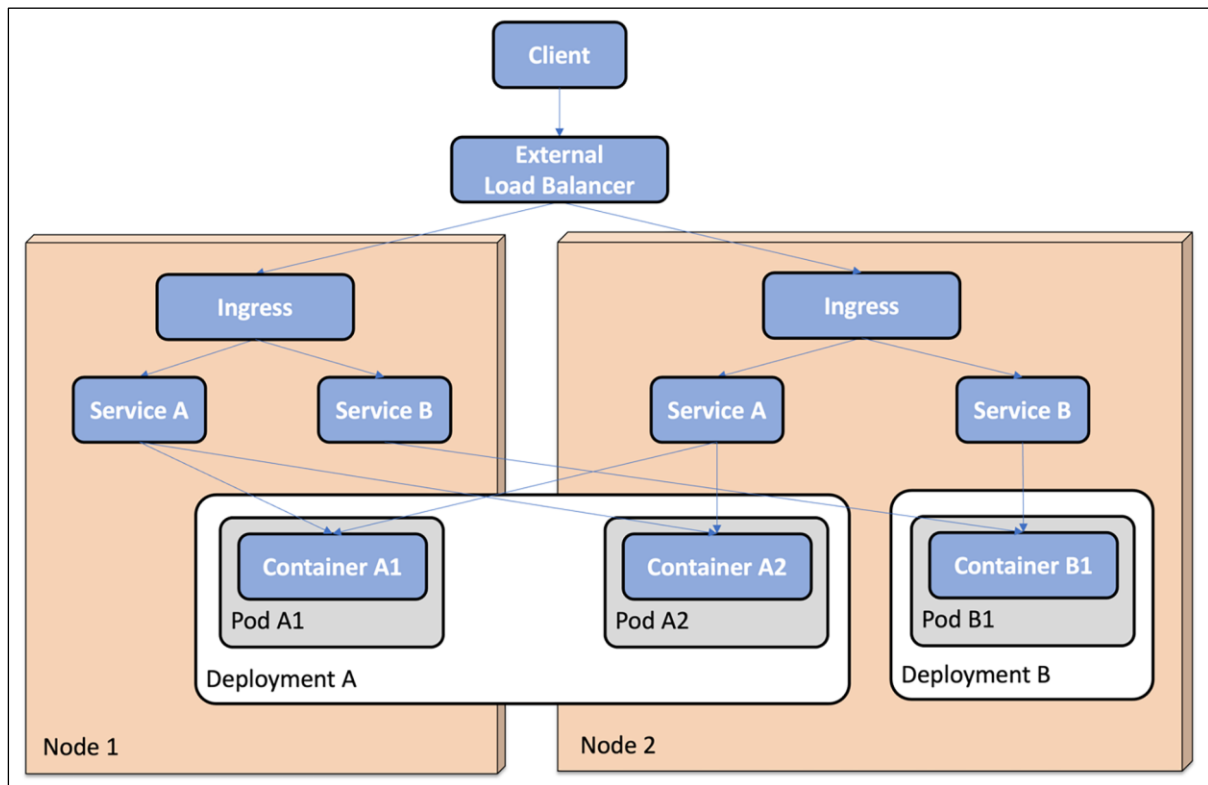
content_type: application/json

93 bytes

Encoding: string

{"eventType": "DELETE", "key": 113, "data": null, "eventCreatedAt": "2023-04-23T09:11:51.21215301Z"}

Chapter 15: Introduction to Kubernetes



```
bash
$ kubectl config get-contexts
CURRENT NAME CLUSTER AUTHINFO NAMESPACE
* handson-spring-boot-cloud handson-spring-boot-cloud handson-spring-boot-cloud first-attempts
minikube minikube minikube default
$
```

```
bash
$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
handson-spring-boot-cloud          Ready    control-plane  40m   v1.26.1
$
```

```
bash
$ kubectl get pods --all-namespaces
NAMESPACE      NAME                                READY   STATUS    RESTARTS   AGE
ingress-nginx  ingress-nginx-admission-create-88v7k  0/1     Completed  0           41m
ingress-nginx  ingress-nginx-admission-patch-j2xws   0/1     Completed  1           41m
ingress-nginx  ingress-nginx-controller-77669ff58-qfj5j 1/1     Running    0           41m
kube-system    coredns-787d4945fb-wbp2r             1/1     Running    0           63m
kube-system    etcd-handson-spring-boot-cloud         1/1     Running    0           63m
kube-system    kube-apiserver-handson-spring-boot-cloud 1/1     Running    0           63m
kube-system    kube-controller-manager-handson-spring-boot-cloud 1/1     Running    0           63m
kube-system    kube-proxy-prrjn                      1/1     Running    0           63m
kube-system    kube-scheduler-handson-spring-boot-cloud 1/1     Running    0           63m
kube-system    metrics-server-5f8fcc9bb7-llns9       1/1     Running    0           40m
kube-system    storage-provisioner                   1/1     Running    1 (62m ago) 63m
$
```

```
bash
$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/nginx-deploy-58bfb487f9-l5hkr  1/1     Running    0           86s

NAME              TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
service/kubernetes ClusterIP      10.96.0.1    <none>        443/TCP    8m29s

NAME              READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-deploy  1/1     1             1           86s

NAME              DESIRED   CURRENT   READY   AGE
replicaset.apps/nginx-deploy-58bfb487f9  1         1         1       86s
$
```

```
kubectl
$ kubectl get pod --watch
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deploy-58bfb487f9-cgc2v      1/1     Running    0           51s
nginx-deploy-58bfb487f9-cgc2v      1/1     Terminating  0           54s
nginx-deploy-58bfb487f9-knhfp      0/1     Pending     0           0s
nginx-deploy-58bfb487f9-knhfp      0/1     Pending     0           0s
nginx-deploy-58bfb487f9-knhfp      0/1     ContainerCreating  0           0s
nginx-deploy-58bfb487f9-cgc2v      0/1     Terminating  0           55s
nginx-deploy-58bfb487f9-cgc2v      0/1     Terminating  0           55s
nginx-deploy-58bfb487f9-cgc2v      0/1     Terminating  0           55s
nginx-deploy-58bfb487f9-knhfp      1/1     Running     0           2s
$
```

```
8. bash
$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/nginx-deploy-59b8c5f7cd-mt6pg   1/1     Running   0           46s
pod/nginx-deploy-59b8c5f7cd-pmns9   1/1     Running   0           12s
pod/nginx-deploy-59b8c5f7cd-vtstj   1/1     Running   0           12s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-deploy        3/3     3             3           46s

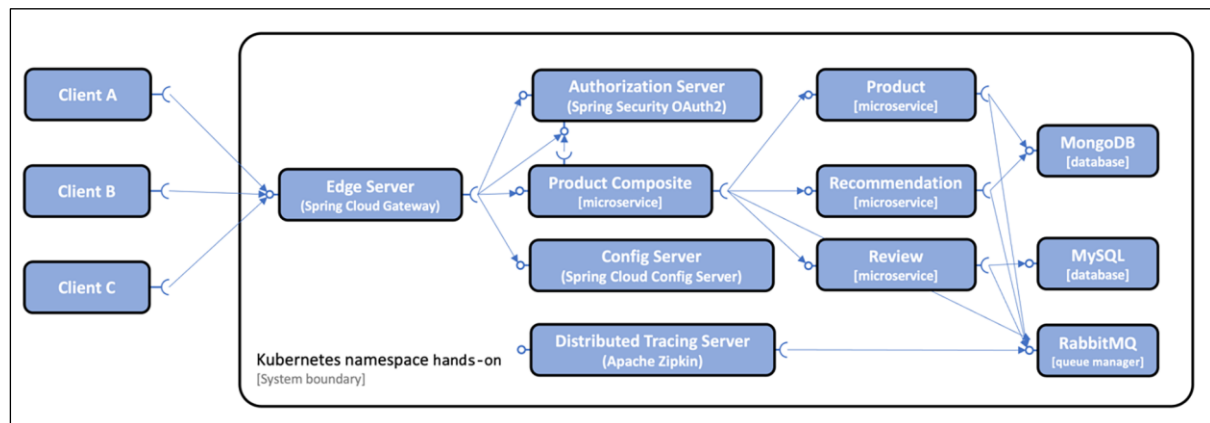
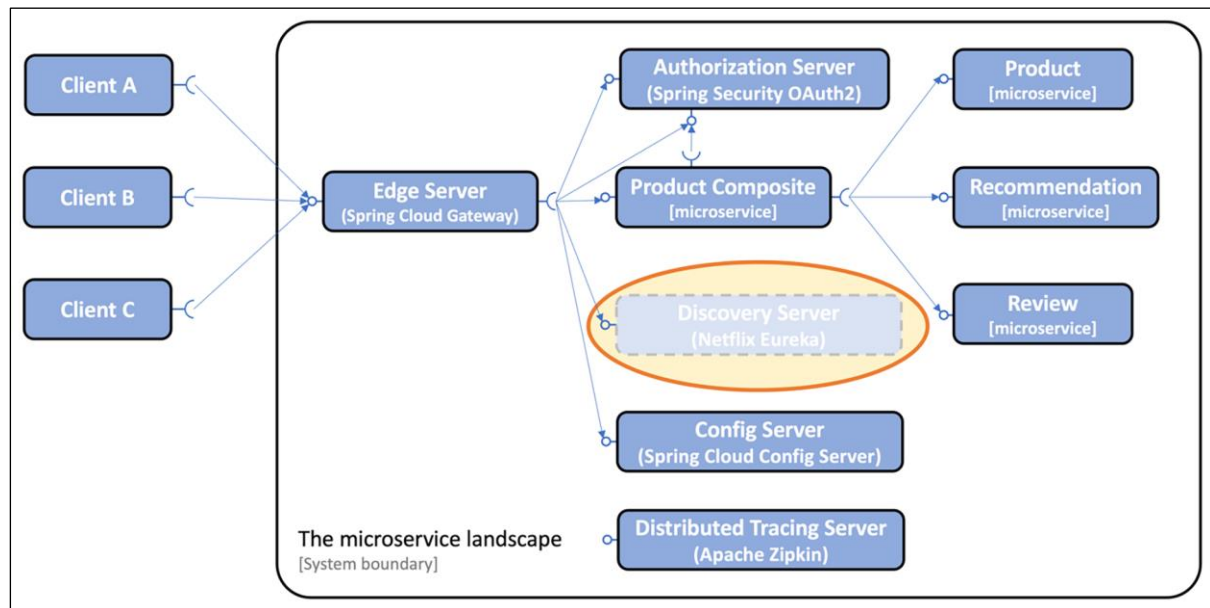
NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/nginx-deploy-59b8c5f7cd 3         3         3       46s
$
```

```
bash
$ kubectl get svc
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
nginx-service   NodePort    10.100.230.194 <none>       80:30080/TCP    30s
$
```



```
bash
$ kubectl run -i --rm --restart=Never \
  curl-client --image=curlimages/curl \
  --command -- curl -s 'http://nginx-service:80'
...
<h1>Welcome to nginx!</h1>
...
pod "curl-client" deleted
$
```

Chapter 16: Deploying Our Microservices to Kubernetes



```
-bash
```

NAME	VERSION	REPOSITORY	STATUS
common	1.0.0	file:///././common	ok
rabbitmq	1.0.0	file:///././components/rabbitmq	ok
mongodb	1.0.0	file:///././components/mongodb	ok
mysql	1.0.0	file:///././components/mysql	ok
config-server	1.0.0	file:///././components/config-server	ok
gateway	1.0.0	file:///././components/gateway	ok
auth-server	1.0.0	file:///././components/auth-server	ok
product	1.0.0	file:///././components/product	ok
recommendation	1.0.0	file:///././components/recommendation	ok
review	1.0.0	file:///././components/review	ok
product-composite	1.0.0	file:///././components/product-composite	ok
zipkin-server	1.0.0	file:///././components/zipkin-server	ok

```
$
```

kubectl					⌂ %1
config-server-6b9f6c6988-slt9n	0/1	ContainerCreating	0	5s	
product-76bb56d7fc-7qrgf	0/1	ContainerCreating	0	5s	
mongodb-d466986f-xtlsj	0/1	ContainerCreating	0	6s	
auth-server-5c58bbcb9b-c9npg	0/1	Running	0	6s	
rabbitmq-6c7b9bf547-c44dj	0/1	Running	0	7s	
gateway-7bdcf586df-pm9l4	0/1	Running	0	7s	
product-composite-f45fccdf5-46dpt	0/1	Running	0	7s	
mongodb-d466986f-xtlsj	1/1	Running	0	8s	
review-777ff7d5b4-g7zrj	0/1	Running	0	8s	
product-76bb56d7fc-7qrgf	0/1	Running	0	9s	
mysql-5846f7fb7c-kxxd7	1/1	Running	0	9s	
zipkin-server-586bcccd96-lgrj7	0/1	Running	0	9s	

```

bash
⌂ %1

"hands-on/auth-server:latest"
"hands-on/config-server:latest"
"hands-on/gateway:latest"
"registry.hub.docker.com/library/mongo:6.0.4"
"registry.hub.docker.com/library/mysql:8.0.32"
"hands-on/product-service:latest"
"hands-on/product-composite-service:latest"
"registry.hub.docker.com/library/rabbitmq:3.11.8-management"
"hands-on/recommendation-service:latest"
"hands-on/review-service:latest"
"registry.hub.docker.com/openzipkin/zipkin:2.24.0"
$

```

```

bash
⌂ %1

Start Tests: Mon May 1 16:41:35 CEST 2023
HOST=localhost
PORT=30443
USE_K8S=true
SKIP_CB_TESTS=false
Wait for: curl -k https://localhost:30443/actuator/health...
DONE, continues...
ACCESS_TOKEN=eyJraWw...
Test OK (HTTP Code: 200)
.
.
.
End, all tests OK: Mon May 1 16:42:03 CEST 2023
$

```



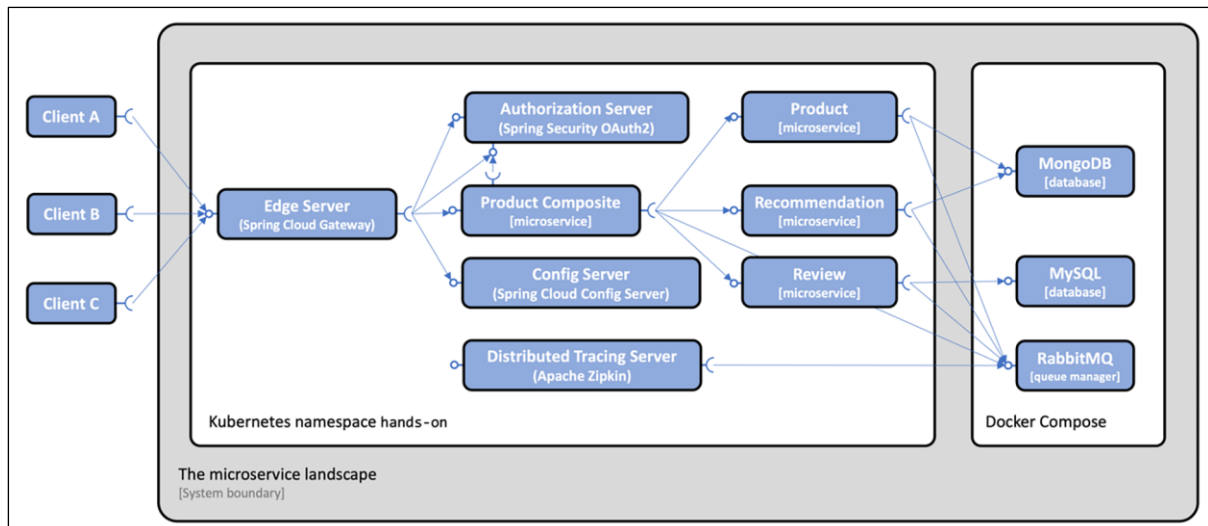
```
bash
2023-05-02T14:12:30.877Z INFO [product,,] 1 ---
[ionShutdownHook] o.s.b.w.embedded.netty.GracefulShutdown :
Commencing graceful shutdown. Waiting for active requests to complete
.
.
.
2023-05-02T14:12:35.245Z INFO [product,,] 1 ---
[ netty-shutdown] o.s.b.w.embedded.netty.GracefulShutdown :
Graceful shutdown complete
```

```
bash
2023-05-02T14:38:51.191Z INFO [product,,] 1 ---
[ionShutdownHook] o.s.b.w.embedded.netty.GracefulShutdown :
Commencing graceful shutdown. Waiting for active requests to complete
.
.
.
2023-05-02T14:39:01.205Z INFO [product,,] 1 ---
[ionShutdownHook] o.s.c.support.DefaultLifecycleProcessor :
Failed to shut down 1 bean with phase value 2147482623 within timeout
of 10000ms: [webServerGracefulShutdown]
2023-05-02T14:39:01.207Z INFO [product,,] 1 ---
[ netty-shutdown] o.s.b.w.embedded.netty.GracefulShutdown :
Graceful shutdown aborted with one or more requests still active
```

```
bash
HTTP/1.1 200    15.06 secs:    771 bytes ==> GET  /product-composite/1?delay=15
HTTP/1.1 500    13.76 secs:    205 bytes ==> GET  /product-composite/1?delay=15
HTTP/1.1 200    15.12 secs:    771 bytes ==> GET  /product-composite/1?delay=15
```

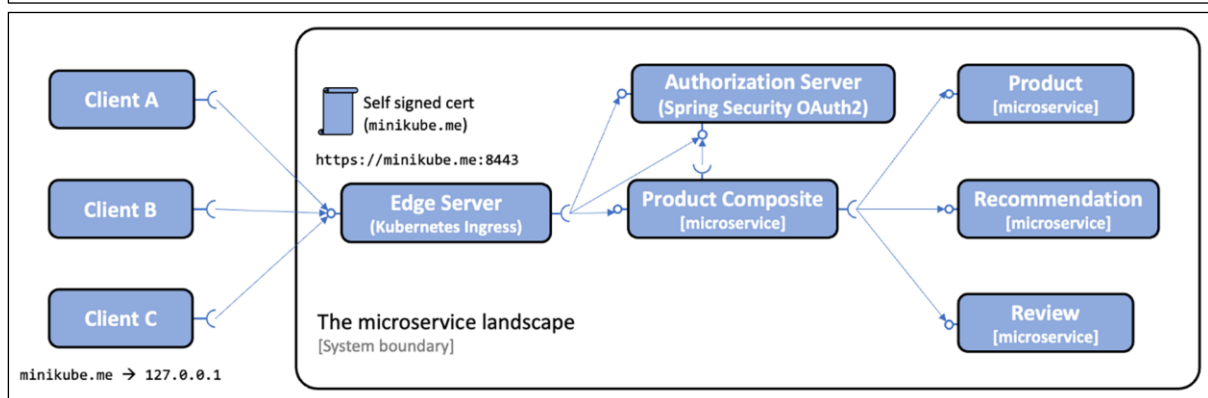
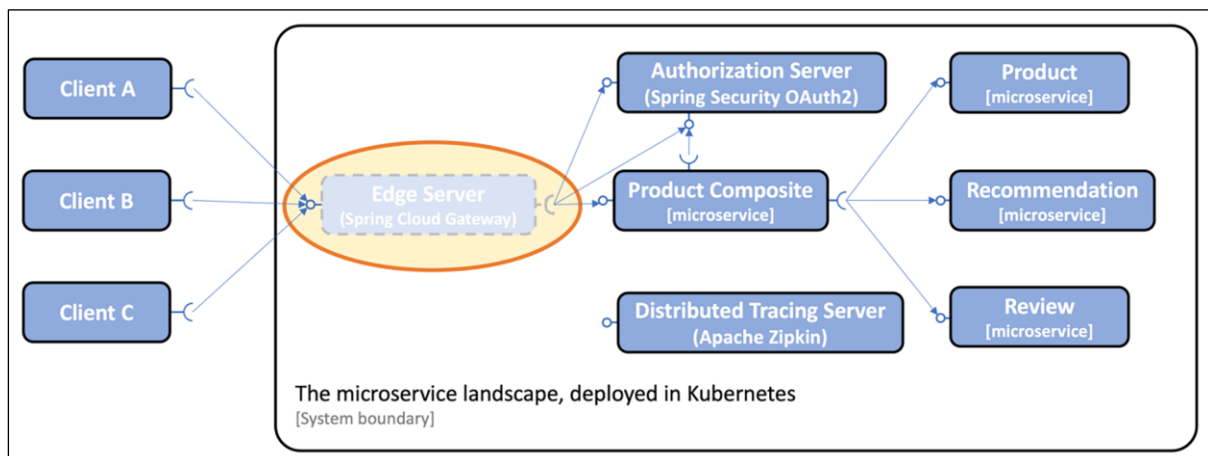
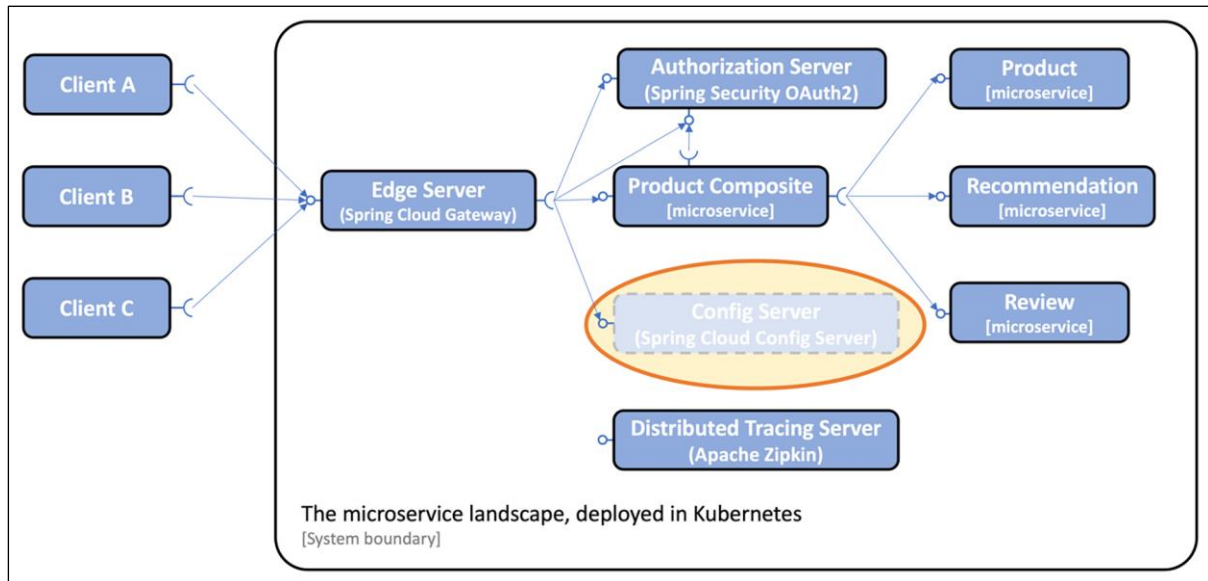
```
bash
{
  "status": "UP"
}
$
```

```
root@81676feda95e: /application  1
{
  "status": "UP",
  "components": {
    "mongo": {
      "status": "UP",
      "details": {
        "maxWireVersion": 17
      }
    },
    "rabbit": {
      "status": "UP",
      "details": {
        "version": "3.11.8"
      }
    },
    "readinessState": {
      "status": "UP"
    }
  }
}
$
```

```
root@81676feda95e: /application
"hands-on/auth-server:v1"
"hands-on/config-server:v1"
"hands-on/gateway:v1"
"hands-on/product-service:v1"
"hands-on/product-composite-service:v1"
"hands-on/recommendation-service:v1"
"hands-on/review-service:v1"
"registry.hub.docker.com/openzipkin/zipkin:2.24.0"
$
```

Chapter 17: Implementing Kubernetes Features to Simplify the System Landscape



NAME	READY	STATUS	RESTARTS	AGE
cert-manager-64f9f45d6f-2dmld	1/1	Running	0	10h
cert-manager-cainjector-56bbdd5c47-krzpw	1/1	Running	0	10h
cert-manager-webhook-d4f4545d7-zrhvf	1/1	Running	0	10h
\$				

```
bash
EVENT      NAME      READY  SECRET  AGE
ADDED      tls-certificate      tls-certificate  0s
...
MODIFIED   tls-certificate  False  tls-certificate  0s
...
MODIFIED   tls-certificate  True   tls-certificate  1s
```

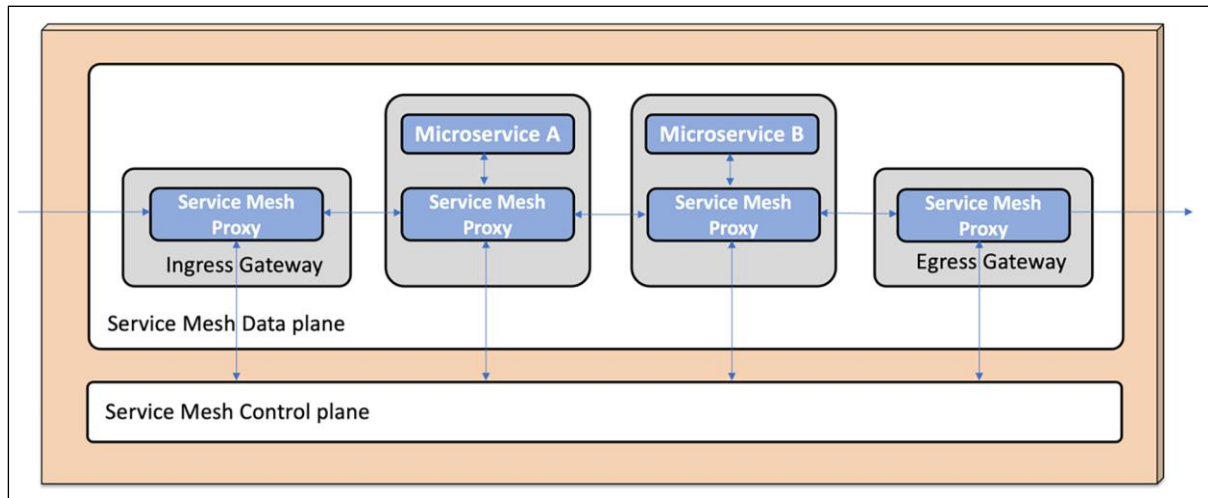
```
bash
Start Tests: Sun May 7 07:48:30 CEST 2023
HOST=minikube.me
PORT=8443
USE_K8S=true
SKIP_CB_TESTS=false
Wait for: curl -k https://minikube.me:8443/actuator/health... DONE, continues...
ACCESS_TOKEN=eyJra
Test OK (HTTP Code: 202, )
...
End, all tests OK: Sun May 7 07:48:58 CEST 2023
$
```

```
bash
Not After:      2023-08-05T05:42:49Z
Not Before:     2023-05-07T05:42:49Z
Renewal Time:   2023-07-06T05:42:49Z
```

```
bash
0s      Normal    Issuing    certificate/tls-certificate
Renewing certificate as renewal was scheduled at 2023-05-07 05:55:40 +0000 UTC
...
0s      Normal    Issuing    certificate/tls-certificate
The certificate has been successfully issued
```

```
Start Tests: Sun May 7 08:13:43 CEST 2023
HOST=localhost
PORT=8443
USE_K8S=false
SKIP_CB_TESTS=false
Restarting the test environment...
$ docker-compose down --remove-orphans
$ docker-compose up -d
...
:: Container chapter17-auth-server-1      Healthy      6.1s
...
:: Container chapter17-zipkin-server-1    Started      0.8s
...
Wait for: curl -k https://localhost:8443/actuator/health... ,
  retry #1 , retry #2 DONE, continues...
ACCESS_TOKEN=eyJra...
Test OK (HTTP Code: 202, )
...
We are done, stopping the test environment...
$ docker-compose down
...
:: Container chapter17-auth-server-1      Removed      1.3s
...
:: Container chapter17-zipkin-server-1    Removed      2.6s
...
End, all tests OK: Sun May 7 08:14:43 CEST 2023
$
```

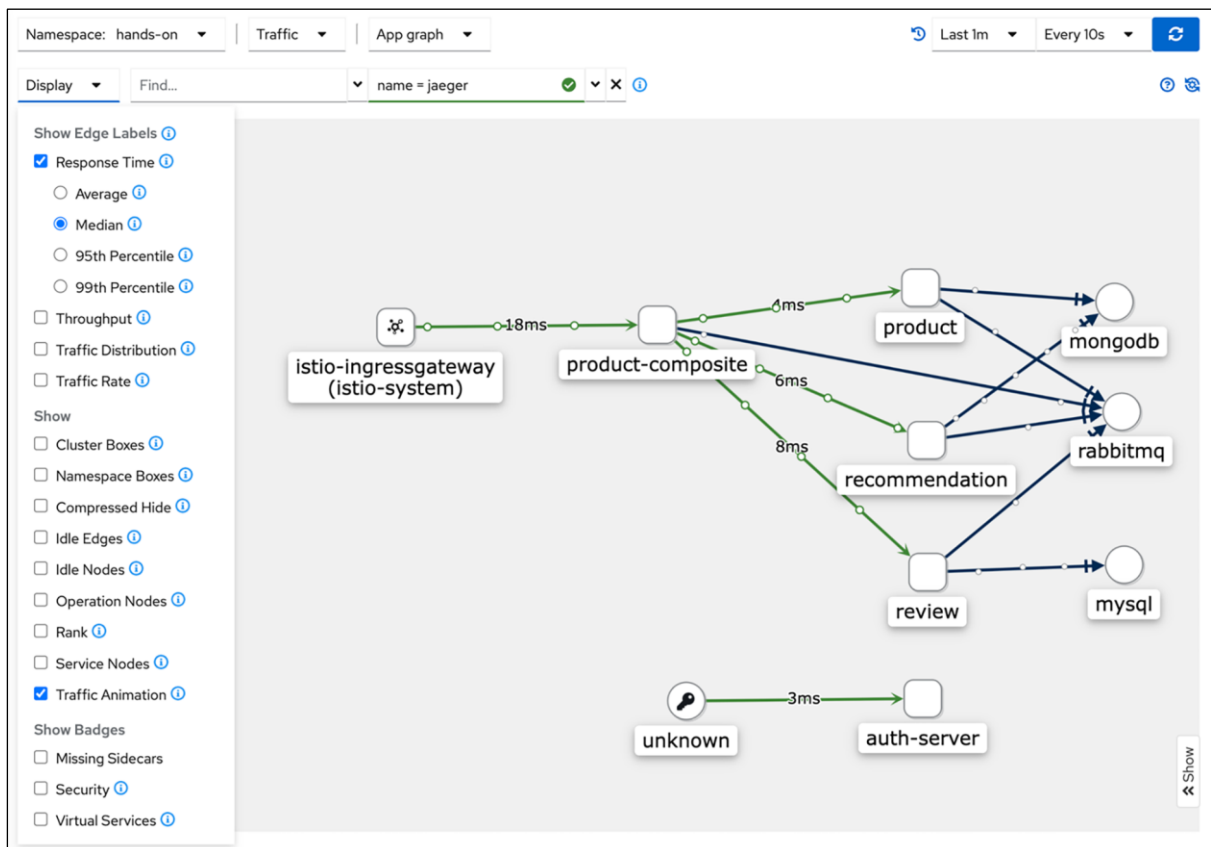
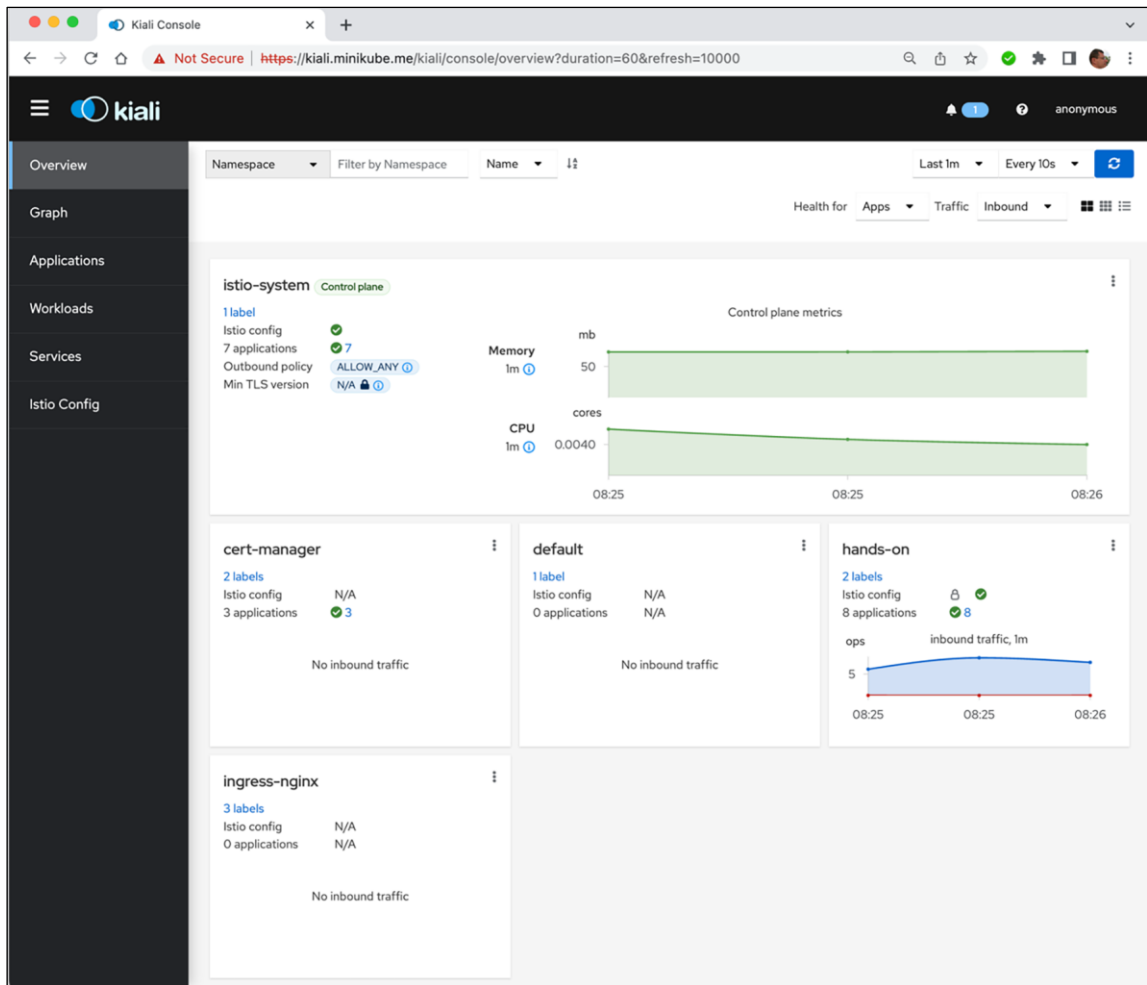
Chapter 18: Using a Service Mesh to Improve Observability and Management

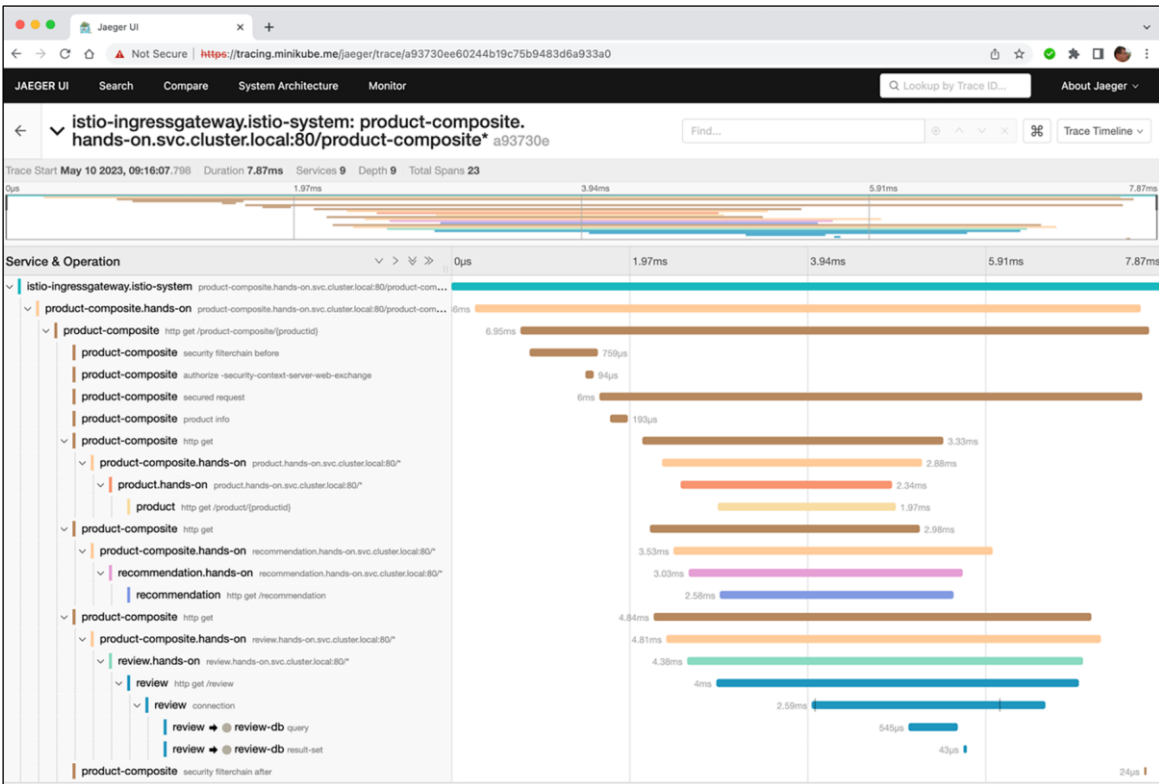
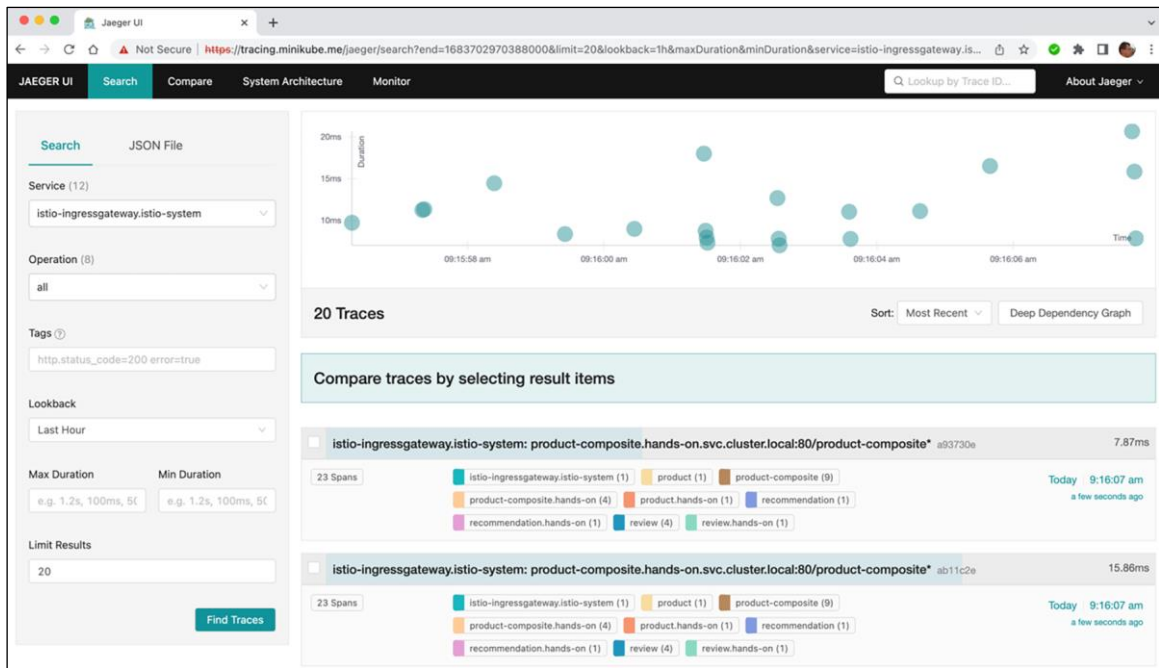


```
bash
NAME                                READY   STATUS    RESTARTS   AGE
auth-server-cf57b9bff-l2wnf         2/2    Running   0           39s
mongodb-6f4b595bbf-kvwnm            1/1    Running   0           39s
mysql-5dd488c7d-l7lvz               1/1    Running   0           39s
product-5454d77458-dvrtz            2/2    Running   0           39s
product-composite-59fb54568c-t8czm   2/2    Running   0           39s
rabbitmq-9cbbb894-wlkgv             1/1    Running   0           39s
recommendation-96984995d-mpvq9       2/2    Running   0           39s
review-5f965dcd76-dsgsm             2/2    Running   0           39s
$
```

```
bash
Start Tests: Tue May 9 18:12:07 CEST 2023
...
Wait for: curl -k https://health.minikube.me/actuator/health...
DONE, continues...
...
Test OK (HTTP Code: 200)
...
End, all tests OK: Tue May 9 18:12:33 CEST 2023
$
```

```
bash
** SIEGE 4.1.6
** Preparing 1 concurrent users for battle.
The server is now under siege...
HTTP/1.1 200    0.08 secs:    771 bytes ==> GET    /product-composite/1
HTTP/1.1 200    0.03 secs:    771 bytes ==> GET    /product-composite/1
HTTP/1.1 200    0.03 secs:    771 bytes ==> GET    /product-composite/1
HTTP/1.1 200    0.05 secs:    771 bytes ==> GET    /product-composite/1
HTTP/1.1 200    0.03 secs:    771 bytes ==> GET    /product-composite/1
```



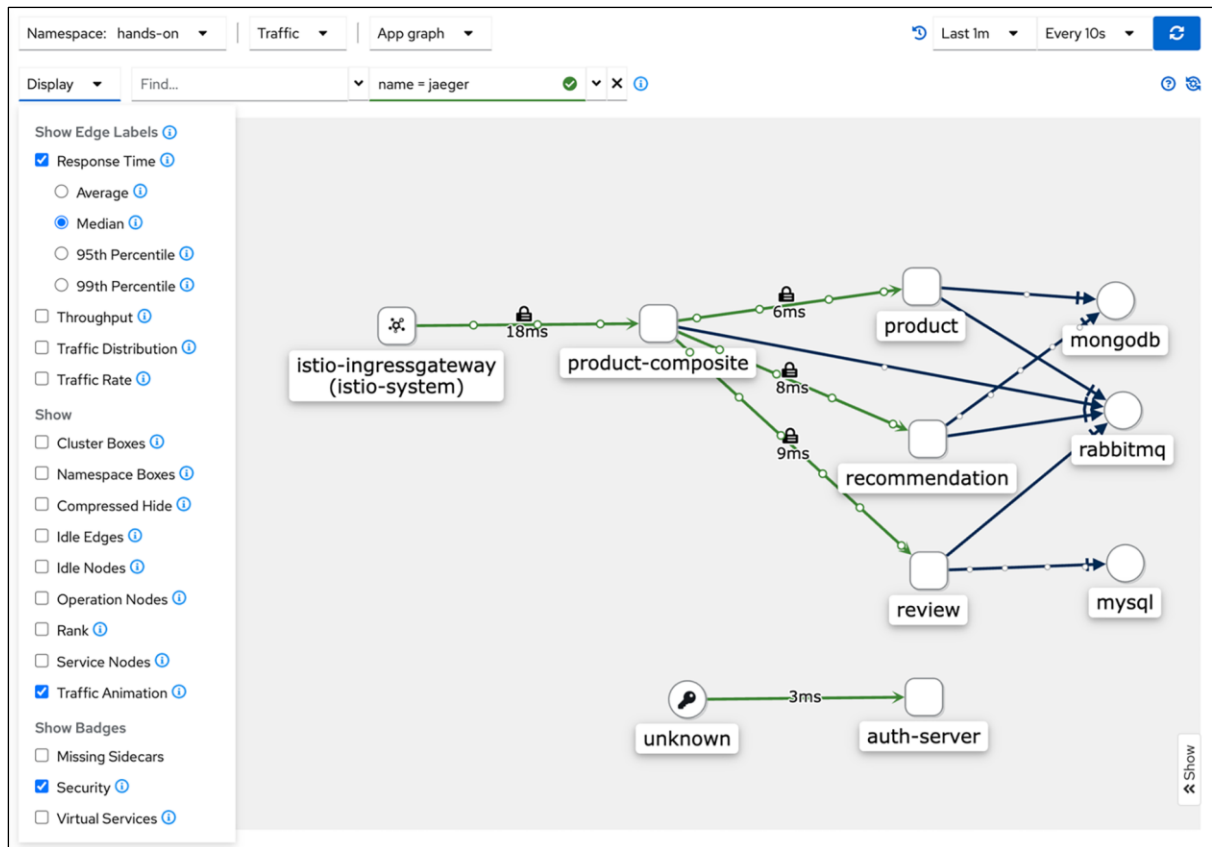


bash

Owner: SERIALNUMBER=my-sn, CN=minikube.me, OU=my-ou, O=my-org, OID.2.5.4.15=my-pc, STREET=my-address, L=my-locality, ST=my-province, C=my-country

Issuer: CN=hands-on-ca

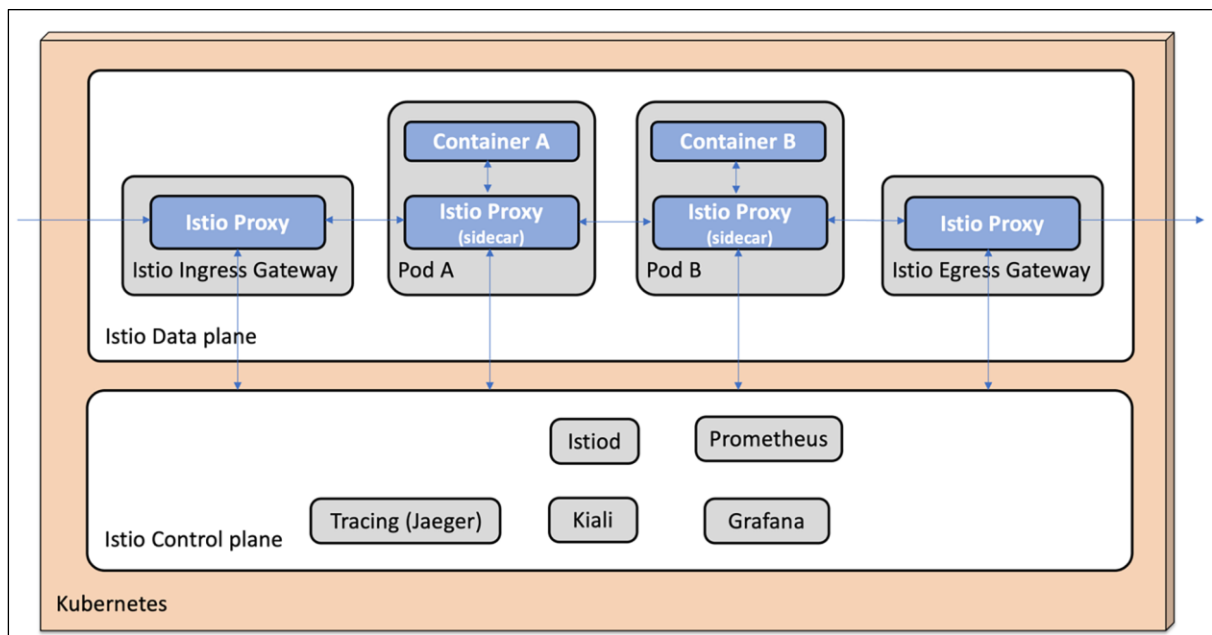
\$



```

bash
HTTP/1.1 200 0.03 secs: 771 bytes ==> GET /product-composite/1
HTTP/1.1 200 1.02 secs: 771 bytes ==> GET /product-composite/1
HTTP/1.1 200 0.03 secs: 771 bytes ==> GET /product-composite/1
HTTP/1.1 200 1.05 secs: 771 bytes ==> GET /product-composite/1
HTTP/1.1 200 0.02 secs: 771 bytes ==> GET /product-composite/1
HTTP/1.1 200 0.03 secs: 771 bytes ==> GET /product-composite/1
HTTP/1.1 200 0.01 secs: 771 bytes ==> GET /product-composite/1

```

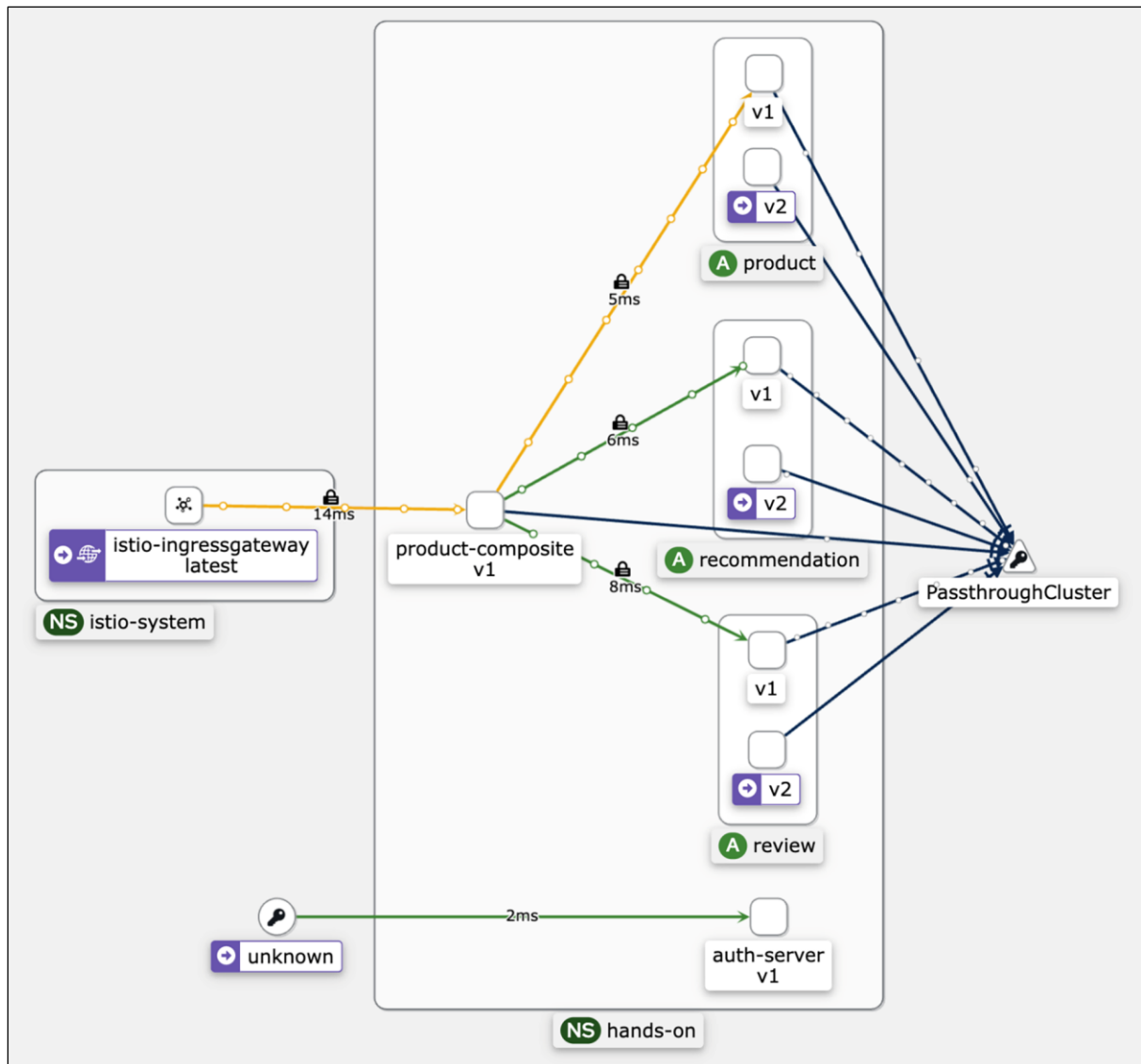



```
bash
{"timestamp":"2023-05-10T07:57:51.042+00:00","path":"/product-composite/1",
"status":500,"error":"Internal Server Error","message":"Did not observe any
item or terminal signal within 2000ms in 'onErrorResume' (and no fallback
has been configured)","requestId":"c1414130-16156"}
real    0m2.272s
```

```
bash
{"productId":1,"name":"Fallback product1",...}
real    0m0.236s
$
```

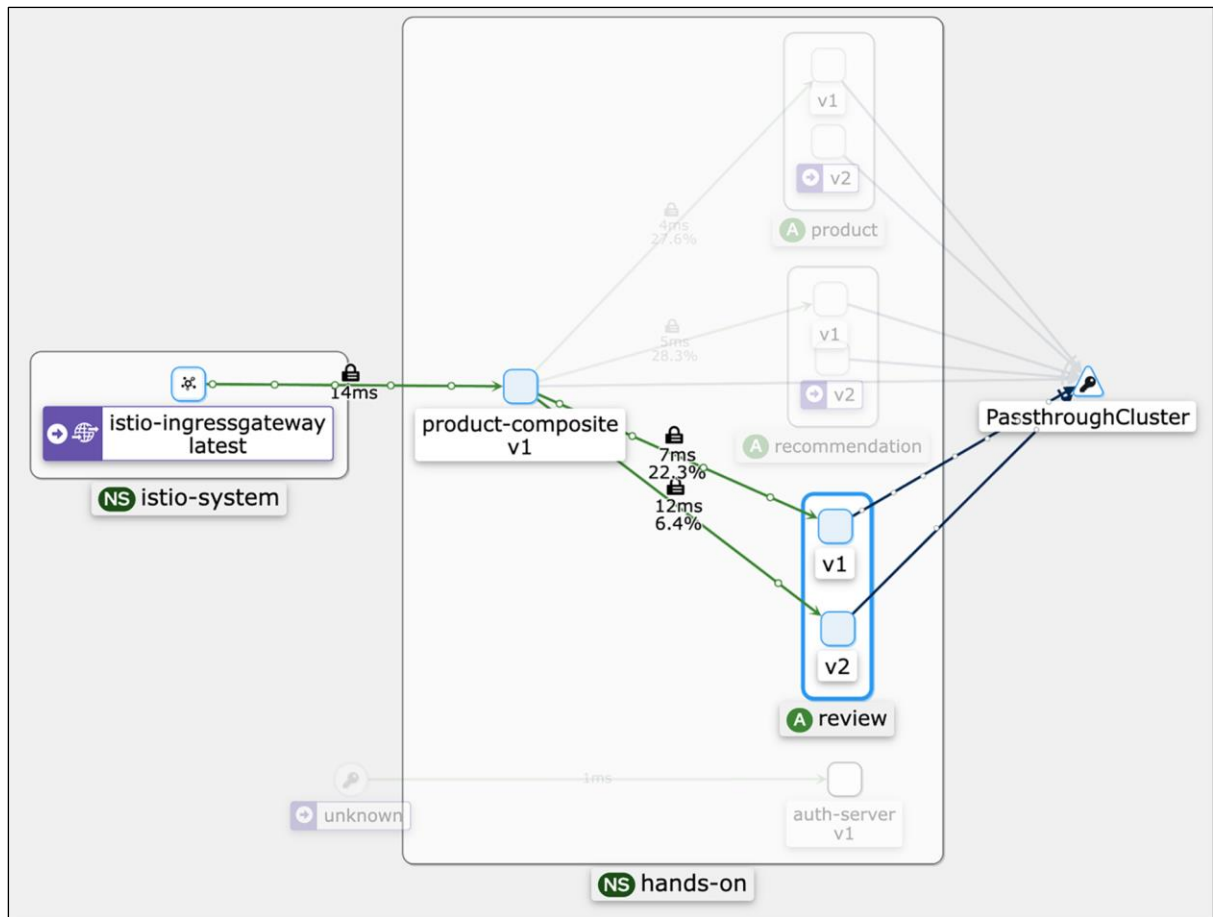
```
bash
NAME                                READY   STATUS    RESTARTS   AGE
auth-server-78c4d44b55-m12rg        2/2     Running   0           47s
product-composite-546597978c-j2cdc   2/2     Running   0           47s
product-v1-76dd48574-hlfgt          2/2     Running   0           47s
product-v2-5ff8575567-s6j2b         2/2     Running   0           47s
recommendation-v1-8fc6456fd-dv92b    2/2     Running   0           47s
recommendation-v2-655b4c4cc7-l7jxz   2/2     Running   0           47s
review-v1-5f5f74b49d-bpprh          2/2     Running   0           47s
review-v2-5d86f74c6f-f7rgm          2/2     Running   0           47s
$
```

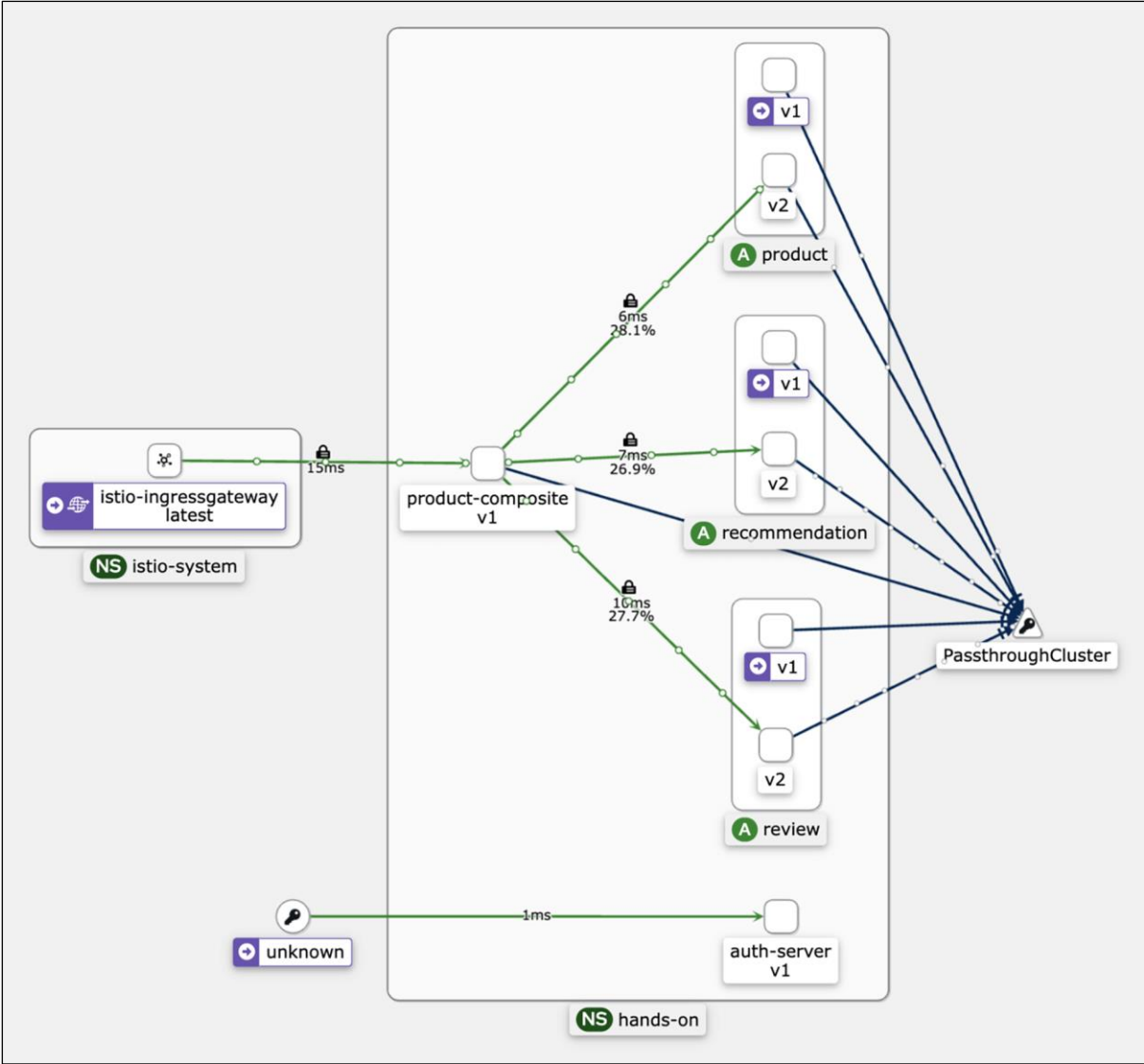
```
bash
Start Tests: Wed May 10 10:07:21 CEST 2023
...
Wait for: curl -k https://health.minikube.me/actuator/health...
DONE, continues...
...
Test OK (HTTP Code: 200)
...
End, all tests OK: Wed May 10 10:07:48 CEST 2023
$
```

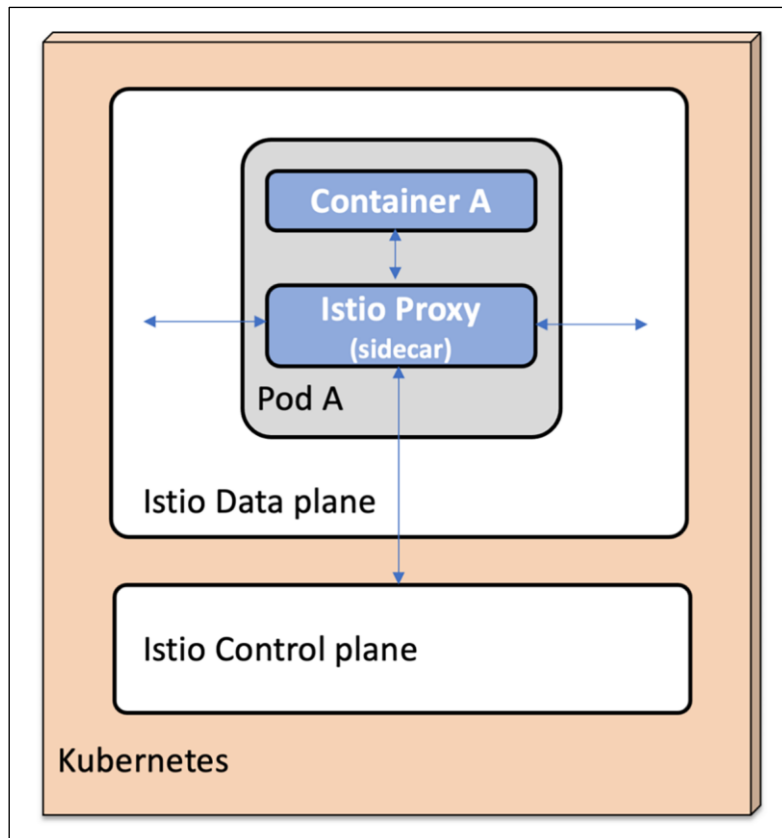


```
bash
{
  "cmp": "product-composite-546597978c-j2cdc/10.244.0.26:80",
  "pro": "product-v1-76dd48574-hlfgt/10.244.0.28:80",
  "rev": "review-v1-5f5f74b49d-bpprh/10.244.0.29:80",
  "rec": "recommendation-v1-8fc6456fd-dv92b/10.244.0.30:80"
}
$
```

```
bash
{
  "cmp": "product-composite-546597978c-j2cdc/10.244.0.26:80",
  "pro": "product-v2-5ff8575567-s6j2b/10.244.0.27:80",
  "rev": "review-v2-5d86f74c6f-f7rgm/10.244.0.31:80",
  "rec": "recommendation-v2-655b4c4cc7-l7jxz/10.244.0.33:80"
}
$
```







```
bash
handson-spring-boot-cloud
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
$
```

```
bash
✓ No issues found when checking the cluster.
Istio is safe to install or upgrade!
  To get started, check out https://istio.io/latest/docs/setup/getting-started/
$
```

```

NAME                READY    UP-TO-DATE    AVAILABLE    AGE
grafana              1/1      1              1            46s
istio-egressgateway  1/1      1              1           3m45s
istio-ingressgateway 1/1      1              1           3m45s
istiod               1/1      1              1           4m1s
jaeger               1/1      1              1           62s
kiali                1/1      1              1           72s
prometheus           1/1      1              1           54s
$

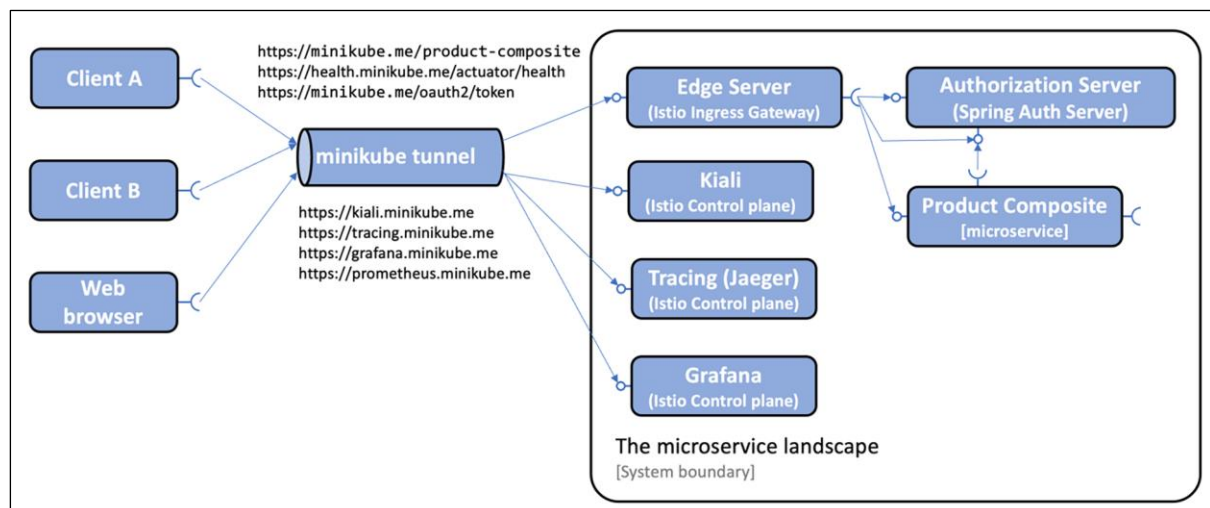
```

```

NAME                TYPE              DATA    AGE
hands-on-certificate kubernetes.io/tls 3         59s

NAME                READY    SECRET              AGE
hands-on-certificate True      hands-on-certificate 65s
$

```

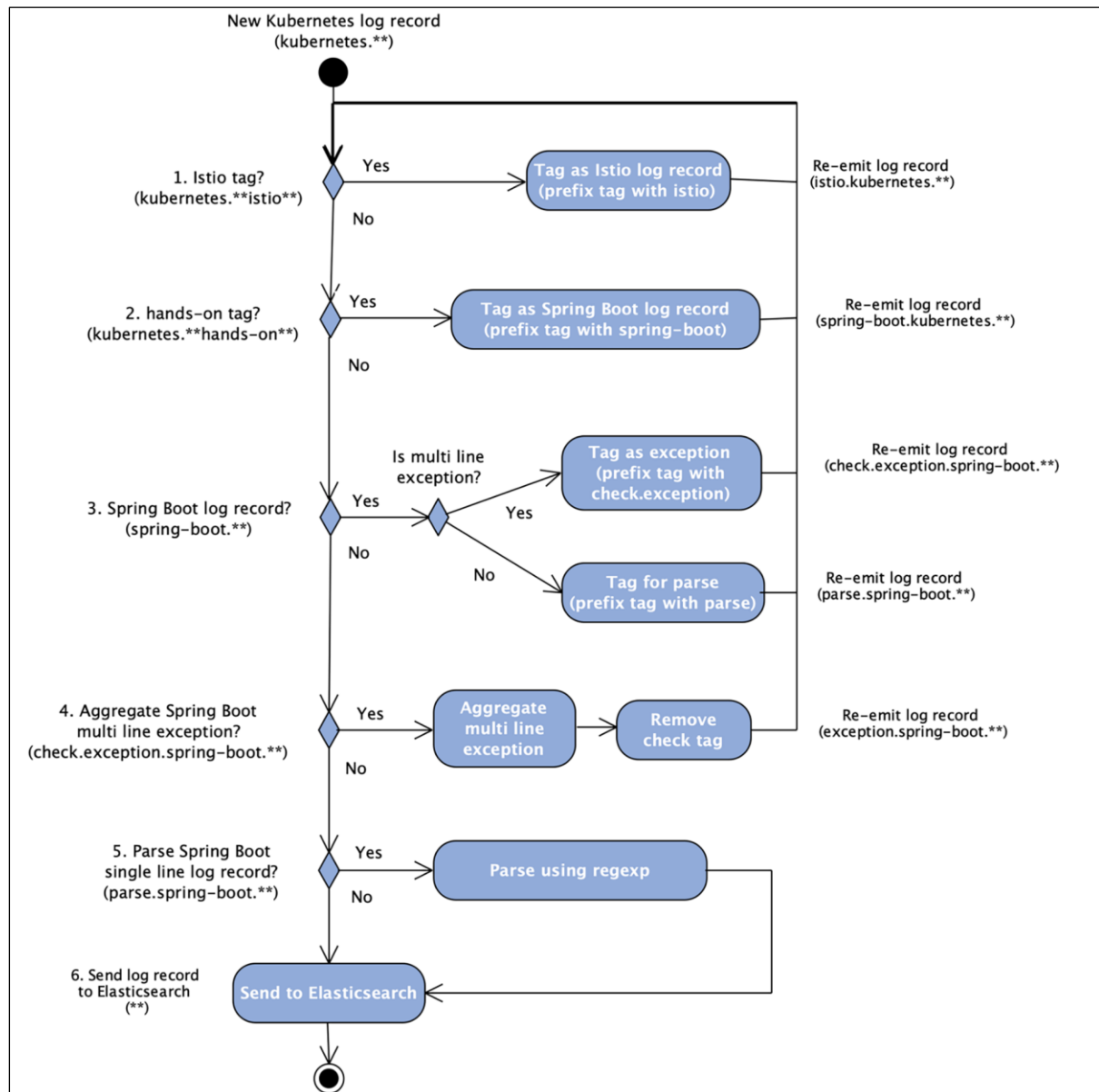


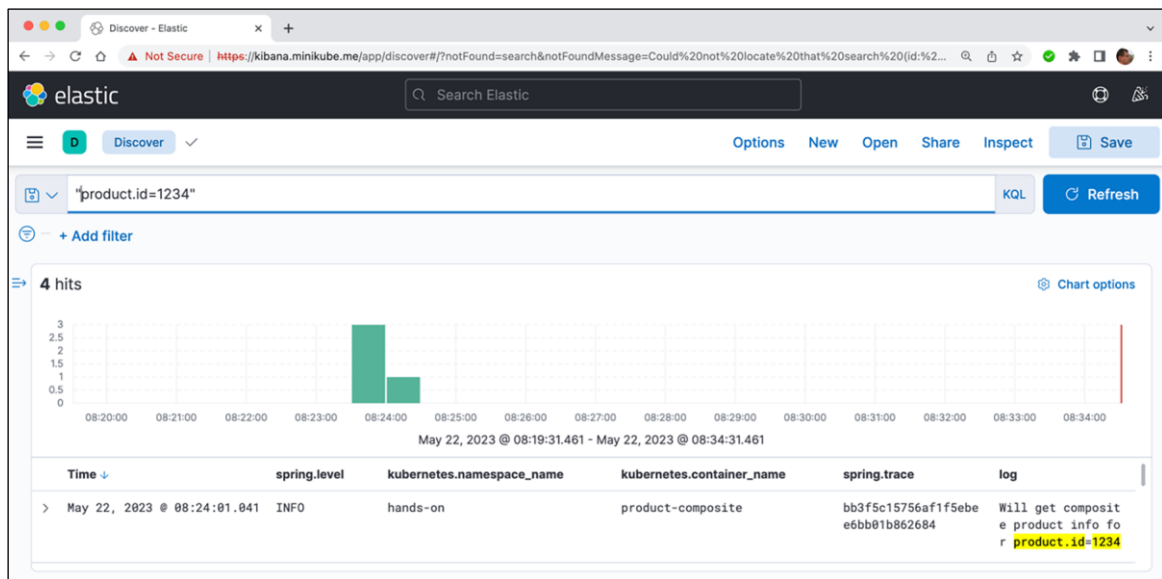
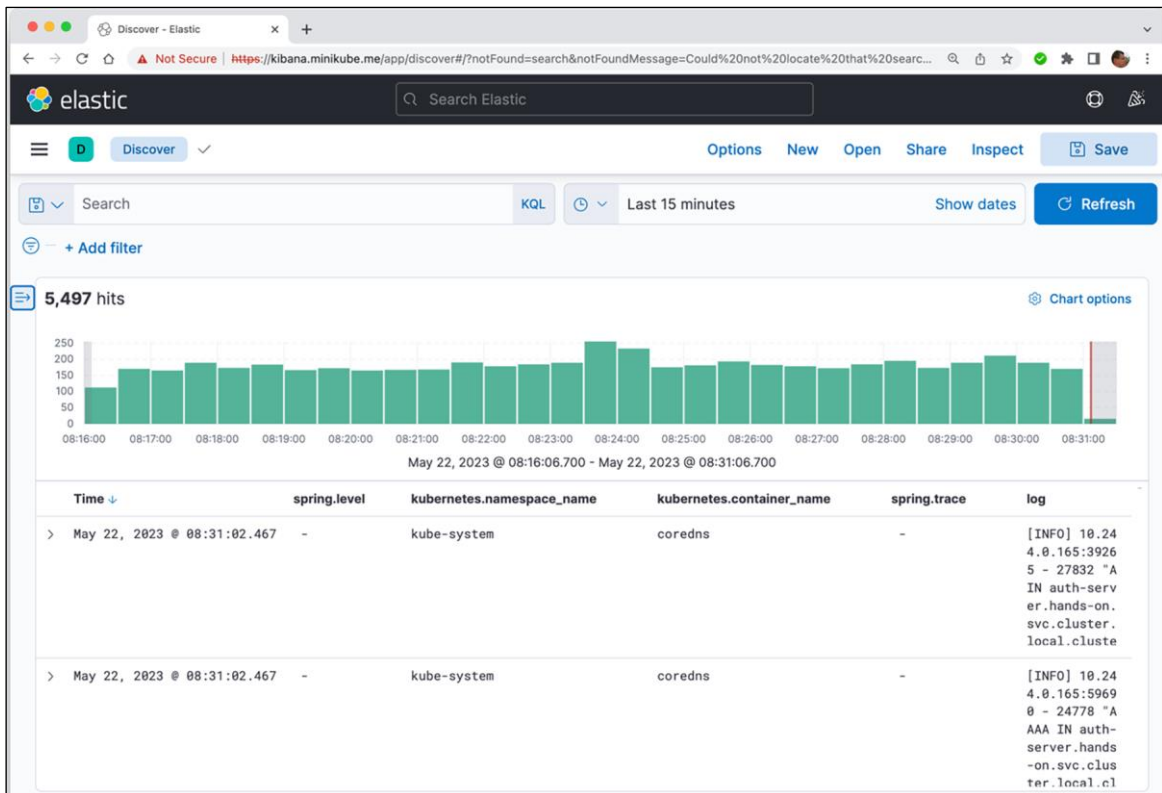
```

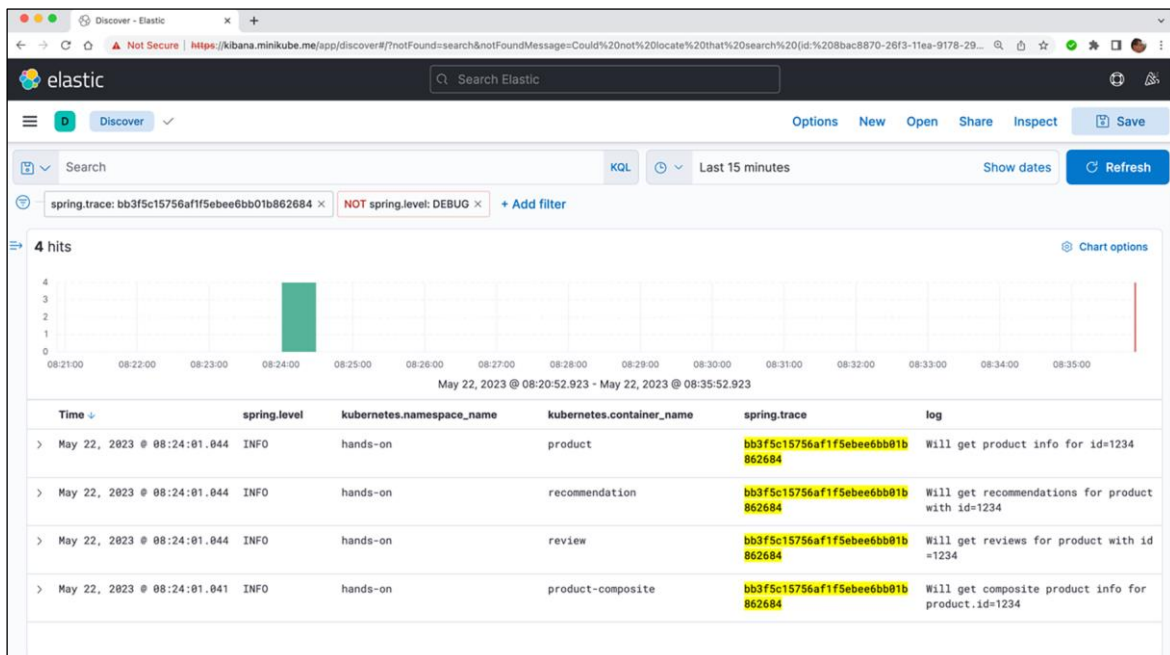
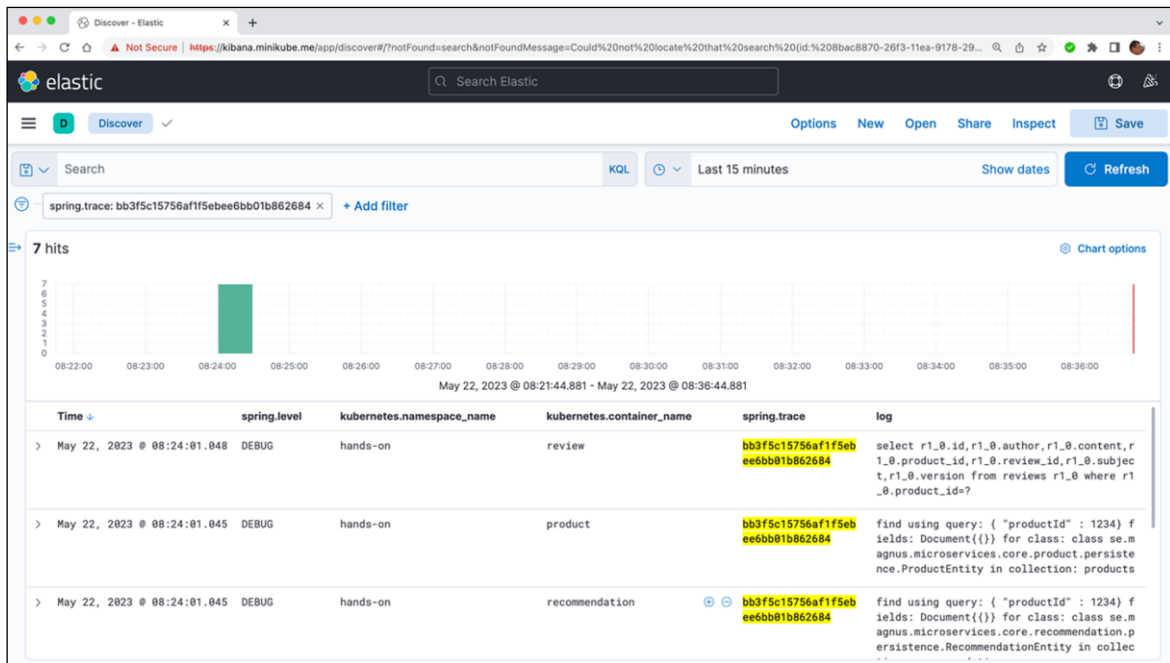
$ cat /etc/hosts
...
127.0.0.1 minikube.me grafana.minikube.me kiali.minikube.me
prometheus.minikube.me tracing.minikube.me kibana.minikube.me
elasticsearch.minikube.me mail.minikube.me health.minikube.me
$

```

Chapter 19: Centralized Logging with the EFK Stack

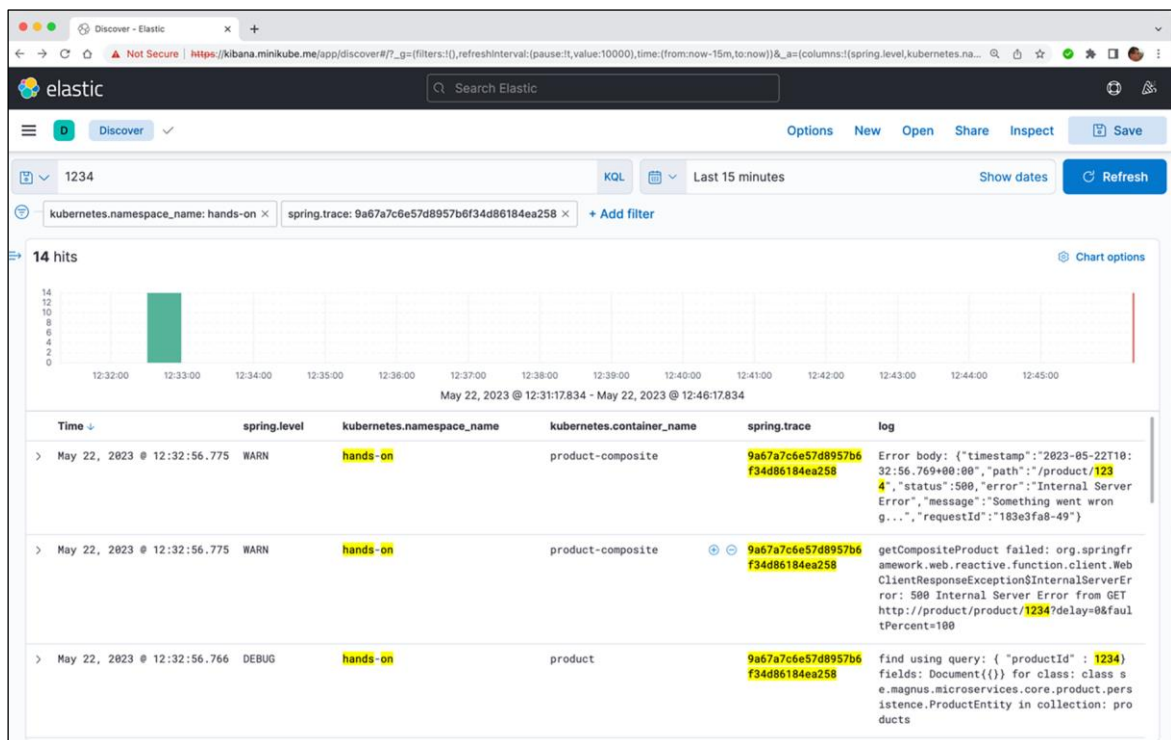
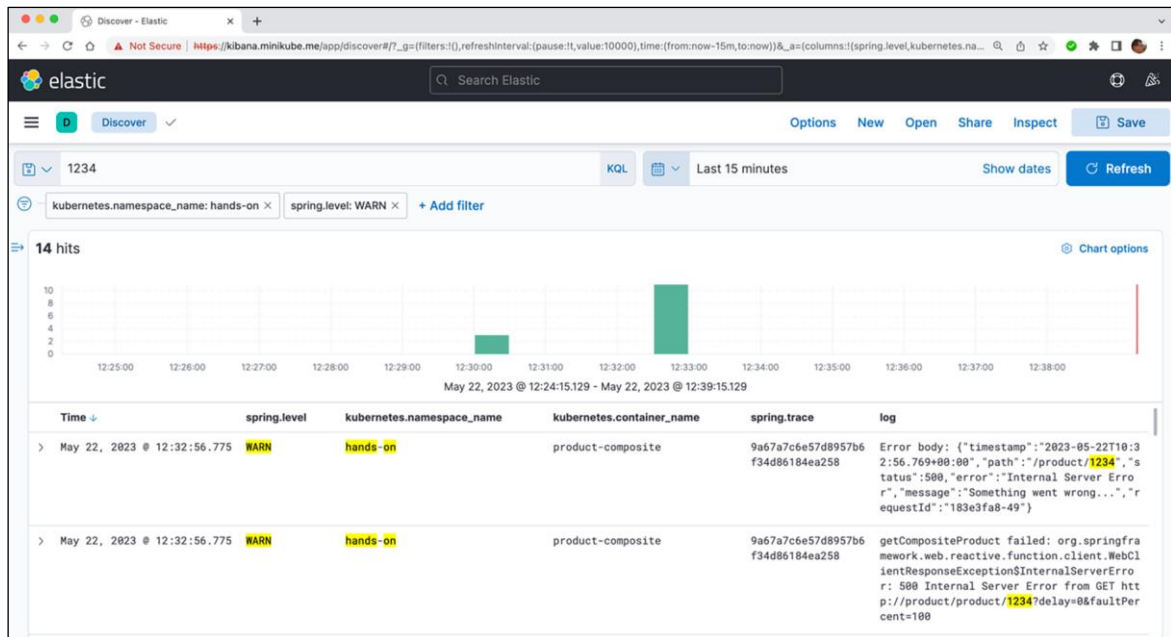


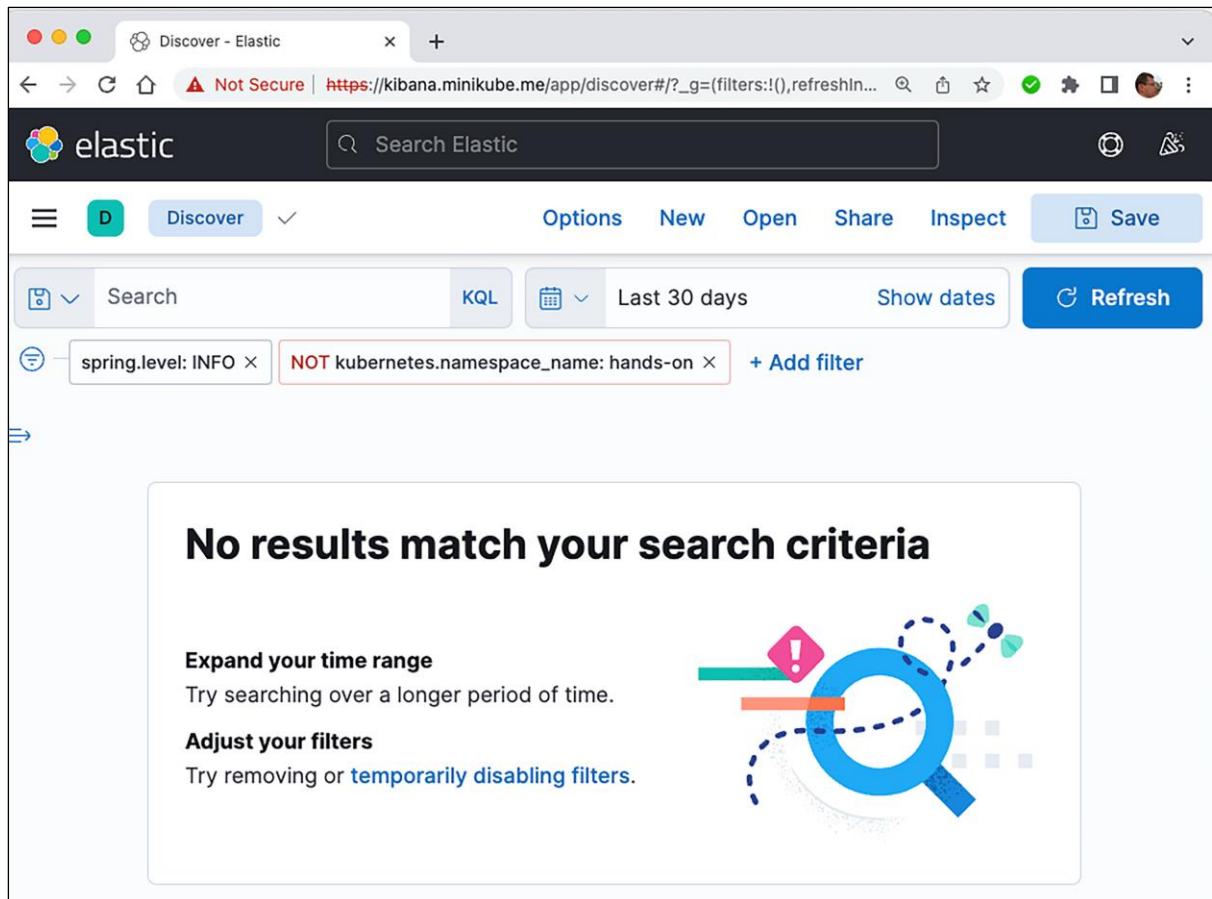




```
bash

{"timestamp": "2023-05-22T08:54:10.848+00:00",
  "path": "/product-composite/1234",
  "status": 500,
  "error": "Internal Server Error",
  "message": "500 Internal Server Error from GET http://product/product/1234?delay=0&faultPercent=100",
  "requestId": "64980619-101"
}
```





```
bash
Start Tests: Thu May 18 09:27:34 CEST 2023
...
Wait for: curl -k https://health.minikube.me/actuator/health...
DONE, continues...
...
Test OK (HTTP Code: 200)
...
End, all tests OK: Thu May 18 09:28:01 CEST 2023
$
```

Discover - Elastic

Not Secure | [https://kibana.minikube.me/app/discover/#/?_g=\(filters:!\(\),refreshInterval:\(...\)\)](https://kibana.minikube.me/app/discover/#/?_g=(filters:!(),refreshInterval:(...)))

elastic Search Elastic

Discover

Options New Open Share Inspect Save

Search KQL Last 1 hour Show dates Refresh

kubernetes.namespace_name: hands-on x spring.level: INFO x + Add filter

9 hits Chart options

log	Will get composite product info for product.id=1234
spring.class	s.m.m.c.p.s.ProductCompositeServiceImpl
spring.level	INFO
spring.pid	1
spring.service	product-composite
spring.span	bed8522970670d65
spring.thread	parallel-2
spring.trace	55d35b33a9a539250fc4a8af208fa8ae

Time	spring.level	kubernetes.namespace_name	kubernetes.container_name	spring.trace	log
May 24, 2023 @ 16:44:19.173	WARN	hands-on	product-composite	f4861c653b9238b39dffb78a0ff69f9	getCompositeProduct failed: se.ma gnus.api.exceptions.InvalidInputE xception: Invalid productId: -1

Expanded document View surrounding documents View single document

Lens - Elastic

Not Secure | [https://kibana.minikube.me/app/lens/#/?_g=\(filters:!\(\),refreshInterval:\(pause:1f,value:10000\),time:\(from:now-15m,to:now\)\)](https://kibana.minikube.me/app/lens/#/?_g=(filters:!(),refreshInterval:(pause:1f,value:10000),time:(from:now-15m,to:now)))

elastic Search Elastic

Visualize Library Create Inspect Download as CSV Save

Search KQL Last 15 minutes Show dates Refresh

+ Add filter

logstash-*

Search field names

Filter by type 0

Records

Available fields 37

- @timestamp
- docker.container_id.keyword
- kubernetes.container_image_id.keyword
- kubernetes.container_image.keyword
- kubernetes.container_name.keyword

Drop some fields here to start

Lens is a new tool for creating visualization

Make requests and give feedback

Bar vertical stacked

logstash-*

Horizontal axis

Add or drag-and-drop a field

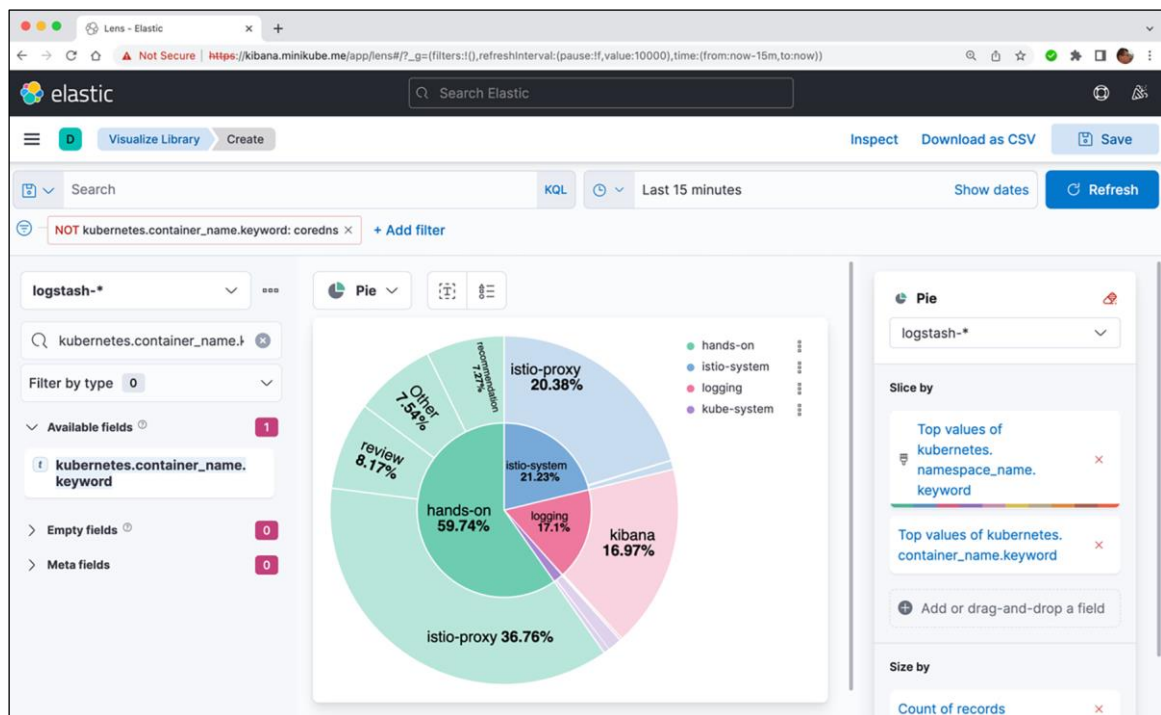
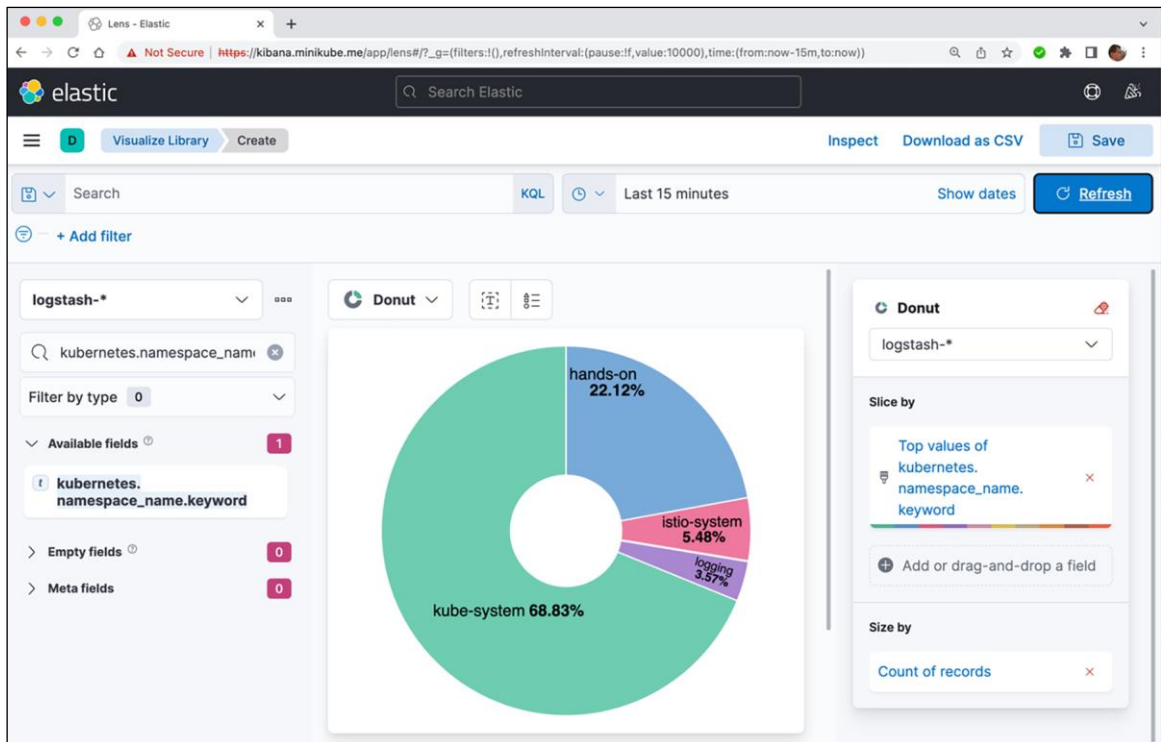
Vertical axis

Add or drag-and-drop a field

Break down by

Add or drag-and-drop a field

Add layer



Lens - Elastic

Not Secure | [https://kibana.minikube.me/app/lens/#/?_g=\(filters:!\(\),refreshInterval:\(pa...](https://kibana.minikube.me/app/lens/#/?_g=(filters:!(),refreshInterval:(pa...)

elastic Search Elastic

Save Lens visualization

Title

hands-on-visualization

Description

This is my first visualization in Kibana

Add to dashboard

☐ Existing

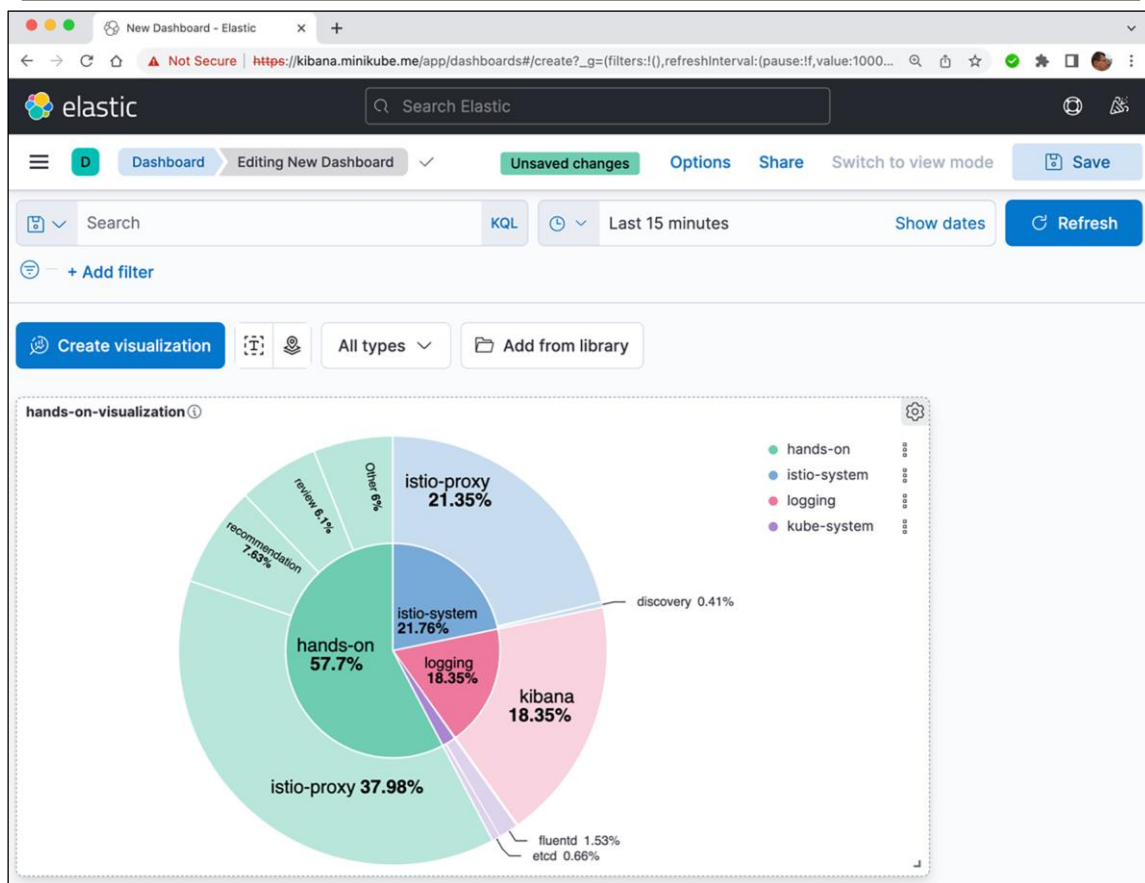
Search dashboards...

☒ New

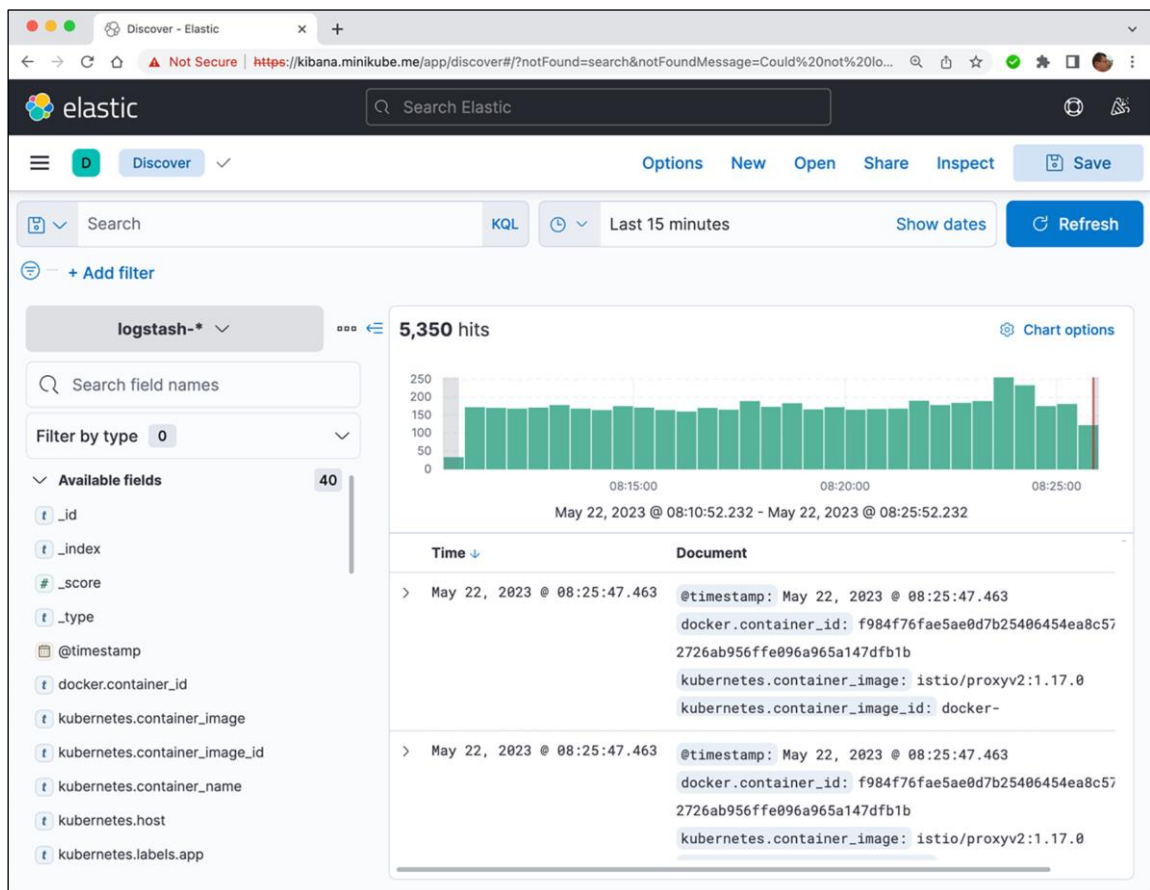
☐ None

☐ Add to library ⓘ

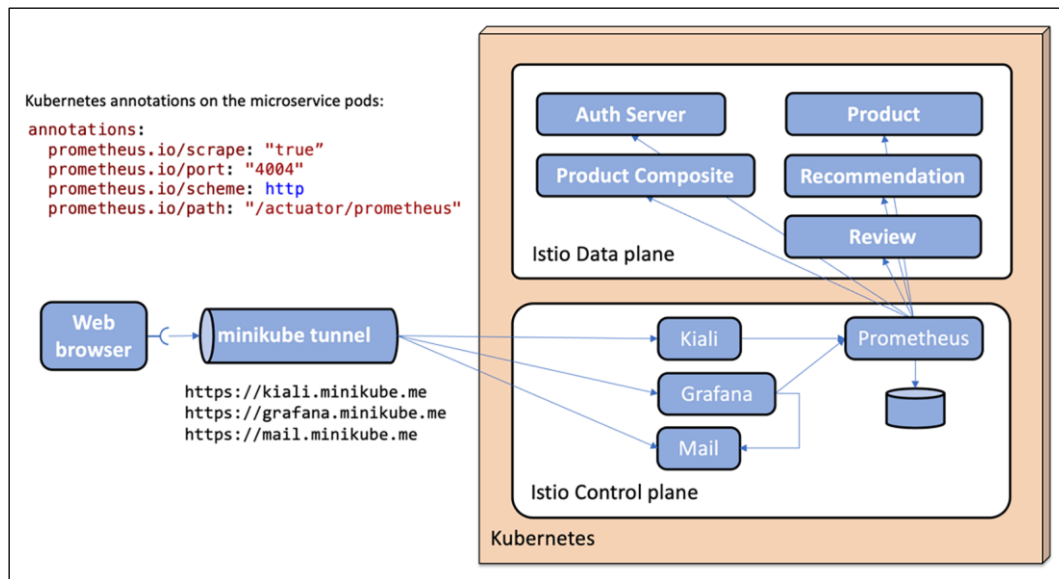
Cancel Save and go to Dashboard



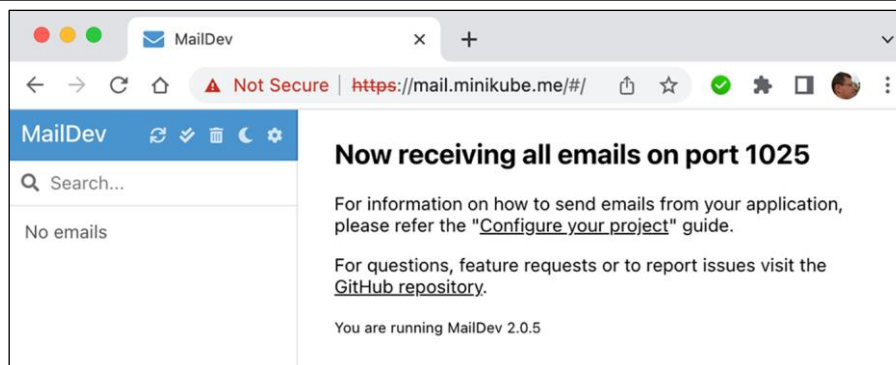

```
bash
{
  "productId": 1234,
  "name": "product name 1234",
  "weight": 1234,
  "recommendations": [],
  "reviews": [],
  "serviceAddresses": {
    "cmp": "product-composite-59fb54568c-p8kzd/10.244.0.166:80",
    "pro": "product-5454d77458-9b565/10.244.0.156:80",
    "rev": "",
    "rec": ""
  }
}
```

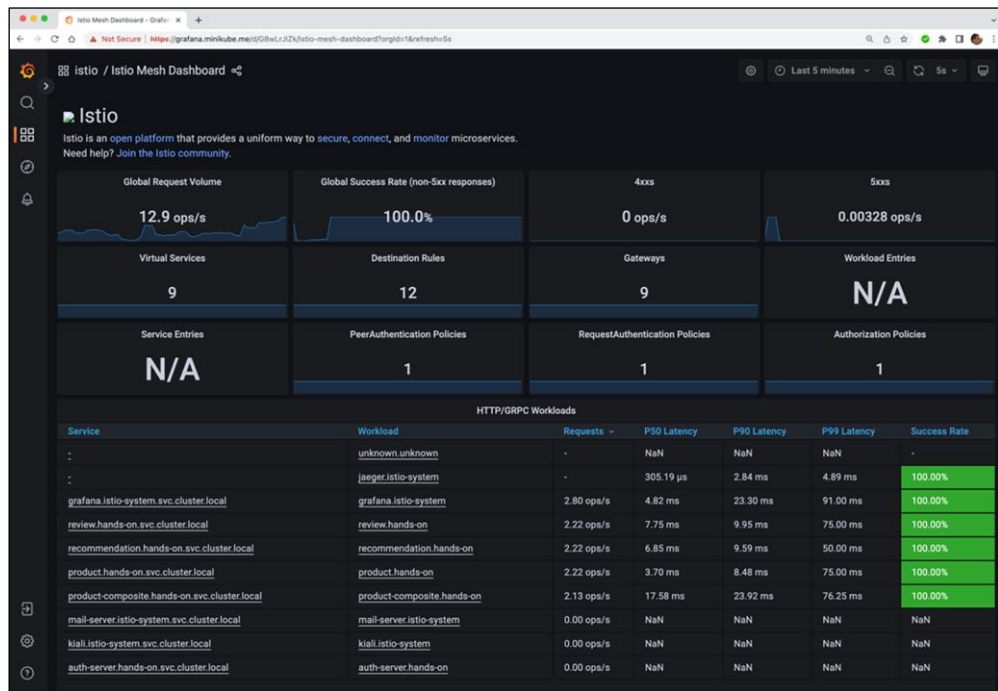
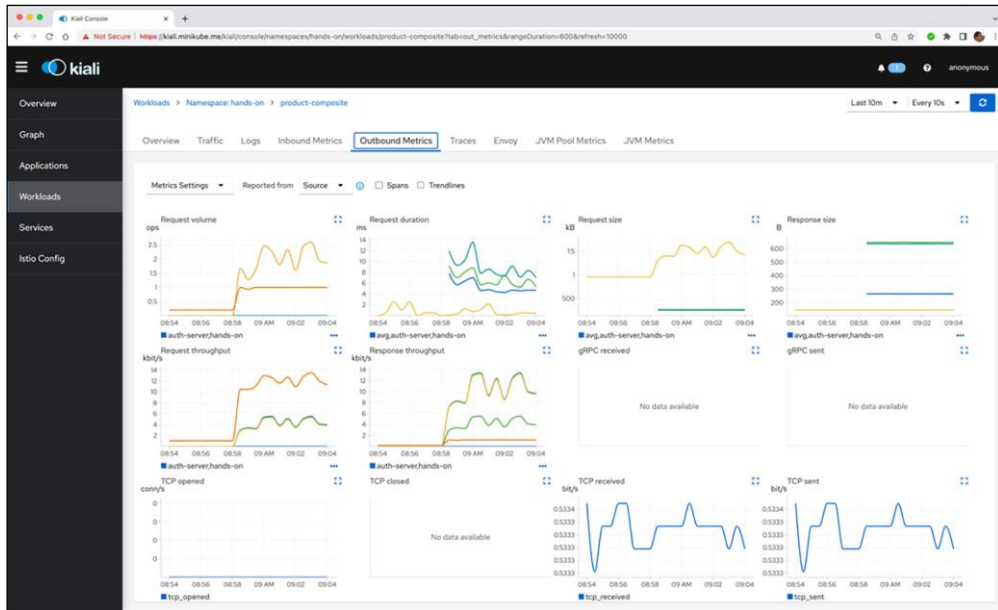


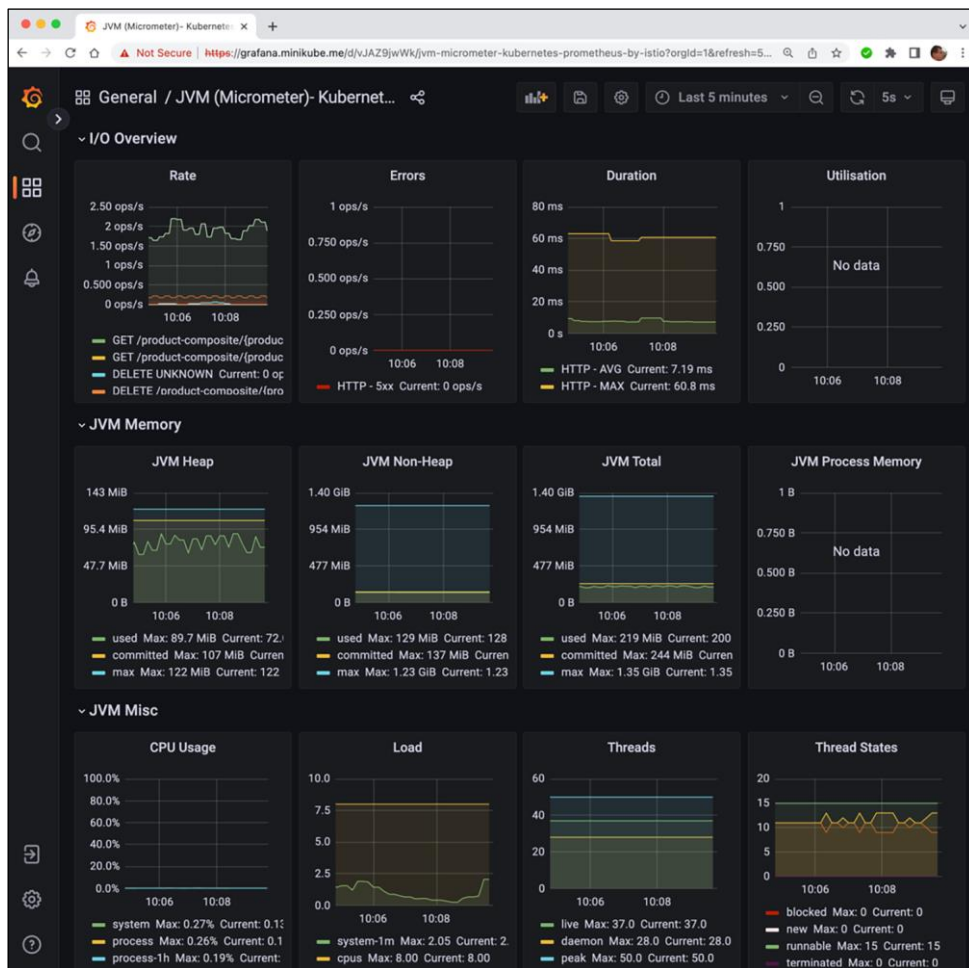
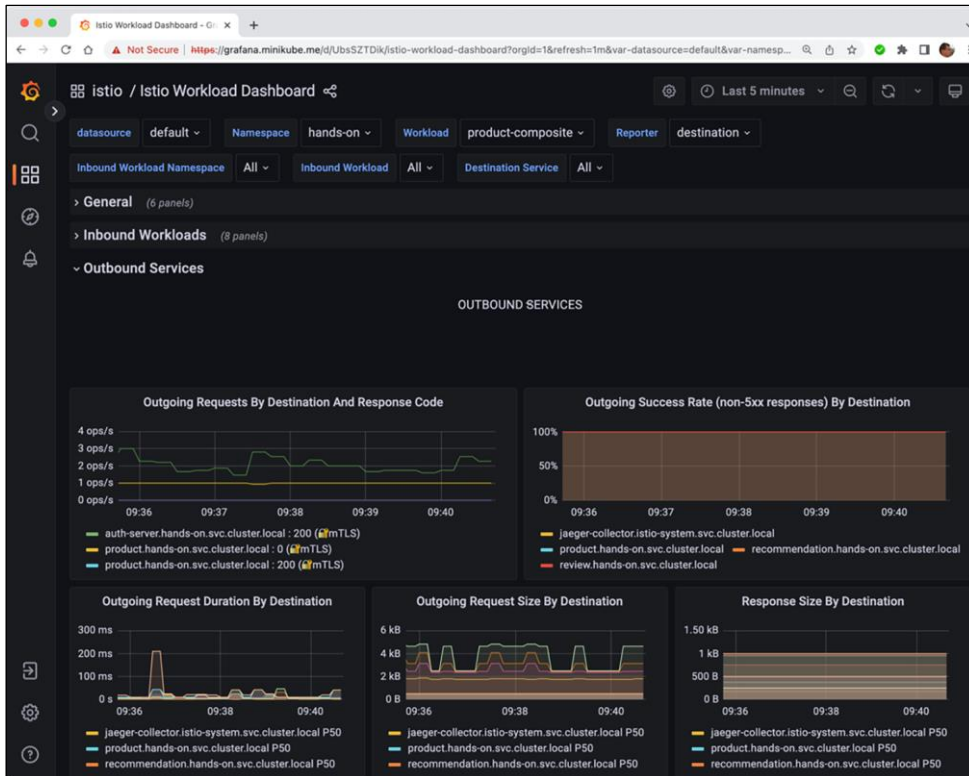
Chapter 20: Monitoring Microservices



```
Start Tests: Sat May 27 17:02:59 CEST 2023  
...  
Wait for: curl -k https://health.minikube.me/actuator/health...  
  
DONE, continues...  
...  
Test OK (HTTP Code: 200)  
...  
End, all tests OK: Sat May 27 17:03:26 CEST 2023  
$
```







```

...
resilience4j_retry_calls_total{application="product-composite",kind="successful_with_retry",name="product",} 0.0
resilience4j_retry_calls_total{application="product-composite",kind="failed_with_retry",name="product",} 0.0
resilience4j_retry_calls_total{application="product-composite",kind="successful_without_retry",name="product",} 11978.0
resilience4j_retry_calls_total{application="product-composite",kind="failed_without_retry",name="product",} 6.0
...
resilience4j_circuitbreaker_state{application="product-composite",name="product",state="open",} 0.0
resilience4j_circuitbreaker_state{application="product-composite",name="product",state="forced_open",} 0.0
resilience4j_circuitbreaker_state{application="product-composite",name="product",state="half_open",} 0.0
resilience4j_circuitbreaker_state{application="product-composite",name="product",state="closed",} 1.0
$

```

New dashboard - Grafana

Not Secure | <https://grafana.minikube.me/dashboard/new?orgId=1>

New dashboard

(Prometheus)

Query patterns Raw query Give feedback Run queries Explain Builder Beta Code

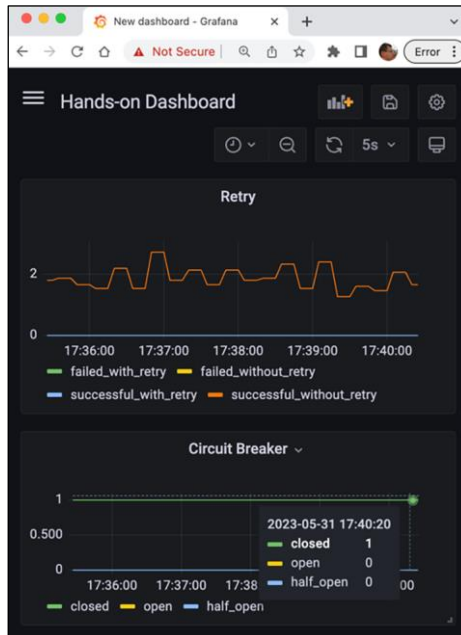
Metric: resilience4j_circuitbreaker_state Labels: state = closed

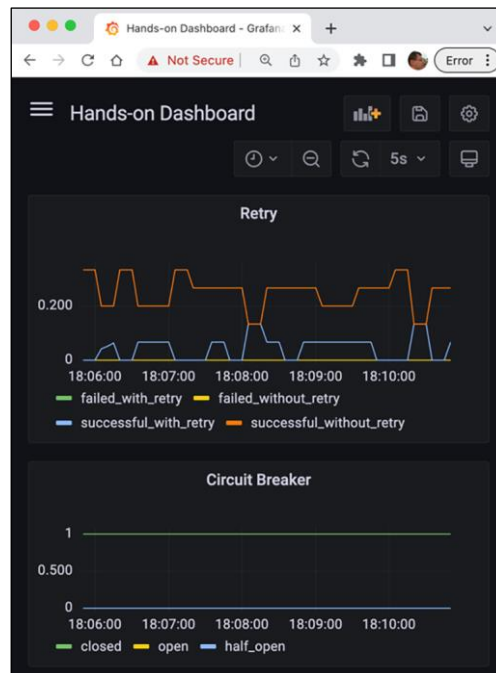
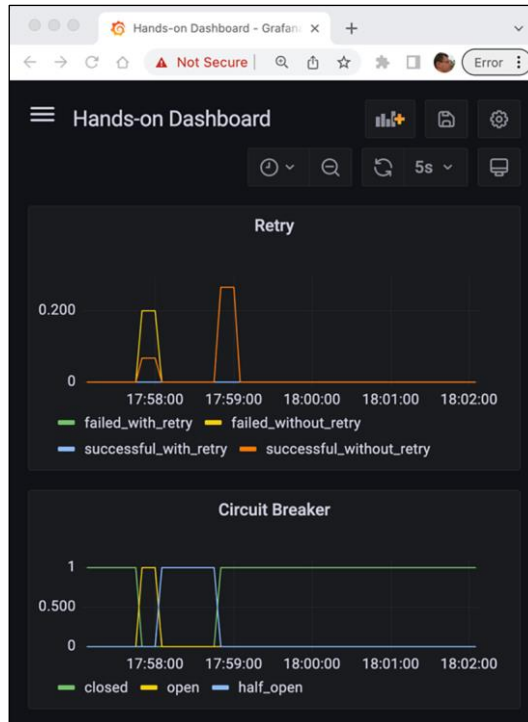
+ Operations

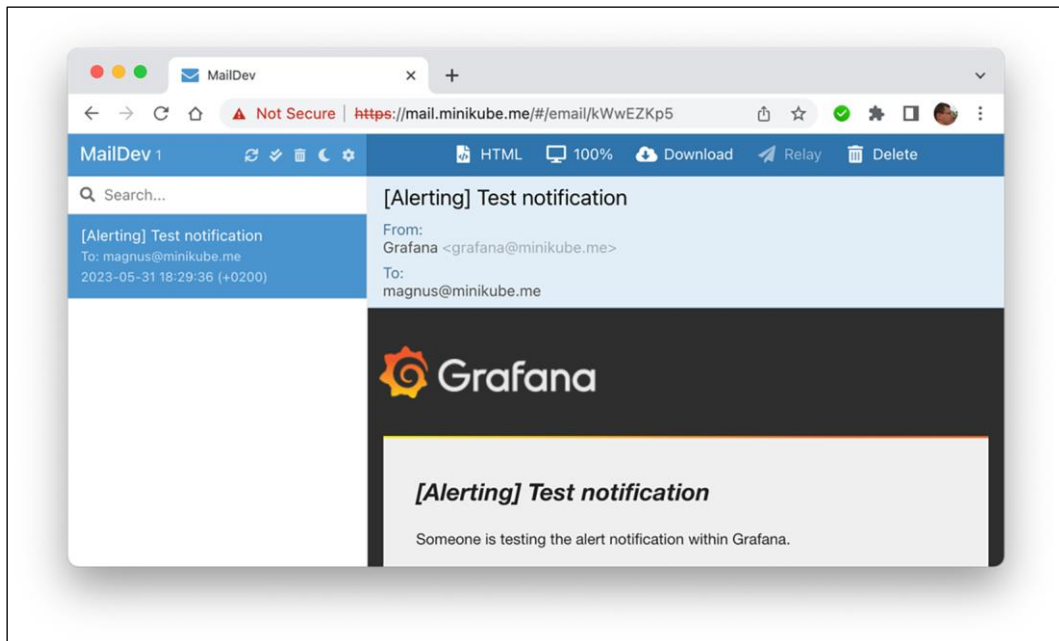
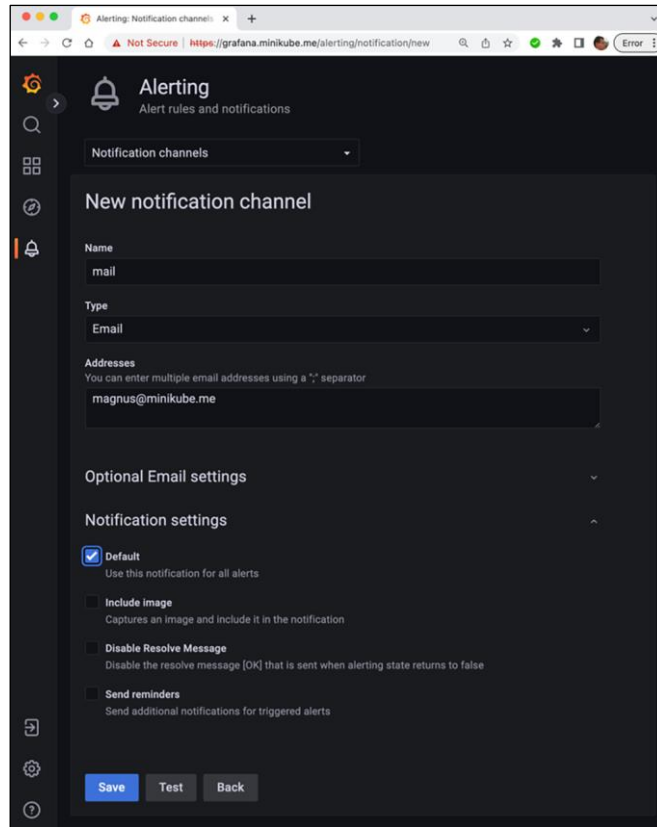
Raw query: resilience4j_circuitbreaker_state{state="closed"}

Options

Legend: {{state}} Min step: auto Format: Time series Type: Range Instant Exemplars: ☐







Query3

Transform0

Alert0

Rule

Name

Circuit Breaker alert

Evaluate every

10s

For

0m

Conditions

WHEN

max ()

OF

query (A, 10s, now)

IS BELOW

0.5

Alert list

Search options

Panel options

Title

Circuit Breaker Alerts

Description

Transparent background

Panel links

Repeat options

Options

Show

Recent state changes

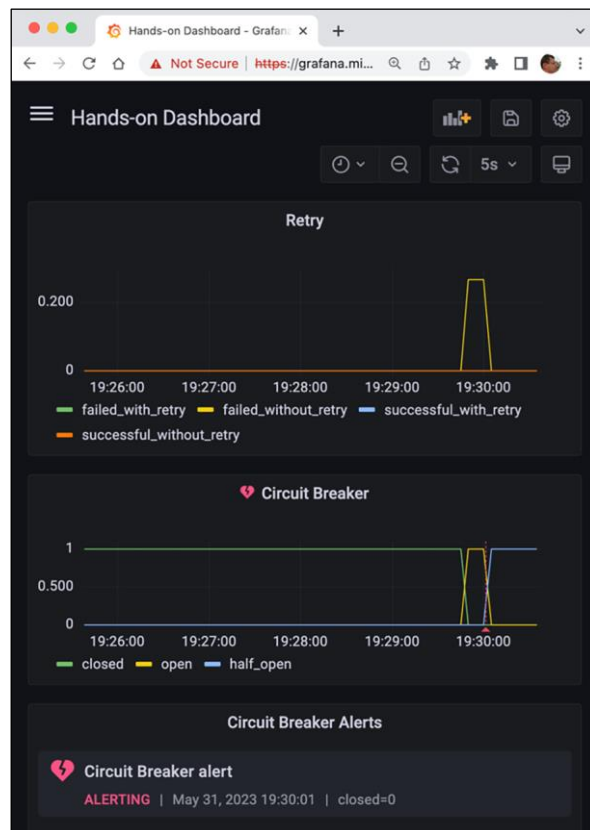
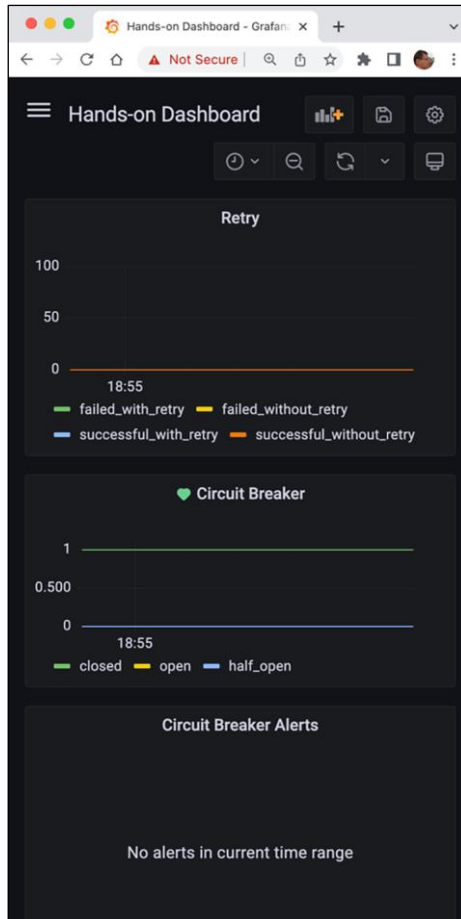
Max items

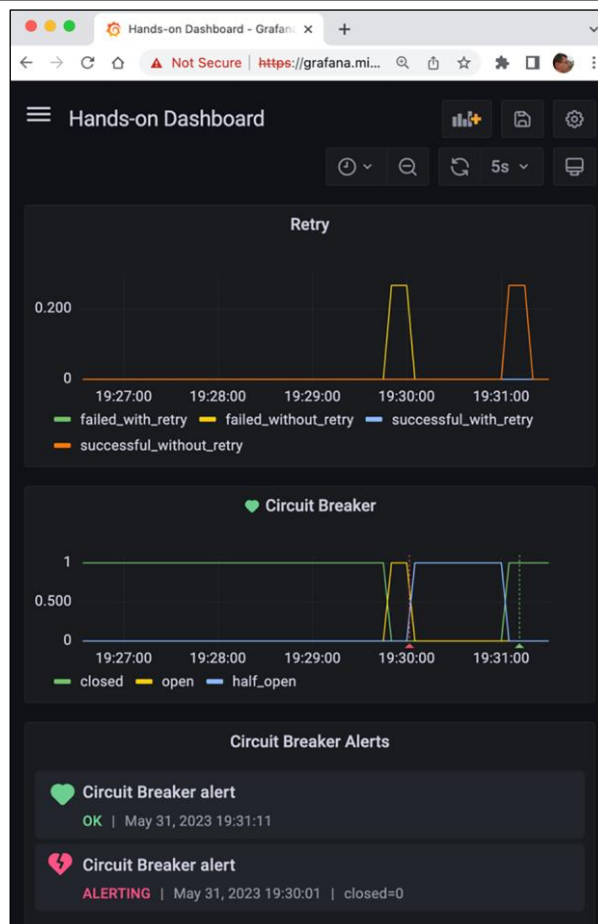
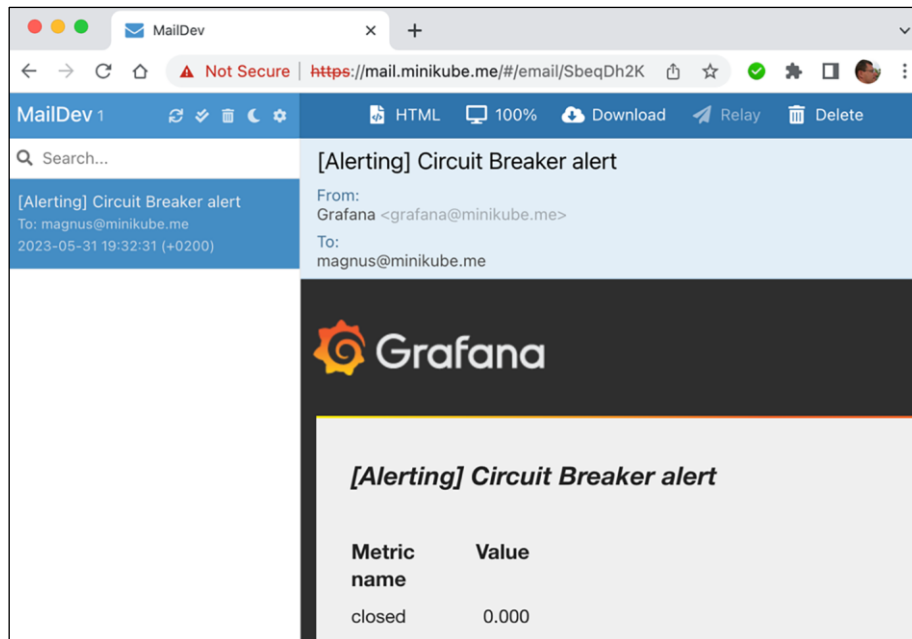
10

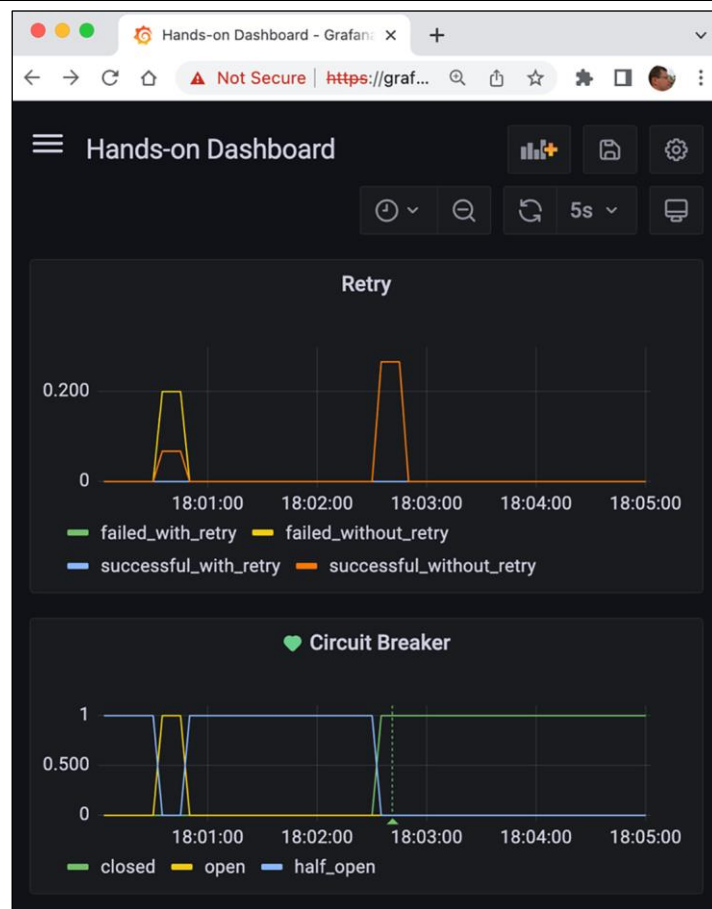
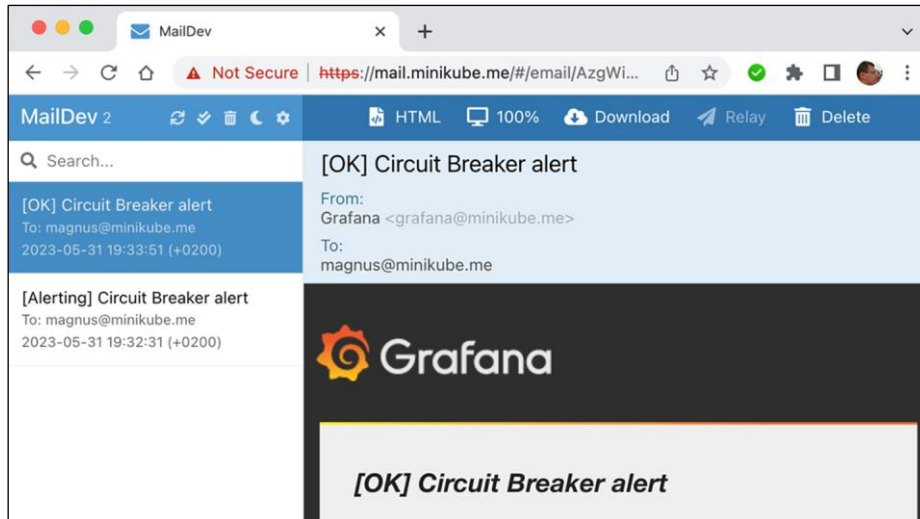
Sort order

Alphabetical (asc)

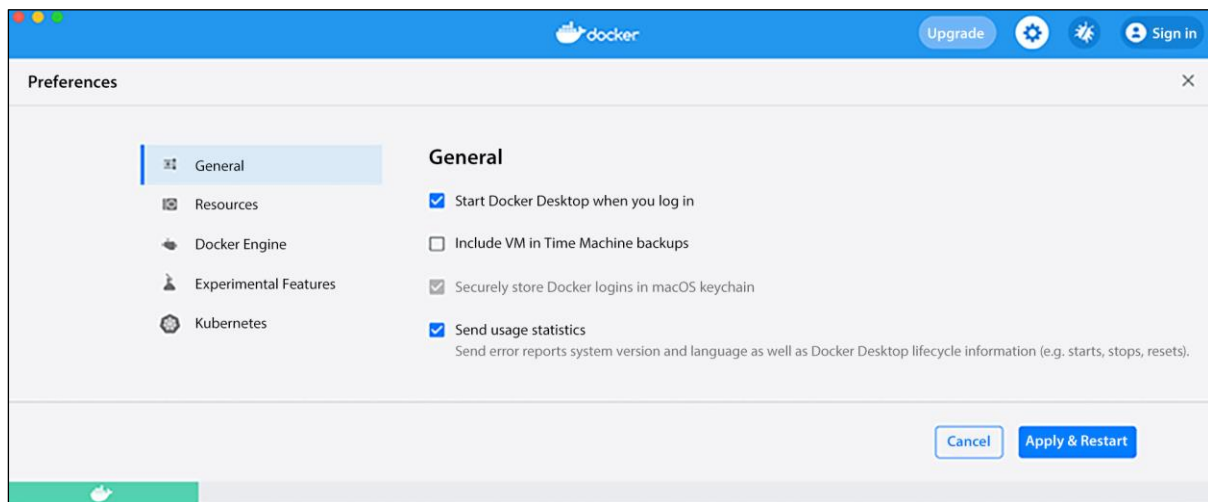
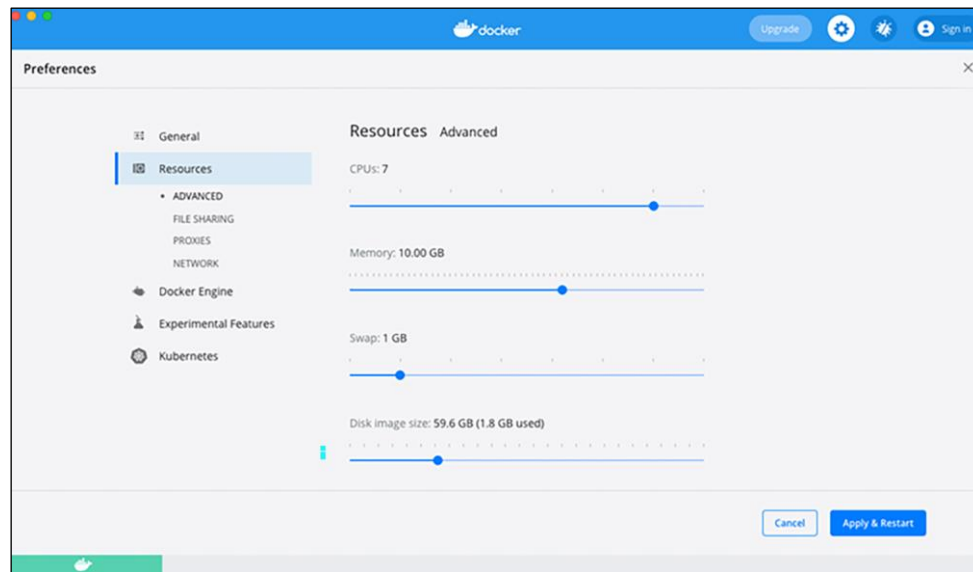
Alerts from this dashboard

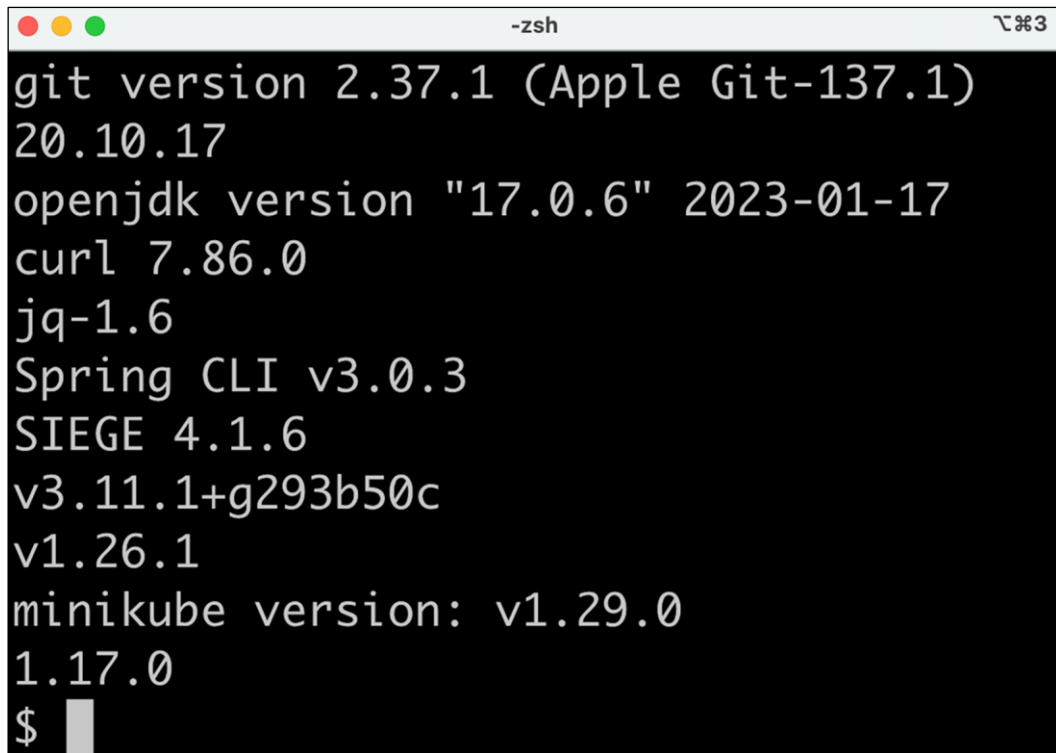






Chapter 21: Installation Instructions for macOS

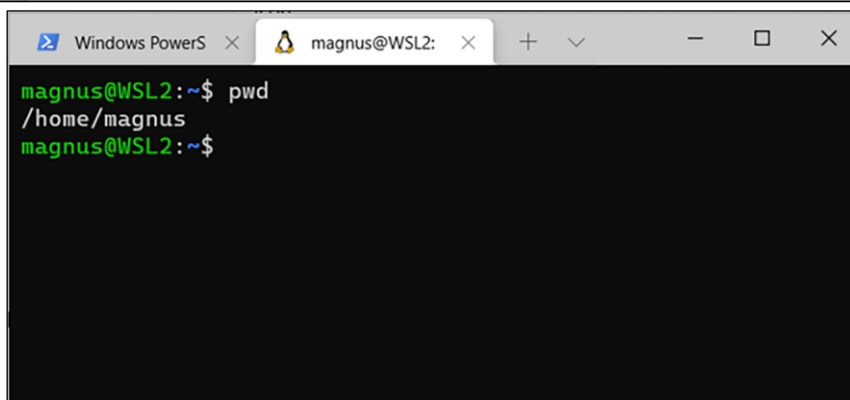
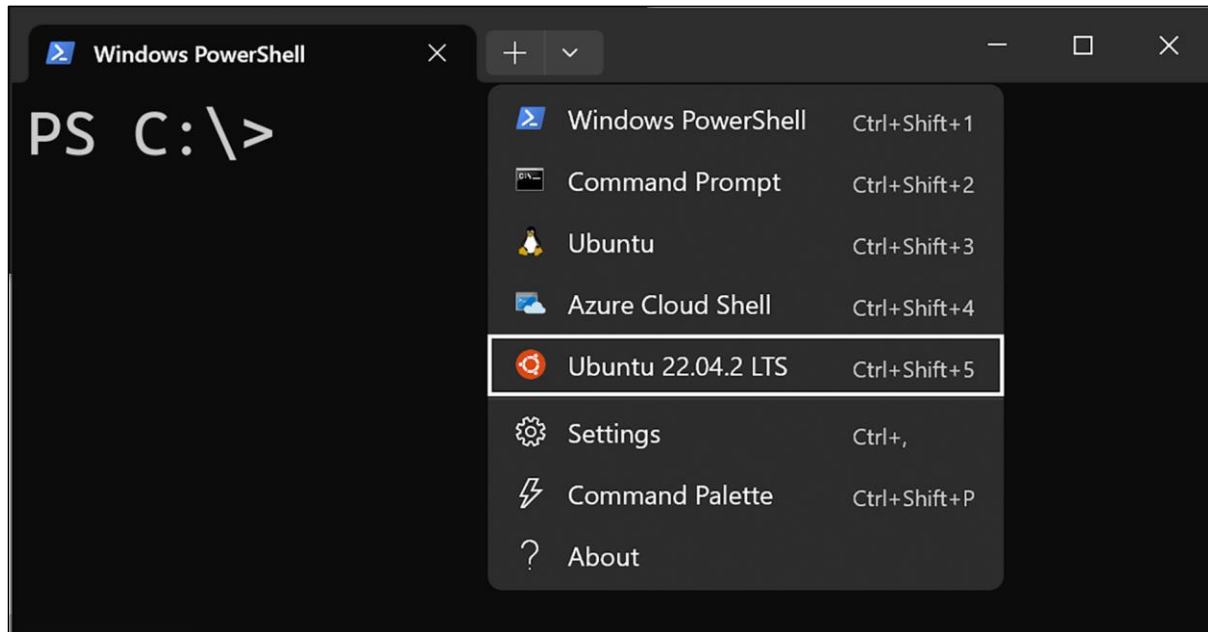




```
git version 2.37.1 (Apple Git-137.1)
20.10.17
openjdk version "17.0.6" 2023-01-17
curl 7.86.0
jq-1.6
Spring CLI v3.0.3
SIEGE 4.1.6
v3.11.1+g293b50c
v1.26.1
minikube version: v1.29.0
1.17.0
$
```

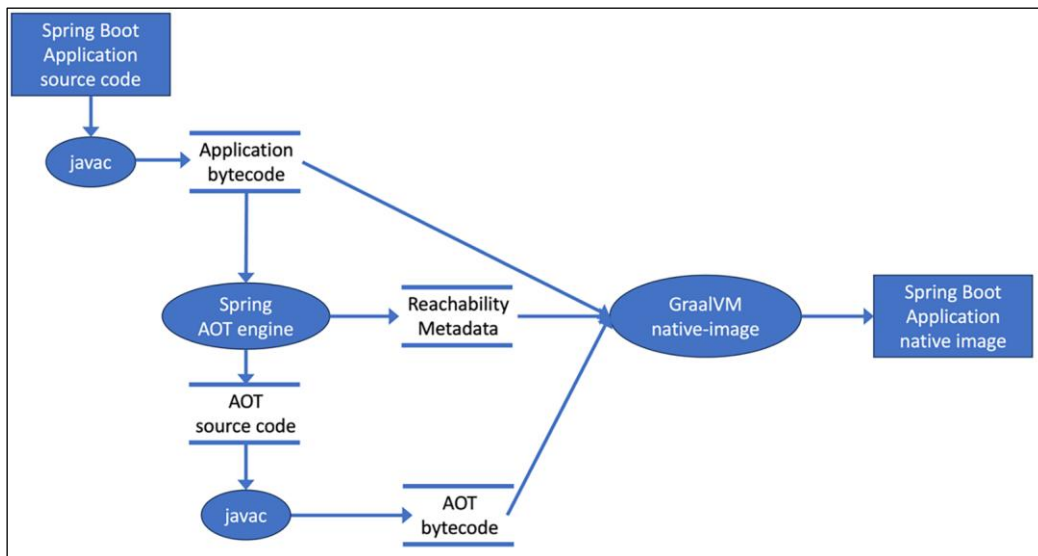
A terminal window with a title bar containing three colored dots (red, yellow, green) on the left, the text "-zsh" in the center, and a keyboard icon with "⌘3" on the right. The terminal has a black background with white text. It displays the output of a command that lists the versions of various installed tools. The tools and their versions are: git (2.37.1, Apple Git-137.1), openjdk ("17.0.6", 2023-01-17), curl (7.86.0), jq (1.6), Spring CLI (v3.0.3), SIEGE (4.1.6), v3.11.1+g293b50c, v1.26.1, minikube (v1.29.0), and 1.17.0. The prompt "\$" is followed by a white cursor bar.

Chapter 22: Installation Instructions for Microsoft Windows with WSL 2 and Ubuntu




```
Windows PowerShell  magnus@wsl2: ~  
git version 2.34.1  
20.10.21  
openjdk version "17.0.6" 2023-01-17  
curl 7.81.0  
jq-1.6  
Spring CLI v3.0.3  
SIEGE 4.0.7  
v3.11.2+g912ebc1  
v1.26.1  
minikube version: v1.29.0  
1.17.0  
magnus@wsl2: ~$
```

Chapter 23: Native Compiled Java Microservices



```
-bash
hands-on/native-product-composite-service latest ...
hands-on/native-recommendation-service latest ...
hands-on/native-product-service latest ...
hands-on/native-review-service latest ...
$
```

```
bash
recommendation-1 | ... Started RecommendationServiceApplication in 5.809 seconds
product-composite-1 | ... Started ProductCompositeServiceApplication in 6.696 seconds
product-1 | ... Started ProductServiceApplication in 5.611 seconds
review-1 | ... Started ReviewServiceApplication in 6.901 seconds
$
```

```
bash
Start Tests: Sat Jun 17 09:36:19 CEST 2023
...
Wait for: curl -k https://health.minikube.me/actuator/health...
DONE, continues...
...
Test OK (HTTP Code: 200)
...
End, all tests OK: Sat Jun 17 09:36:40 CEST 2023
$
```

```
bash
CONTAINER ID   NAME                                CPU %   MEM USAGE ...
c2fd8c3cf276   chapter23-review-1                 0.14%   307.7MiB ...
ecb84348619f   chapter23-recommendation-1         0.13%   251.4MiB ...
d276f9d9c4bf   chapter23-product-composite-1      0.16%   293.6MiB ...
0aa87dcc0145   chapter23-product-1                0.19%   245.2MiB ...
8a37a5918b2f   chapter23-jaeger-collector.istio-system-1 0.13%   198.4MiB ...
6ee062ef5073   chapter23-gateway-1                0.12%   232.9MiB ...
efc90cd09472   chapter23-auth-server-1            0.20%   164.2MiB ...
32cf0ffa6870   chapter23-mongodb-1                1.17%   164.7MiB ...
7ead773af325   chapter23-mysql-1                  0.85%   394.6MiB ...
b46099e0b2ab   chapter23-rabbitmq-1               7.96%   122.8MiB ...
$
```

```
bash
product-composite-1 | ... Started ProductCompositeServiceApplication in 0.482 seconds
product-1 | ... Started ProductServiceApplication in 0.205 seconds
recommendation-1 | ... Started RecommendationServiceApplication in 0.198 seconds
review-1 | ... Started ReviewServiceApplication in 0.269 seconds
$
```

```
bash
CONTAINER ID   NAME                                CPU %     MEM USAGE ...
bc0db067daef   chapter23-review-1                 0.03%     98.16MiB ...
5dd4faceb8ab   chapter23-product-1                0.04%     81.8MiB ...
f2add10d08de   chapter23-recommendation-1         0.03%     87.82MiB ...
b5fa5e14d4a6   chapter23-product-composite-1      0.02%     131.8MiB ...
aedcef0006a5   chapter23-jaeger-collector.istio-system-1 0.58%     202.4MiB ...
29051d43ede6   chapter23-gateway-1                0.22%     238.7MiB ...
960630b2ecf0   chapter23-auth-server-1            0.39%     160.1MiB ...
2ba3e34a60a8   chapter23-mongodb-1                3.11%     164.4MiB ...
ab50716bc791   chapter23-rabbitmq-1               0.72%     119.7MiB ...
d505bd4e5713   chapter23-mysql-1                  1.33%     397.3MiB ...
$
```

```
bash
2023-06-17T08:20:50.759Z ... Started ProductCompositeServiceApplication in 0.442 seconds
$
```

```
bash
hands-on/auth-server:latest
hands-on/native-product-service:latest
hands-on/native-product-composite-service:latest
hands-on/native-recommendation-service:latest
hands-on/native-review-service:latest
$
```